

Project Data Page:

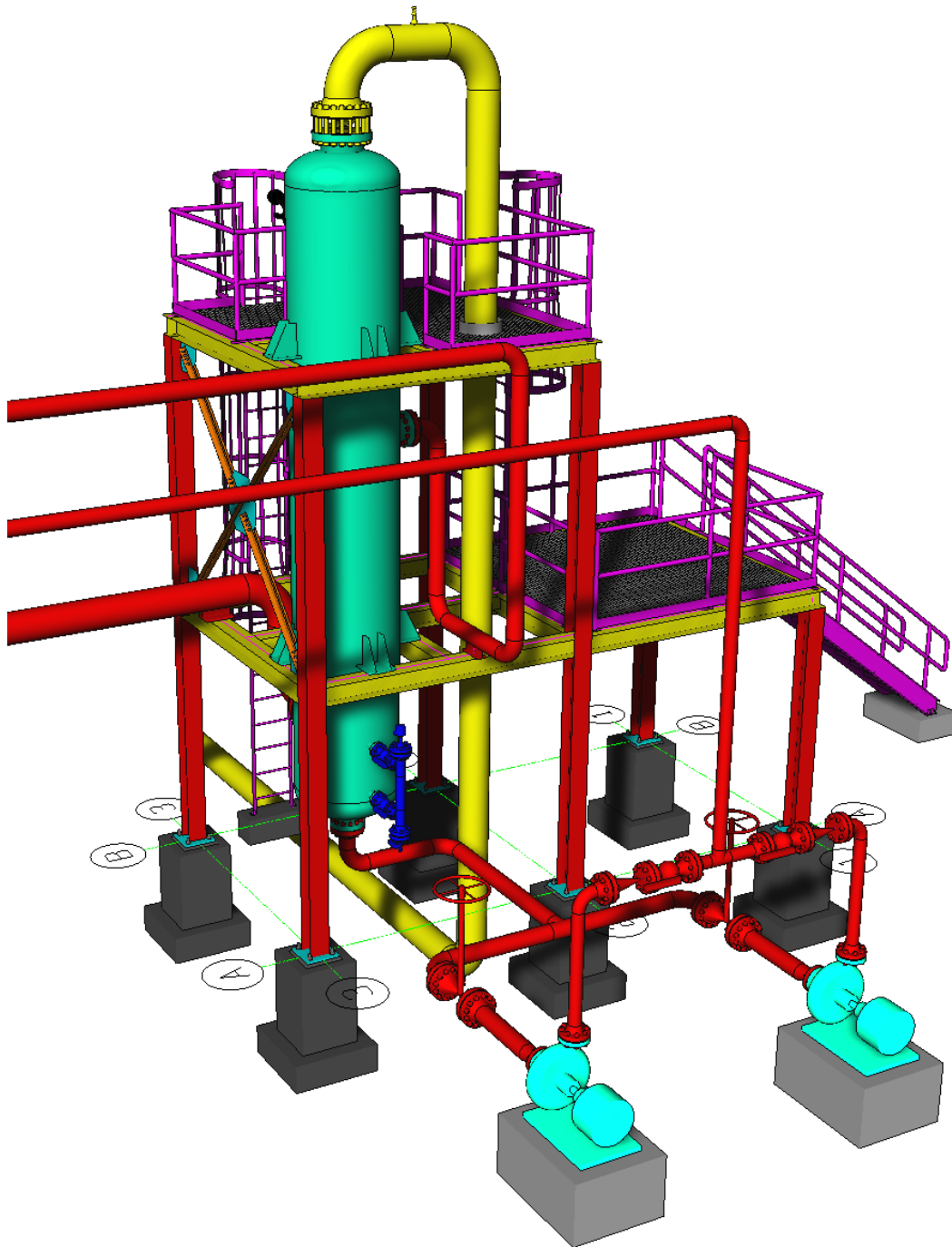




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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2023

Analysis Performed by : SL User

Job File : ...\\135-CBEU.pvdb

Date of Analysis : Sep 9, 2026 9:31am

PV Elite 27, January 2025



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FileName : 135-C_BEU

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Vessel Design Summary:

Step: 35

9:32am

Sep 9, 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2023

Diameter Spec : 440.000 mm ID
Vessel Design Length, Tangent to Tangent 4089.85 mm
Specified Datum Line Distance 50.00 mm
Shell Side Design Temperature 280 °C
Channel Side Design Temperature 160 °C
Shell Side Design Pressure 520.000 KPa.
Channel Side Design Pressure 6400.000 KPa.
Shop Shell Side Test Pressure 2548.008 KPa.
Field Shell Side Test Pressure 676.000 KPa.
Shop Channel Side Test Pressure 11016.894 KPa.
Field Channel Side Test Pressure 8320.000 KPa.
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Normal ized	Impact Tested
Shell	SA-516 70	...	...	K02700	No	No
Head	SA-516 70	...	...	K02700	No	No
Flange	SA-266 2	...	...	K03506	No	No
Nozzle	SA-266 2	...	...	K03506	No	No
Nozzle	SA-106 B	...	...	K03006	No	No
Re-Pad	SA-516 70	...	...	K02700	No	No
Nozzle Flg	SA-105	...	...	K03504	No	No
Tubes	SA-179	...	...	K01200	No	No
Tubesheet	SA-266 2	...	...	K03506	No	No
Flg Bolting	SA-193 B7	...	<= 64	G41400	No	No
Hrz Bolting	SA-307 A	...	6 <= t <= 100		No	No
Saddles	SA-516 70	...	...	K02700	No	No

- Normalized is determined based on the UCS-66 material curve selection and Figure UCS-66.
- Impact Tested is based on material selection and material data properties.

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Channel head	6403.950	0.00	6640.000	3.0000	No	No
Channel shell	6403.950	0.00	7768.000	3.0000	N/A	No
Channel flng	6404.166	0.00	7768.000	3.0000	N/A	No
Shell flng	520.000	101.30	5390.000	3.0000	N/A	No
Shell	523.950	101.30	5386.000	3.0000	N/A	No
Shell head	523.921	101.30	6531.000	0.0000	No	No

Liquid Level: 425.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element	"To" Elev	Element	Nominal	Finished	Reqd Thk	Reqd Thk	Long	Circ
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Vessel Design Summary:

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Type	mm	Length mm	Thickness mm	Thickness mm	Internal mm	External mm	Eff	Eff
Ellipse	0.0	50.0	16.0	13.6	13.2	...	1.00	1.00
Cylinder	600.0	600.0	16.0	16.0	13.6	...	1.00	1.00
Body Flg	746.7	139.0	123.0	115.0	110.5	110.5	1.00	1.00
Body Flg	989.9	138.0	126.0	118.0	114.2	114.2	1.00	1.00
Cylinder	3989.8	3000.0	12.0	12.0	5.5	6.2	1.00	1.00
Ellipse	4039.9	50.0	12.0	10.2	2.5	1.2	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N4-Tube inlet	WRC-297	0.819	Shell	Passed
Tube outlet	WRC-297	0.819	Shell	Passed
Shell outlet	WRC-297	0.521	Nozzle	Passed
Shell Inlet	WRC-297	0.472	Nozzle	Passed

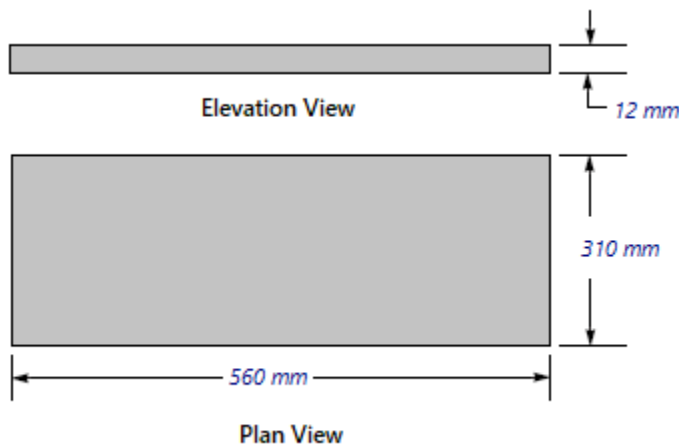
Foundation/Support Summary

Saddle Parameters:

Saddle Width	285.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	600.000 mm
Wear Pad Width	365.000 mm
Wear Pad Thickness	15.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	620.000 mm

Baseplate Length	560.000 mm
Baseplate Thickness	12.000 mm
Baseplate Width	310.000 mm
Number of Ribs (including outside ribs)	3
Rib Thickness	10.000 mm
Web Thickness	10.000 mm
Height of Center Web	390.000 mm
Number of Bolts in Baseplate	4

Baseplate Sketch



Maximum Tensile Bolt Load

4. Kg



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Vessel Design Summary:

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Sep 9, 2026

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	3700.98	Kgf
Maximum Transverse Saddle Shear Load	115.07	Kgf
Maximum Longitudinal Saddle Shear Load	195.41	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	3235.32	Kgf
Maximum Transverse Saddle Shear Load	36.26	Kgf
Maximum Longitudinal Saddle Shear Load	18.92	Kgf

Weights:

Fabricated - Bare w/o removable internals	2474.5	kg.
Shop Test - Fabricated + water (full)	3060.8	kg.
Shipping - Fab. + rem. intls.+ shipping app.	2474.5	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	2674.5	kg.
Empty - Fab. + intls. + details + wghts.	2937.8	kg.
Operating - Empty + operating liquid (no ca)	3496.4	kg.
Field Test - Empty weight + test medium (full)	3193.2	kg.

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Nozzle Summary:

Step: 33 9:32am Sep 9,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N4-Tube inlet	7768	...	...	OK	10.16	OK	Passed
Tube outlet	6510	...	...	OK	9.22	OK	Passed
Shell outlet	5386	...	...	OK	5.50	OK	No Calc[*]
Shell Inlet	5386	OK	...	OK	5.50	OK	Passed
Vent	-0.0069	...	...	Failed	5.50	OK	No Calc[*]
Drain	-0.0069	...	...	Failed	5.50	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 5386 Nozzle : Shell outlet [Shellside]
 Minimum MAWP Nozzles : 6510 Nozzle : Tube outlet [Tubeside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
20	N4-Tube inlet	350.000	90.000	399.350
20	Tube outlet	350.000	270.000	304.659
50	Shell outlet	3889.850	270.000	159.320
50	Shell Inlet	1339.850	90.000	206.361
50	Vent	2539.850	90.000	93.890
50	Drain	2539.850	270.000	93.890

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 32 9:32am Sep 9,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
Vent	1.612 in	Actual	LWN	2.375	9.690	100.00	12.00	214.08	150
Drain	1.612 in	Actual	WNF	2.375	9.690	100.00	12.00	214.08	150
Shell outlet	3.000 in	80	LWN	3.500	7.620	...	...	234.54	600
Shell Inlet	4.000 in	80	WNF	4.500	8.560	...	...	187.55	300
Tube outlet	6.000 in	80	WNF	6.625	10.973	...	...	241.69	600
N4-Tube inlet	8.000 in	80	WNF	8.625	12.700	...	...	281.72	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Vent	SA-106 B	12.000	20.000	10.000	12.000	...
Drain	SA-106 B	12.000	20.000	10.000	12.000	...
Shell outle	SA-106 B	7.620	10.000	...	...	...
Shell Inlet	SA-106 B	8.560	10.000	...	...	...
Tube outlet	SA-266 2	14.000	15.000	...	...	...
N4-Tube inl	SA-266 2	14.000	15.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Vent	2489.850	90.0	200.00	0.00	Shell
Drain	2489.850	270.0	200.00	0.00	Shell
Shell outlet	3839.850	270.0	200.00	0.00	Shell
Shell Inlet	1289.850	90.0	150.00	0.00	Shell
Tube outlet	300.000	270.0	200.00	0.00	Channel shell
N4-Tube inlet	300.000	90.0	200.00	0.00	Channel shell

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Bill of Material:

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 16.0mm THK X 440.0mm ID X 50.0mm	SA-516 70
1	WELD NECK FLANGE: 123.0mm THK X 670.0mm OD	SA-266 2
1	WELD NECK FLANGE: 126.0mm THK X 670.0mm OD	SA-266 2
1	ELLIPTICAL HEAD: 2.0 X 1, 12.0mm THK X 425.0mm ID X 50.0mm	SA-516 70
1	INSULATION: 50mm X 80mm THK	
1	INSULATION: 600mm X 80mm THK	Hot
1	INSULATION: 135mm X 80mm THK	Hot Insulation
8	BAFFLES: 390.0mm DIA X 5.0mm THK	SA-516 70
6	TIE RODS: 25mm DIA X 2500mm	SA-193 B7
1	TUBESHEET: 670.0mm X 102.0mm THK	SA-266 2
106	TUBES: 3000.0mm X 25.4mm DIA X 2.1mm THK	SA-179
1	INSULATION: 95mm X 80mm THK	Hot insulation
2	SADDLE: 285.0mm X 120 DEG	SA-516 70
1	INSULATION: 3000mm X 100mm THK	Hot
1	INSULATION: 150mm X 100mm THK	Hot
1	GASKET: 495.0mm OD X 469.0mm ID	...
20	BODY FLANGE BOLTS: 38.1mm DIA	SA-193 B7
40	NUTS FOR BODY FLANGE BOLTS: 38.1mm DIA	...
2	CYLINDRICAL SEGMENT 16.0mm THK X 549.1mm X 600.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 284.1mm X 600.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm THK X 533.7mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm THK X 529.9mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm THK X 318.7mm X 3000.0mm	SA-516 70
1	NAMEPLATE	...



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MDMT Summary:

Step: 34 9:32am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Shell flng [11]	B	-23	-104	-29	0.150	12.000	1.00	No
Shell [8]	B	-23	-104	-29	0.278	12.000	1.00	No
Shell head [10]	B	-28	-104	-29	0.245	10.200	1.00	No
Shell head [7]	B	-23	-104	-29	0.208	12.000	1.00	No
Left Saddle [24]	B	-23	-104	-29	0.278	12.000	1.00	No
Sdl 2 Fr40 [24]	B	-29	-104	-29	0.278	10.000	1.00	No
Shell outlet [1]	B	-29	-104	-29	0.048	6.668	1.00	No
Shell Inlet [1]	B	-8	-104	-29	0.051	7.490	1.00	No
Nozzle Flg [5]	A	-18	-104					
Vent [1]	B	-29	-104	-29	0.016	9.690	1.00	No
Nozzle Flg [5]	A	-18	-104					
Drain [1]	B	-8	-104	-29	0.016	9.690	1.00	No
Nozzle Flg [5]	A	-18	-104					
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No
Tubesheet: SS[13]	B	0	-39		0.455	25.500	1.00	No
Warmest MDMT:		0	-39					
Channel flng [11]	B	-14	-32	-29	0.677	16.000	1.00	No
Channel head [10]	B	-19	-20	-29	0.980	13.600	1.00	No
Channel head [7]	B	-14	-23	-29	0.833	16.000	1.00	No
Channel shell [8]	B	-14	-23	-29	0.833	16.000	1.00	No
N4-Tube inlet [1]	B	-5	-29	-29	0.587	11.113	1.00	No
Nozzle Flg [5]	A	-18	-39					
Tube outlet [1]	B	-14	-29	-29	0.819	16.000	1.00	No
Nozzle Flg [5]	A	-18	-38					
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No
Tubesheet: CS[14]	B	0	-39		0.455	25.500	1.00	No
Bolting [21]		-48						
Warmest MDMT:		0	-20					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-39.0	-12.0	Pass
Channel/Tube	-29.0	-12.0	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material



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MDMT Summary:

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[22] - Nozzle Flange Bolting Material
[23] - Stiffening Ring to Shell Weld
[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 9:31am Sep 9,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning: Channel flng and Shell flng

There are mating flanges in this model that have different design pressures. This will generate different bolt loads. If a tubesheet is in between these flanges, PV Elite will compute the mating flange loads for you. If there is not a tubesheet physically separating these flanges, you must enter in the mating flange loads into the flange dialog.

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

Step: 1 9:31am Sep 9,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	520	KPa.
Channel Side Design Pressure	6400	KPa.
Shell Side Design Temperature	280.0	°C
Channel Side Design Temperature	160.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



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Input Echo:

Step: 1 9:31am Sep 9,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	Channel head
Distance "FROM" to "TO"	50 mm
Inside Diameter	440 mm
Element Thickness	13.6 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	6400 KPa.
Design Temperature Internal Pressure	160 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230.4 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	400 mm
Liquid Density	0.0009984 kg./cm ³



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Input Echo:

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Element From Node	10
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	50 mm
Thickness of Insulation	80 mm
Density	0.000352 kg./cm ³

Element#	2/6
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Channel shell
Distance "FROM" to "TO"	600 mm
Inside Diameter	440 mm
Element Thickness	16 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	6400 KPa.
Design Temperature Internal Pressure	160 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	400 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	600 mm
Thickness of Insulation	80 mm
Density	0.000352 kg./cm ³

Element From Node	20
Detail Type	Nozzle
Detail ID	N4-Tube inlet
Dist. from "FROM" Node / Offset dist	300 mm
Nozzle Diameter	8 in.
Nozzle Schedule	80
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	90.031 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2

Element From Node	20
Detail Type	Nozzle
Detail ID	Tube outlet
Dist. from "FROM" Node / Offset dist	300 mm
Nozzle Diameter	6 in.
Nozzle Schedule	80



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Input Echo:

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Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	49.766	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element From Node	20	
Detail Type	Weight	
Detail ID	Partiotion weig	
Dist. from "FROM" Node / Offset dist	250	mm
Miscellaneous Weight	100	Kgf
Offset from Element Centerline	0	mm

Element#	3/6	
Element From Node	30	
Element To Node	40	
Element Type	Flange	
Description	Channel flng	
Distance "FROM" to "TO"	139	mm
Flange Inside Diameter	440	mm
Element Thickness	115	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	123	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	6400	KPa.
Design Temperature Internal Pressure	160	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	217.8	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	425	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	135	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³

Element#	4/6
Element From Node	40



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Input Echo:

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Element To Node	50
Element Type	Flange
Description	Shell flng
Distance "FROM" to "TO"	138 mm
Flange Inside Diameter	440 mm
Element Thickness	118 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	126 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	280 °C
Design External Pressure	101.3 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	40
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	95 mm
Thickness of Insulation	80 mm
Density	0.000352 kg./cm³

Element#	5/6
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	3000 mm
Inside Diameter	440 mm
Element Thickness	12 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	280 °C
Design External Pressure	101.3 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm²
Allowable Stress, Operating	136.8 N./mm²
Allowable Stress, Hydrotest	234 N./mm²
Material Density	0.00775 kg./cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	208.8 N./mm²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	620 mm
Width of Saddle	285 mm
Height of Saddle at Bottom	600 mm



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Input Echo:

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Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	365	mm
Thickness of Wear Plate	15	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	50	
Detail Type	Saddle	
Detail ID	Sdl 2 Fr40	
Dist. from "FROM" Node / Offset dist	2480	mm
Width of Saddle	285	mm
Height of Saddle at Bottom	600	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	365	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	400	mm
Liquid Density	0.0009984	kg./cm³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Shell outlet	
Dist. from "FROM" Node / Offset dist	2850	mm
Nozzle Diameter	3	in.
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	8.6222	Kgf
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Shell Inlet	
Dist. from "FROM" Node / Offset dist	300	mm
Nozzle Diameter	4	in.
Nozzle Schedule	80	
Nozzle Class	300	



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Input Echo:

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Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	12.06	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Vent	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	1.612	in.
Nozzle Schedule	XXS	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	6.1172	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Drain	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	1.612	in.
Nozzle Schedule	XXS	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	6.3818	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Weight	
Detail ID	Fire proofing1	
Dist. from "FROM" Node / Offset dist	620	mm
Miscellaneous Weight	100	Kgf
Offset from Element Centerline	500	mm

Element From Node	50	
Detail Type	Weight	
Detail ID	Fire proofing 2	
Dist. from "FROM" Node / Offset dist	2480	mm
Miscellaneous Weight	100	Kgf
Offset from Element Centerline	500	mm

Element#	6/6	
Element From Node	60	
Element To Node	70	
Element Type	Elliptical	
Description	Shell head	
Distance "FROM" to "TO"	50	mm
Inside Diameter	425	mm
Element Thickness	10.2	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	520	KPa.
Design Temperature Internal Pressure	280	°C
Design External Pressure	101.3	KPa.
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	



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Input Echo:

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Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	400 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	150 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

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XY Coordinate Calculations:

Step: 2 9:31am

Sep 9,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Channel head		50	...	50	...
Channel shell		650	...	600	...
Channel flng		796.675	...	139	...
Shell flng		1039.85	...	138	...
Shell		4039.85	...	3000	...
Shell head		4089.85	...	50	...

PV Elite includes an 1/8 inch (3.175mm) raised face and gasket thicknesses for girth flanges and tubesheet thicknesses where applicable in the Tangent to Tangent length calculation. The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact.

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Flg Calc [Int P]: Channel flng Flng: 11 9:31am Sep 9,2026

Flange Input Data Values

Description: Channel flng :

Channel flng

Flange Type/Sketch	Integral Weld Neck		
Design Pressure	P	6404.17	KPa.
Design Temperature		160	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	440.000	mm
Flange Outside Diameter	A	670.000	mm
Flange Thickness	t	115.0000	mm
Thickness of Hub at Small End	go	16.0000	mm
Thickness of Hub at Large End	gl	23.0000	mm
Length of Hub	h	24.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	590.000	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		20	
Flange Face Outside Diameter	Fod	506.000	mm
Flange Face Inside Diameter	Fid	440.000	mm
Flange Facing Sketch	1, Code Sketch 1a		
Gasket Outside Diameter	Go	495.000	mm
Gasket Inside Diameter	Gi	469.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Column for Gasket Seating	2, Code Column II		
Gasket Thickness	tg	4.5000	mm
Length of Partition Gasket	lp	425.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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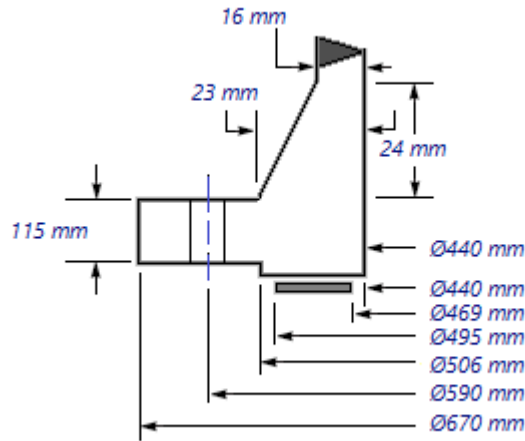
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Flg Calc [Int P]: Channel flng

Flng: 11

9:31am

Sep 9, 2026



ASME Code, Section VIII Division 1, 2023

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (6404 \cdot (440/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 6404) + Ca$$

$$= 13.6460 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 13) / (223 + 0.6 \cdot 13)$$

$$= 7772.371 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 16) / (220 + 0.6 \cdot 16)$$

$$= 9615.993 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	112.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	446.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	20.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	13.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	52.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $bo = N/2$	6.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	482.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H]:

$$= \pi/4 \cdot G^2 \cdot P_{eq}$$

$$= \pi/4 \cdot 482^2 \cdot 6404$$

$$= 119238.086 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 482 \cdot 3 \cdot 6404$$

$$+ 2 \cdot 425 \cdot 5 \cdot 3 \cdot 6404$$

$$= 46458.398 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 446^2 \cdot 6404$$

$$= 102027.859 \text{ Kgf}$$



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Flg Calc [Int P]: Channel flng Flng: 11 9:31am Sep 9,2026

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 119238 - 102028$$

$$= 17210.229 \text{ Kgf}$$

Operating Bolt Load [Wm1]:

$$= \max(H + H_p + H'_p, 0)$$

$$= \max(119238 + 46458 + 0, 0)$$

$$= 165696.469 \text{ Kgf}$$

Gasket Seating Bolt Load [Wm2]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.42 \cdot \pi \cdot 482 + 68.9 \cdot 5 \cdot 425$$

$$= 83360.320 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max(W_{m1}/S_b, W_{m2}/S_a)$$

$$= \max(165696/172, 83360/172)$$

$$= 94.476 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmx]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 112/(3 + 0.5)$$

$$= 268.200 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 590 \cdot \sin(\pi / 20)$$

$$= 92.296 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 92.3 / (2 \cdot 38.1 + 112) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	94.476	181.290	
Radial Distance between Hub and Bolts:	50.800	52.000	
Radial Distance between Bolts and Edge:	38.100	40.000	
Circ. Spacing between the Bolts:	82.550	92.296	268.200

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 181 \cdot 172 / (68.9 \cdot \pi (495 + 469))$$

$$= 14.933 \text{ mm [Note: Exceeds actual gasket width, 13.000]}$$

Flange Design Bolt Load, Gasket Seating [W]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 181$$

$$= 317955.00 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_{m1} - H$$

$$= 165696 - 119238$$

$$= 46458.39 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$



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Flg Calc [Int P]: Channel flng Flng: 11 9:31am Sep 9,2026

$$= (590 - 482) / 2$$

$$= 53.9245 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= (R + g1 + hg) / 2$$

$$= (52 + 20 + 53.9) / 2$$

$$= 62.9623 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= R + (g1 / 2)$$

$$= 52 + (20 / 2)$$

$$= 62.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	102028.	62.0000	1.0000	6326.
Face Pressure, Mt	17210.	62.9623	1.0000	1084.
Gasket Load, Mg	46458.	53.9245	1.0000	2505.
Gasket Seating, Matm	317955.	53.9245	1.0000	17146.

Total Moment for Operation, Mop 9915. Kg-m.
Total Moment for Gasket seating, Matm 17146. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 76.145 mm
Hub Ratio, h/h0 = HL / H0 0.315
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.538

Flange Factors for Integral Flange:

Factor F 0.875
Factor V 0.357
Factor f 1.167

Factors from Figure 2-7.1 K = 1.502
T = 1.710 U = 5.432
Y = 4.944 Z = 2.591
e = 0.0115 1/mm d = 0.19578E+06 mm³

Stress Factors Alpha = 2.288
Beta = 2.717 Gamma = 1.338
Delta = 7.176 Lambda = 8.514

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g1^2)$$

$$= (1.17 \cdot 9915 / 446) / (8.51 \cdot 20^2)$$

$$= 74.71 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g1^2)$$

$$= (1.17 \cdot 17146 / 446) / (8.51 \cdot 20^2)$$

$$= 129.20 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.72 \cdot 9915 / 446) / (8.51 \cdot 112^2)$$

$$= 5.55 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.72 \cdot 17146 / 446) / (8.51 \cdot 112^2)$$

$$= 9.59 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (4.94 \cdot 9915 / (112^2 \cdot 446)) - 2.59 \cdot 5.55$$



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$$= 71.54 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (4.94 \cdot 17146 / (112^2 \cdot 446)) - 2.59 \cdot 9.59$$

$$= 123.72 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (74.7 + \max(5.55, 71.5)) / 2$$

$$= 73.13 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (129 + \max(9.59, 124)) / 2$$

$$= 126.46 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 165696 / 181$$

$$= 89.63 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (83360 / 181)$$

$$= 45.09 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	74.71	207.00	129.20	207.00
Radial Flange	5.55	138.00	9.59	138.00
Tangential Flange	71.54	138.00	123.72	138.00
Maximum Average	73.13	138.00	126.46	138.00
Bolting	89.63	172.00	45.09	172.00

Minimum Required Flange Thickness 110.465 mm

Estimated M.A.W.P. (Operating) 7772 KPa.

Estimated Finished Weight of Flange at given Thk. 184.0 kg.

Estimated Unfinished Weight of Forging at given Thk 216.0 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	394.03	497.82	0.792	394.03	500.02	0.788
Tangential Flange	St	373.37	373.37	1.000	373.37	375.01	0.996
Tangential Flange	Sto	116.43	248.91	0.468	116.43	250.01	0.466
Combined	Sh/f + St	707.44	746.74	0.947	707.44	750.03	0.943
Combined	Sr + St	402.31	746.74	0.539	402.31	750.03	0.536
Combined	Sh + Sto	510.45	746.74	0.684	510.45	750.03	0.681
Radial Flange	Sr	28.94	497.82	0.058	28.94	500.02	0.058
Maximum Stress Ratio				1.000			0.996



Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.5/1.27)

= 1.1795

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)

= max(0.25; 0.42/1.21) if h/ho < 0.35, else max (0.25; 0.28/0.77)

= 0.3485

In this case, the hub ratio h/ho, 0.3152 was < 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	519.1 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	275.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.260
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Root Area of a Single Bolt	Abs	9.0645 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	217.8 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.257 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$$

$$= \pi/4 (495^2 - 469^2) + (0.5 \cdot 425 \cdot 5)$$

$$= 207.5 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208 \cdot 207 / (20 \cdot 9.06)$$

$$= 238.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(238; 508)$$

$$= 238.0 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(238; 290)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290; 519)$$

$$= 290.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 290 \geq 140 (207 / (20 \cdot 9.06))$$

$$:: 290.0 \geq 160.2 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 290 \geq (97 \cdot 207 + \pi/4 \cdot 6404 \cdot 469^2) / (0.7 \cdot 9.06 \cdot 20)$$

$$:: 290.0 \geq 245.8 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 290 \leq 275 (207 / (20 \cdot 9.06))$$

$$:: 290.0 \leq 314.7 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(\text{thetagmax} / \text{thetafmax})$$

$$:: 290 \leq 519 (1/0.26)$$

$$:: 290.0 \leq 2023.0 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 290 \cdot 0.26 \cdot 9.06 \cdot 38.1$$

$$= 265.5 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 101 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange Rigidity Based on Required Thickness [ASME]:



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Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 17146 / 1 \cdot 9807 \cdot 0.36 / (7.65 \cdot 201334 \cdot 13^2 \cdot 76.1 \cdot 0.3)$
 $= 0.526$ (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 9915 / 1 \cdot 9807 \cdot 0.36 / (7.65 \cdot 193199 \cdot 13^2 \cdot 76.1 \cdot 0.3)$
 $= 0.317$ (should be ≤ 1)

Flange Rigidity Based on Given Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 17146 / 1 \cdot 9807 \cdot 0.36 / (8.51 \cdot 201334 \cdot 13^2 \cdot 76.1 \cdot 0.3)$
 $= 0.473$ (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 9915 / 1 \cdot 9807 \cdot 0.36 / (8.51 \cdot 193199 \cdot 13^2 \cdot 76.1 \cdot 0.3)$
 $= 0.285$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.68, Temperature Reduction per Fig. UCS 66.1 = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-32 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

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Flg Calc [Int P]: Shell flng

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Flange Input Data Values

Description: Shell flng :

Shell flng

Flange Type/Sketch	Integral Weld Neck		
Design Pressure	P	520.00	KPa.
Design Temperature		280	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	440.000	mm
Flange Outside Diameter	A	670.000	mm
Flange Thickness	t	118.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	17.0000	mm
Length of Hub	h	20.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	131.80	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	590.000	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		20	
Flange Face Outside Diameter	Fod	506.000	mm
Flange Face Inside Diameter	Fid	440.000	mm
Flange Facing Sketch	1, Code Sketch 1a		
Gasket Outside Diameter	Go	495.000	mm
Gasket Inside Diameter	Gi	469.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Column for Gasket Seating	2, Code Column II		
Gasket Thickness	tg	4.5000	mm
Length of Partition Gasket	lp	425.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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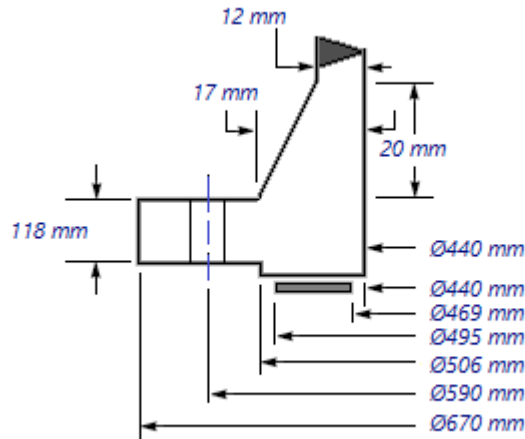
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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (520 \cdot (440/2 + 3)) / (137 \cdot 1 - 0.6 \cdot 520) + Ca$$

$$= 3.8497 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137 \cdot 1 \cdot 9) / (223 + 0.6 \cdot 9)$$

$$= 5390.170 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 12) / (220 + 0.6 \cdot 12)$$

$$= 7288.178 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	115.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	446.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	14.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	9.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	58.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $bo = N/2$	6.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	482.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H]:

$$= \pi/4 \cdot G^2 \cdot P_{eq}$$

$$= \pi/4 \cdot 482^2 \cdot 520$$

$$= 9681.792 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{part} \cdot m_{part} \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 482 \cdot 3 \cdot 520$$

$$+ 2 \cdot 425 \cdot 5 \cdot 3 \cdot 520$$

$$= 3772.290 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 446^2 \cdot 520$$

$$= 8284.371 \text{ Kg}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 9682 - 8284$$

$$= 1397.421 \text{ Kgf}$$

Operating Bolt Load [Wm1]:

$$= \max(H + H_p + H'p, 0)$$

$$= \max(9682 + 3772 + 0, 0)$$

$$= 13454.082 \text{ Kgf}$$

$$= 165696.469 \text{ Kgf, Mating Flange Load Governs}$$

Gasket Seating Bolt Load [Wm2]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.42 \cdot \pi \cdot 482 + 68.9 \cdot 5 \cdot 425$$

$$= 83360.320 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max(W_{m1}/S_b, W_{m2}/S_a)$$

$$= \max(165696/172, 83360/172)$$

$$= 94.476 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmx]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 115/(3 + 0.5)$$

$$= 273.343 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 590 \cdot \sin(\pi / 20)$$

$$= 92.296 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$$= \max(\sqrt{ (Bs / (2a + t)) }, 1)$$

$$= \max(\sqrt{ (92.3 / (2 \cdot 38.1 + 115)) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	94.476	181.290	
Radial Distance between Hub and Bolts:	50.800	58.000	
Radial Distance between Bolts and Edge:	38.100	40.000	
Circ. Spacing between the Bolts:	82.550	92.296	273.343

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 181 \cdot 172 / (68.9 \cdot \pi (495 + 469))$$

$$= 14.933 \text{ mm [Note: Exceeds actual gasket width, 13.000]}$$

Flange Design Bolt Load, Gasket Seating [W]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 181$$

$$= 317955.00 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_{m1} - H$$

$$= 165696 - 9682$$

$$= 156014.69 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:



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$$= (C - G) / 2$$

$$= (590 - 482) / 2$$

$$= 53.9245 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= (R + g1 + hg) / 2$$

$$= (58 + 14 + 53.9) / 2$$

$$= 62.9623 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= R + (g1 / 2)$$

$$= 58 + (14 / 2)$$

$$= 65.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	8284.	65.0000	1.0000	538.
Face Pressure, Mt	1397.	62.9623	1.0000	88.
Gasket Load, Mg	156015.	53.9245	1.0000	8413.
Gasket Seating, Matm	317955.	53.9245	1.0000	17146.

Total Moment for Operation, Mop 9040. Kg-m.
Total Moment for Gasket seating, Matm 17146. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 63.356 mm
Hub Ratio, h/h0 = HL / H0 0.316
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.556

Flange Factors for Integral Flange:

Factor F 0.875
Factor V 0.354
Factor f 1.193

Factors from Figure 2-7.1 K = 1.502
T = 1.710 U = 5.432
Y = 4.944 Z = 2.591
e = 0.0138 1/mm d = 78828.148 mm³

Stress Factors Alpha = 2.588
Beta = 3.117 Gamma = 1.514
Delta = 19.294 Lambda = 20.807

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g1^2)$$

$$= (1.19 \cdot 9040 / 446) / (20.8 \cdot 14^2)$$

$$= 58.12 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g1^2)$$

$$= (1.19 \cdot 17146 / 446) / (20.8 \cdot 14^2)$$

$$= 110.24 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (3.12 \cdot 9040 / 446) / (20.8 \cdot 115^2)$$

$$= 2.25 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (3.12 \cdot 17146 / 446) / (20.8 \cdot 115^2)$$

$$= 4.27 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$$



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$$= (4.94 \cdot 9040 / (115^2 \cdot 446)) - 2.59 \cdot 2.25$$

$$= 68.46 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (4.94 \cdot 17146 / (115^2 \cdot 446)) - 2.59 \cdot 4.27$$

$$= 129.86 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (58.1 + \max(2.25, 68.5)) / 2$$

$$= 63.29 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (110 + \max(4.27, 130)) / 2$$

$$= 120.05 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 165696 / 181$$

$$= 89.63 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (83360 / 181)$$

$$= 45.09 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	58.12	197.70	110.24	207.00
Radial Flange	2.25	131.80	4.27	138.00
Tangential Flange	68.46	131.80	129.86	138.00
Maximum Average	63.29	131.80	120.05	138.00
Bolting	89.63	172.00	45.09	172.00

Minimum Required Flange Thickness 114.224 mm

Estimated M.A.W.P. (Operating) 5390 KPa.

Estimated Finished Weight of Flange at given Thk. 186.6 kg.

Estimated Unfinished Weight of Forging at given Thk 214.5 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

	Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub Sh	291.22	452.57	0.643	291.22	500.02	0.582
Tangential Flange St	339.43	339.43	1.000	339.43	375.01	0.905
Tangential Flange Sto	85.87	226.29	0.379	85.87	250.01	0.343
Combined Sh/f + St	581.07	678.86	0.856	581.07	750.03	0.775
Combined Sr + St	350.59	678.86	0.516	350.59	750.03	0.467
Combined Sh + Sto	377.09	678.86	0.555	377.09	750.03	0.503
Radial Flange Sr	11.16	452.57	0.025	11.16	500.02	0.022
Maximum Stress Ratio			1.000			0.905



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Flg Calc [Int P]: Shell flng

Flng: 12

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Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.53/1.27)

= 1.2052

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.43/1.21) if $h/h0 < 0.35$, else max (0.25; 0.28/0.76)

= 0.3554

In this case, the hub ratio $h/h0$, 0.3157 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	449.6 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	275.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.260
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Flg Calc [Int P]: Shell flng Flng: 12 9:31am Sep 9,2026

Root Area of a Single Bolt	Abs	9.0645 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	198.0 N./mm ²
Single flange rotation at Sfmax	Thetaamax	0.185 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$$

$$= \pi/4 (495^2 - 469^2) + (0.5 \cdot 425 \cdot 5)$$

$$= 207.5 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208 \cdot 207 / (20 \cdot 9.06)$$

$$= 238.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(238; 508)$$

$$= 238.0 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(238; 290)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290; 450)$$

$$= 290.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 290 \geq 140 (207 / (20 \cdot 9.06))$$

$$:: 290.0 \geq 160.2 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 290 \geq (97 \cdot 207 + \pi/4 \cdot 520 \cdot 469^2) / (0.7 \cdot 9.06 \cdot 20)$$

$$:: 290.0 \geq 165.7 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 290 \leq 275 (207 / (20 \cdot 9.06))$$

$$:: 290.0 \leq 314.7 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(thetagmax / thetaamax)$$

$$:: 290 \leq 450 (1/0.18)$$

$$:: 290.0 \leq 2431.6 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 290 \cdot 0.26 \cdot 9.06 \cdot 38.1$$

$$= 265.5 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 87 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange Rigidity Based on Required Thickness [ASME]:



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Flg Calc [Int P]: Shell flng

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Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 17146 / 1 \cdot 9807 \cdot 0.35 / (18.9 \cdot 201334 \cdot 9^2 \cdot 63.4 \cdot 0.3)$
 $= 0.528$ (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 9040 / 1 \cdot 9807 \cdot 0.35 / (18.9 \cdot 185119 \cdot 9^2 \cdot 63.4 \cdot 0.3)$
 $= 0.303$ (should be ≤ 1)

Flange Rigidity Based on Given Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 17146 / 1 \cdot 9807 \cdot 0.35 / (20.8 \cdot 201334 \cdot 9^2 \cdot 63.4 \cdot 0.3)$
 $= 0.480$ (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 9040 / 1 \cdot 9807 \cdot 0.35 / (20.8 \cdot 185119 \cdot 9^2 \cdot 63.4 \cdot 0.3)$
 $= 0.275$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.15, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
Channel head		6404	16	3	440	138	1
Channel shell		6404	16	3	440	138	2
Channel flng		6404.2	123	3	440	138	3
Shell flng		520	126	3	440	131.8	4
Shell		523.95	12	3	440	136.8	5
Shell head		523.92	12	...	425	136.8	6

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
Channel head		6400	6640	8478	13.6	13.2147	1
Channel shell		6400	7768	9616	16	13.6456	2
Channel flng		6400	7768	9616	115	110.465	3
Shell flng		520	5390	7288	118	114.224	4
Shell		520	5386	7288	12	5.5	5
Shell head		520	6531	6592	10.2	2.5	6

Note: Shell Side MAWP governed by a standard Flange!

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:
 For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 1096 KPa.
 Shell Side MAPnc = 1960 KPa.
 Channel Side MAWP = 6510 KPa.
 Channel Side MAPnc = 8478 KPa.

Elements Suitable for Design Internal Pressure.

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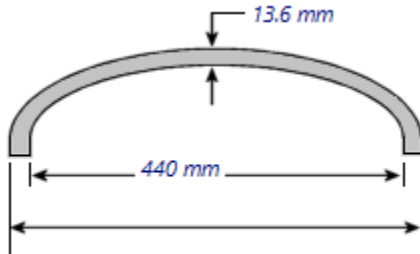
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Element: Channel head

Step: 6 9:31am Sep 9,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Channel head

Elliptical Head Design Information:			
Design Pressure P	6403.950 KPa.	Int. Design Temperature	160 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	50.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	440.000 mm	Actual Stress	133.0 N./mm ²
Spec. Min. Thk t,e, or sa	13.6000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	3156.4 cm ²	SG of Contents	1.000
Element Volume	18756.6 cm ³	Weight of Contents	18.3 Kg
Empty Weight	39.7 Kg	Operating Weight	58.1 Kg
		Comp. All. Ope. Stress	141.7 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency E	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 5.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 2.750 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (6404 \cdot 446 \cdot 0.98) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 6404)$$

$$= 10.2147 + 3.0000 = 13.2147 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 3.950 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 10.6) / (0.98 \cdot 446 + 0.2 \cdot 10.6)$$

$$= 6644 - 3.95 = 6640 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 13.6) / (1 \cdot 440 + 0.2 \cdot 13.6)$$

$$= 8478 \text{ KPa.}$$



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Element: Channel head

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Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (6404 \cdot (0.98 \cdot 446 + 0.2 \cdot 10.6)) / (2 \cdot 1 \cdot 10.6)$$

$$= 133.004 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (6404 \cdot 223) / (138 \cdot 1 - 0.6 \cdot 6404) + 3$$

$$= 13.646 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 3.950 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 13) / (223 + 0.6 \cdot 13)$$

$$= 7772 - 3.95 = 7768 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (446 / (2 \cdot 113)))^2 / 6$$

$$= 0.982418$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 14.493 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 13.6, tr = 10.4, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.98$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-20 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 16, tr = 10.8, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.83$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-23 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6509.507 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.833	Temperature Difference	9 °C
UCS-66(c) red.	No	Computed Min. Temp.	-23 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 5.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 2.750 mm



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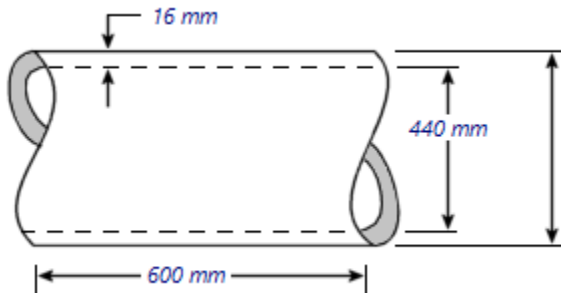
Element: Channel shell

Step: 7 9:31am

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2023



Internal Pressure Results for: Channel shell

Cylindrical Shell Design Information:			
Design Pressure P	6403.950 KPa.	Int. Design Temperature	160 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	600.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	440.000 mm	Actual Stress	113.7 N./mm ²
Spec. Min. Thk t,e, or sa	16.0000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	8897.0 cm ²	SG of Contents	1.000
Element Volume	91248.2 cm ³	Weight of Contents	87.1 Kg
Empty Weight	106.6 Kg	Operating Weight	193.7 Kg
		Comp. All. Ope. Stress	143.9 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency E	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (6404 \cdot 223) / (138 \cdot 1 - 0.6 \cdot 6404) \\
 &= 10.6456 + 3.0000 = 13.6456 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &\text{Less Operating Hydrostatic Head Pressure of 3.950 KPa.} \\
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 13) / (223 + 0.6 \cdot 13) \\
 &= 7772 - 3.95 = 7768 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 16) / (220 + 0.6 \cdot 16) \\
 &= 9616 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (6404 \cdot (223 + 0.6 \cdot 13)) / (1 \cdot 13) \\
 &= 113.701 \text{ N./mm}^2
 \end{aligned}$$



% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 3.509 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 16$, $t_r = 10.8$, $c = 3$ mm, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.83$, Temp. Reduction = 9 °C

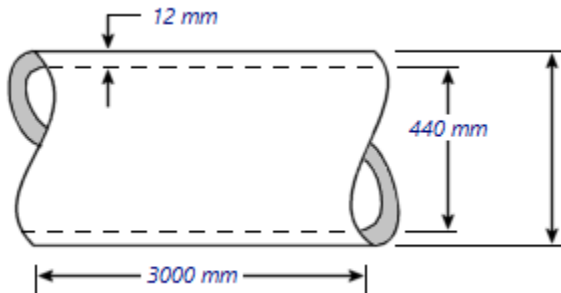
Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-23 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6509.507 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.833	Temperature Difference	9 °C
UCS-66(c) red.	No	Computed Min. Temp.	-23 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %



Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	523.950 KPa.	Int. Design Temperature	280 °C
Material	SA-516 70	Ext. Design Temperature	200 °C
Length	3000.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	136.8 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	440.000 mm	Actual Stress	13.3 N./mm ²
Spec. Min. Thk t _e , or sa	12.0000 mm	Spec. Nominal Thk.	12.000 mm
Surface Area	43731.0 cm ²	SG of Contents	1.000
Element Volume	287960.7 cm ³	Weight of Contents	287.8 Kg
Empty Weight	396.2 Kg	Operating Weight	684.0 Kg
		Comp. All. Ope. Stress	106.8 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency E	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (524 \cdot 223) / (137 \cdot 1 - 0.6 \cdot 524)$$

$$= 0.8561 + 3.0000 = 3.8561 \text{ mm}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 2.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 3.950 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137 \cdot 1 \cdot 9) / (223 + 0.6 \cdot 9)$$

$$= 5390 - 3.95 = 5386 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 12) / (220 + 0.6 \cdot 12)$$

$$= 7288 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$



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Element: Shell

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$$= (524 \cdot (223 + 0.6 \cdot 9)) / (1 \cdot 9)$$

$$= 13.297 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.655 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 12$, $t_r = 2.5$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.28$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1096.044 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.278	Temperature Difference	81 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: Shell

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	464.00	3085.42	51.56	6.6496	0.0005331	51.61

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 51.6) / (3 \cdot 51.6) = 1335 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.214	464.00	3085.42	144.38	6.6496	0.0001133	10.97

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 11) / (3 \cdot 144) = 101 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	464.00	114546.42	51.56	50.0000	0.0004138	40.06

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 40.1) / (3 \cdot 51.6) = 1036 \text{ KPa.}$$



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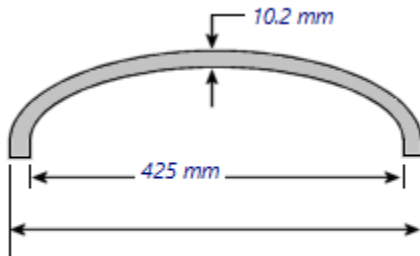
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Element: Shell head

Step: 9 9:31am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2023



Internal Pressure Results for: Shell head

Elliptical Head Design Information:			
Design Pressure P	523.921 KPa.	Int. Design Temperature	280 °C
Material	SA-516 70	Ext. Design Temperature	200 °C
Length	50.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	0.0000 mm	Allowable Stress (ope) S	136.8 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	425.000 mm	Actual Stress	11.0 N./mm ²
Spec. Min. Thk t,e, or sa	10.2000 mm	Spec. Nominal Thk.	12.000 mm
Surface Area	2890.6 cm ²	SG of Contents	1.000
Element Volume	17144.8 cm ³	Weight of Contents	17.0 Kg
Empty Weight	27.6 Kg	Operating Weight	44.6 Kg
		Comp. All. Ope. Stress	106.8 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency E	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 5.313 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 2.656 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (524 \cdot 425 \cdot 1) / (2 \cdot 137 \cdot 1 - 0.2 \cdot 524)$$

$$= 0.8142 + 0.0000 = 0.8142 \text{ mm}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 2.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 3.921 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 137 \cdot 1 \cdot 10.2) / (1 \cdot 425 + 0.2 \cdot 10.2)$$

$$= 6535 - 3.92 = 6531 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$



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Element: Shell head

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$$= (2 \cdot 138 \cdot 1 \cdot 10.2) / (1 \cdot 425 + 0.2 \cdot 10.2)$$

$$= 6592 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (524 \cdot (1 \cdot 425 + 0.2 \cdot 10.2)) / (2 \cdot 1 \cdot 10.2)$$

$$= 10.968 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (524 \cdot 213) / (137 \cdot 1 - 0.6 \cdot 524) + 0$$

$$= 0.816 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 3.921 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137 \cdot 1 \cdot 12) / (213 + 0.6 \cdot 12)$$

$$= 7471 - 3.92 = 7468 \text{ KPa.}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 11.502 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 10.2, tr = 2.5, c = 0 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.25$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-28 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 12, tr = 2.5, c = 0 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.21$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1096.044 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	10.200 mm	Basic MDMT	-28 °C
Stress/Thk. Ratio	0.245	Temperature Difference	76 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 5.313 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 2.656 mm

EXTERNAL PRESSURE RESULTS for: Shell head

Elliptical Head From 60 to 70 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.



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Element: Shell head

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Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
10.200	445.40	43.67	0.0031807	100.47

$$MAEP = B / (K0 * Do / t) = 100 / (0.9 * 43.7) = 2556 \text{ KPa.}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
1.160	445.40	384.00	0.0003617	35.01

$$MAEP = B / (K0 * Do / t) = 35 / (0.9 * 384) = 101 \text{ KPa.}$$

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Pressure Test Results: Shop Test Step: 10 9:31am Sep 9,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1424.858 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	676.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2548.008 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1205.649 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	743.600 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	743.600 KPa.
App 27-4	= M.A.W.P.	1096.044 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	8462.358 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	8320.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	11016.894 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	7160.458 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	9152.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	9152.000 KPa.
App 27-4	= M.A.W.P.	6509.507 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Channel head	228.9	234.0	0.978	11021.21
Channel shell	195.7	234.0	0.836	11021.21
Shell	64.8	234.0	0.277	2552.32
Shell head	53.4	234.0	0.228	2552.17

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N4-Tube inlet	Nozzle	138.00	138.00	1.000
Tube outlet	Nozzle	138.00	138.00	1.000
Shell outlet	Nozzle	118.00	118.00	1.000
Shell Inlet	Nozzle	118.00	118.00	1.000
Vent	Nozzle	118.00	118.00	1.000
Vent	Pad	138.00	136.80	1.009
Drain	Nozzle	118.00	118.00	1.000
Drain	Pad	138.00	136.80	1.009

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Channel head	138.00	138.00	1.000
Channel shell	138.00	138.00	1.000
Channel flng	138.00	138.00	1.000
Shell flng	138.00	131.80	1.047
Shell	138.00	136.80	1.009
Shell head	138.00	136.80	1.009



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Pressure Test Results: Shop Test

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Minimum

1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.10	1.003
Tubesheet Material	138.00	131.80	1.047

Minimum

1.003

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N4-Tube inlet	120.06	225.00	0.534
Tube outlet	111.91	225.00	0.497
Shell outlet	23.54	216.00	0.109
Shell Inlet	25.22	216.00	0.117
Vent	10.49	216.00	0.049
Drain	10.49	216.00	0.049

Elements Suitable for Test Pressure.

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Pressure Test Results: Field Test Step: 11 9:31am Sep 9,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1424.858 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	676.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2548.008 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1205.649 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	743.600 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	743.600 KPa.
App 27-4	= M.A.W.P.	1096.044 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	8462.358 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	8320.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	11016.894 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	7160.458 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	9152.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	9152.000 KPa.
App 27-4	= M.A.W.P.	6509.507 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 520 • 1

= 676.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 6400 • 1

= 8320.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Channel head	172.9	234.0	0.739	8324.31
Channel shell	147.8	234.0	0.632	8324.31
Shell	17.3	234.0	0.074	680.31
Shell head	14.2	234.0	0.061	680.17

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N4-Tube inlet	Nozzle	138.00	138.00	1.000
Tube outlet	Nozzle	138.00	138.00	1.000
Shell outlet	Nozzle	118.00	118.00	1.000
Shell Inlet	Nozzle	118.00	118.00	1.000
Vent	Nozzle	118.00	118.00	1.000
Vent	Pad	138.00	136.80	1.009
Drain	Nozzle	118.00	118.00	1.000
Drain	Pad	138.00	136.80	1.009

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):



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Pressure Test Results: Field Test

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Description	Ambient	Operating	Ratio
Channel head	138.00	138.00	1.000
Channel shell	138.00	138.00	1.000
Channel flng	138.00	138.00	1.000
Shell flng	138.00	131.80	1.047
Shell	138.00	136.80	1.009
Shell head	138.00	136.80	1.009

Minimum

1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.10	1.003
Tubesheet Material	138.00	131.80	1.047

Minimum

1.003

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N4-Tube inlet	90.68	225.00	0.403
Tube outlet	84.52	225.00	0.376
Shell outlet	6.27	216.00	0.029
Shell Inlet	6.72	216.00	0.031
Vent	2.80	216.00	0.013
Drain	2.80	216.00	0.013

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2023

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	467.2	10.6	0.0031512	112.761
20	30	686.667	472	13	0.0043055	117.522
30	40	No Calc	...	112	No Calc	No Calc
40	50	No Calc	...	115	No Calc	No Calc
50	60	3085.42	464	9	0.00053315	51.6123
60	70	No Calc	445.4	10.2	0.0031807	100.471

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	13.6	No Calc	...	2842
20	30	16	No Calc	...	4316
30	40	115	110.465	...	No Calc
40	50	118	114.224	101.3	No Calc
50	60	12	6.21368	101.3	1335
60	70	10.2	1.15988	101.3	2556

Minimum

1335

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	686.667	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	3085.42	114546	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Flg Calc [Ext P]: Shell flng

Step: 13

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Flange Input Data Values

Description: Shell flng :

Shell flng

Flange Type/Sketch	Integral Weld Neck		
Design Pressure	P	-101.30	KPa.
Design Temperature		200	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	440.000	mm
Flange Outside Diameter	A	670.000	mm
Flange Thickness	t	118.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	17.0000	mm
Length of Hub	h	20.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.01	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.01	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	590.000	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		20	
Flange Face Outside Diameter	Fod	506.000	mm
Flange Face Inside Diameter	Fid	440.000	mm
Flange Facing Sketch	1, Code Sketch 1a		
Gasket Outside Diameter	Go	495.000	mm
Gasket Inside Diameter	Gi	469.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Column for Gasket Seating	2, Code Column II		
Gasket Thickness	tg	4.5000	mm
Length of Partition Gasket	lp	425.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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FileName : 135-C_BEU

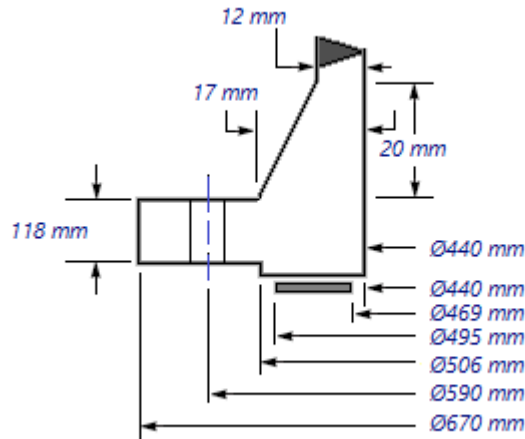
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Flg Calc [Ext P]: Shell flng

Step: 13

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Bolted Flange Calculation Results:

Corroded Flange Thickness, $t_c = T - c_i$	115.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	446.000 mm
Corroded Large Hub, $g_{lCor} = g_l - c_i$	14.000 mm
Corroded Small Hub, $g_{oCor} = g_o - c_i$	9.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g_{lCor}$	58.000 mm
Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $b_o = N/2$	6.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	482.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H]:

$$= \pi/4 \cdot G^2 \cdot P_{eq}$$

$$= \pi/4 \cdot 482^2 \cdot -101$$

$$= -1886.088 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 482 \cdot 3 \cdot -101$$

$$+ 2 \cdot 425 \cdot 5 \cdot 3 \cdot -101$$

$$= -734.871 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 446^2 \cdot -101$$

$$= -1613.859 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= -1886 - -1614$$

$$= -272.228 \text{ Kgf}$$

Operating Bolt Load [Wm1]:

$$= \max(H + H_p + H'_p, 0)$$

$$= \max(-1886 + -735 + 0, 0)$$

$$= 0.000 \text{ Kgf}$$

Gasket Seating Bolt Load [Wm2]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.42 \cdot \pi \cdot 482 + 68.9 \cdot 5 \cdot 425$$

$$= 83360.320 \text{ Kgf}$$



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Flg Calc [Ext P]: Shell flng

Step: 13

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Required Bolt Area [Am]:

$$= \max(Wm1/Sb, Wm2/Sa)$$

$$= \max(0/172, 83360/172)$$

$$= 47.530 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmx]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 115/(3 + 0.5)$$

$$= 273.343 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 590 \cdot \sin(\pi / 20)$$

$$= 92.296 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 92.3 / (2 \cdot 38.1 + 115) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	47.530	181.290	
Radial Distance between Hub and Bolts:	50.800	58.000	
Radial Distance between Bolts and Edge:	38.100	40.000	
Circ. Spacing between the Bolts:	82.550	92.296	273.343

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 181 \cdot 172 / (68.9 \cdot \pi (495 + 469))$$

$$= 14.933 \text{ mm} [\text{Note: Exceeds actual gasket width, 13.000 }]$$

Flange Design Bolt Load, Gasket Seating [W]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 181$$

$$= 317955.00 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (590 - 482) / 2$$

$$= 53.9245 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= (R + g1 + hg) / 2$$

$$= (58 + 14 + 53.9) / 2$$

$$= 62.9623 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= R + (g1 / 2)$$

$$= 58 + (14 / 2)$$

$$= 65.0000 \text{ mm}$$

Summary of Moments for External Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	-1614.	11.0754	1.0000	-18.
Face Pressure, Mt	-272.	9.0377	1.0000	-2.
Gasket Seating, Matm	317955.	53.9245	1.0000	17146.



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Flg Calc [Ext P]: Shell flng

Step: 13

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Total Moment for Operation, Mop -20. Kg-m.
Total Moment for Gasket seating, Matm 17146. Kg-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 63.356 mm
Hub Ratio, $h/h_o = HL / H_o$ 0.316
Thickness Ratio, $g_1/g_o = (g_1Cor/g_oCor)$ 1.556

Flange Factors for Integral Flange:

Factor F 0.875
Factor V 0.354
Factor f 1.193

Factors from Figure 2-7.1

$T = 1.710$
 $Y = 4.944$
 $e = 0.0138 \text{ 1/mm}$
 $K = 1.502$
 $U = 5.432$
 $Z = 2.591$
 $d = 78828.148 \text{ mm}^3$

Stress Factors

$Beta = 3.117$
 $Delta = 19.294$
 $Alpha = 2.588$
 $Gamma = 1.514$
 $Lambda = 20.807$

Longitudinal Hub Stress, Operating [SHo]:

$= (f \cdot Mop / B_{cor}) / (L \cdot g_1^2)$
 $= (1.19 \cdot 20.3 / 446) / (20.8 \cdot 14^2)$
 $= 0.13 \text{ N./mm}^2$

Longitudinal Hub Stress, Seating [SHa]:

$= (f \cdot Matm / B_{cor}) / (L \cdot g_1^2)$
 $= (1.19 \cdot 17146 / 446) / (20.8 \cdot 14^2)$
 $= 110.24 \text{ N./mm}^2$

Radial Flange Stress, Operating [SRo]:

$= (\beta \cdot Mop / B_{cor}) / (L \cdot t^2)$
 $= (3.12 \cdot 20.3 / 446) / (20.8 \cdot 115^2)$
 $= 0.01 \text{ N./mm}^2$

Radial Flange Stress, Seating [SRa]:

$= (\beta \cdot Matm / B_{cor}) / (L \cdot t^2)$
 $= (3.12 \cdot 17146 / 446) / (20.8 \cdot 115^2)$
 $= 4.27 \text{ N./mm}^2$

Tangential Flange Stress, Operating [STo]:

$= (Y \cdot Mo / (t^2 \cdot B_{cor})) - Z \cdot SRo$
 $= (4.94 \cdot 20.3 / (115^2 \cdot 446)) - 2.59 \cdot 0.0051$
 $= 0.15 \text{ N./mm}^2$

Tangential Flange Stress, Seating [STa]:

$= (y \cdot Matm / (t^2 \cdot B_{cor})) - Z \cdot SRa$
 $= (4.94 \cdot 17146 / (115^2 \cdot 446)) - 2.59 \cdot 4.27$
 $= 129.86 \text{ N./mm}^2$

Average Flange Stress, Operating [SAo]:

$= (SHo + \max(SRo, STo)) / 2$
 $= (0.13 + \max(0.0051, 0.15)) / 2$
 $= 0.14 \text{ N./mm}^2$

Average Flange Stress, Seating [SAa]:

$= (SHa + \max(SRa, STa)) / 2$
 $= (110 + \max(4.27, 130)) / 2$
 $= 120.05 \text{ N./mm}^2$

Bolt Stress, Operating [BSo]:

$= Wm_1 / Ab$
 $= 0 / 181$
 $= 0.00 \text{ N./mm}^2$



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Flg Calc [Ext P]: Shell flng

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Bolt Stress, Seating [BSa]:

= (Wm2 / Ab)

= (83360/181)

= 45.09 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	0.13	207.01	110.24	207.00
Radial Flange	0.01	138.01	4.27	138.00
Tangential Flange	0.15	138.01	129.86	138.00
Maximum Average	0.14	138.01	120.05	138.00
Bolting	0.00	172.01	45.09	172.00

Minimum Required Flange Thickness 114.224 mm
 Stress Check: OK for External Pressure -101.3 KPa.
 Estimated Finished Weight of Flange at given Thk. 186.6 kg.
 Estimated Unfinished Weight of Forging at given Thk 214.5 kg.

Flange Rigidity Based on Required Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
 = $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 = $52.14 \cdot 17146 / 1 \cdot 9807 \cdot 0.35 / (18.9 \cdot 201334 \cdot 9^2 \cdot 63.4 \cdot 0.3)$
 = 0.528 (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
 = $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 = $52.14 \cdot 20.3 / 1 \cdot 9807 \cdot 0.35 / (18.9 \cdot 191213 \cdot 9^2 \cdot 63.4 \cdot 0.3)$
 = 0.000 (should be ≤ 1)

Flange Rigidity Based on Given Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
 = $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 = $52.14 \cdot 17146 / 1 \cdot 9807 \cdot 0.35 / (20.8 \cdot 201334 \cdot 9^2 \cdot 63.4 \cdot 0.3)$
 = 0.480 (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
 = $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 = $52.14 \cdot 20.3 / 1 \cdot 9807 \cdot 0.35 / (20.8 \cdot 191213 \cdot 9^2 \cdot 63.4 \cdot 0.3)$
 = 0.000 (should be ≤ 1)

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Element and Detail Weights:

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	39.7442	18756.6	32.2922	19427.9	3.97442
20	30	106.59	91248.2	87.1739	93753.8	10.659
30	40	196.387	22869.4	191.63	22969.7	19.6387
40	50	199.022	22729.9	194.298	22813.4	19.9022
50	60	396.206	287961	299.126	300489	39.6206
60	70	27.6419	17144.8	27.6419	17144.8	2.76419
Total		965	460709.59	832	476598.03	96



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Liqd	18.3085	-36.6667	20	...	Liquid: 10
10	Insl	12.8239	25	...	...	Insulation
20	Liqd	87.0667	300	20	...	Liquid: 20
20	Insl	29.3347	300	...	...	Ins: 40
20	Nozl	99.0341	300	316.837	...	N4-Tube inlet
20	Nozl	54.7425	300	-293.165	...	Tube outlet
20	Wght	100	250	...	...	Partiotion weig
30	Liqd	22.6138	69.5	7.5	...	Liquid: 30
30	Insl	9.23394	67.5	...	...	Insulation
40	Insl	9.14346	47.5	...	...	Insulation
50	Sadl	91.18	620	396.5	...	Left Saddle
50	Sadl	81.9948	2480	399	...	Sdl 2 Fr40
50	Liqd	287.785	1500	20	...	Liquid: 40
50	Insl	187.327	1500	...	...	Ins: 40
50	Nozl	9.48441	2850	-256.83	...	Shell outlet
50	Nozl	13.2662	300	268.59	...	Shell Inlet
50	Nozl	6.72896	1500	240.472	...	Vent
50	Nozl	7.01999	1500	-240.472	...	Drain
50	Wght	100	620	-500	...	Fire proofing1
50	Wght	100	2480	-500	...	Fire proofing 2
60	Liqd	16.9634	85.4167	12.5	...	Liquid: 50
60	Insl	15.4424	75	...	...	Ins: 40
30	FTsh	259.886	195.5	...	...	Tubesheet
30	Tube	458.771	1644.5	...	...	Tubes
40	Baf1	28.6874	327	...	...	Baffles
40	Baf1	28.6874	654	...	...	Baffles
40	Baf1	28.6874	981	...	...	Baffles
40	Baf1	28.6874	1308	...	...	Baffles
40	Baf1	28.6874	1635	...	...	Baffles
40	Baf1	28.6874	1962	...	...	Baffles
40	Baf1	28.6874	2289	...	...	Baffles
40	Baf1	29.4359	2616	...	...	Baffles

Total Weight of Each Detail Type:

Saddles 173.2
Liquid 432.7



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Element and Detail Weights: Step: 14 9:31am Sep 9,2026

Insulation	263.3
Nozzles	190.3
Weights	300.0
Exchanger Components	948.9
Liquid in Tubes	125.9

Sum of the Detail Weights 2434.3 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	1062.1	1062.1	1062.1	1062.1	1062.1	1062.1
Saddles	173.2	173.2	173.2	173.2	173.2	173.2
Nozzles	190.3	190.3	190.3	190.3	190.3	190.3
Wld Weights	100.0	100.0	100.0	100.0	100.0	100.0
Exchanger	948.9	948.9	948.9	948.9	948.9	948.9
Insulation	...	...	...	...	263.3	263.3 *
Empty Weights	...	...	...	200.0	200.0	...
Ope Weights	...	...	...	...	...	200.0
Ope. Liquid	...	...	...	...	...	432.7
Tube Ope Lqd	...	...	...	...	...	125.9
Test Liquid	...	460.4	...	...	...	...
Tube Tst Lqd	...	125.9	...	...	...	...
Totals	2474.5	3060.8	2474.5	2674.5	2937.8	3496.4

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	2474.5 kg.
Shop Test Wt.	- Fabricated weight + water (full)	3060.8 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	2474.5 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	2674.5 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	2937.8 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	3496.4 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	3349.7 kg.
Field Test Wt.	- Empty weight + test medium (full)	3193.2 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	125973.3 cm ³
Weight of Ope Liq in tubes :	125.9 kg.
Weight of Water in tubes :	125.9 kg.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	3156.36	2789.75
20	30	8896.99	8293.8
30	40	4960.54	2031.98
40	50	4955.08	2018.16
50	60	43731	41469



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Element and Detail Weights:

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60	70	2890.61	2625.54
----	----	---------	---------

Total 68590.6 59228.3 cm²

Total 73.8 63.8 ft²

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Nozzle Flange MAWP:

Step: 15 9:31am Sep 9,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Flange Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	UG-44	Max Pressure 50%	DNV
N4-Tube inlet	8968.1	10210.0	160	600	GR 1.1	3074.3	8968.1	8968.1	8968.1
Tube outlet	8968.1	10210.0	160	600	GR 1.1	3584.7	8968.1	8968.1	8968.1
Shell outlet	6308.1	9930.0	280	600	GR 2.1	5319.8	4142.3	6308.1	4142.3
Shell Inlet	4064.0	5110.0	280	300	GR 1.1	4328.7	1767.4	4064.0	1767.4
Vent	1096.0	1960.0	280	150	GR 1.1	...	...	...	...
Drain	1096.0	1960.0	280	150	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 1096.044 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 1960.006 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 8968.055 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 10210.003 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N4-Tube inlet	244.602	0.50	807	1530	6509.5	8968.1	Pass
Tube outlet	192.151	0.50	454	1147	6505.6	8968.1	Pass
Shell outlet	111.125	0.50	129	612	1092.1	4142.3	Pass
Shell Inlet	138.176	0.50	202	765	1096.0	1767.4	Pass

Evaluating Flange on Nozzle [N4-Tube inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(807) + 4 \cdot \text{abs}(1530)244.602 \leq \pi \cdot 244.602^3 [(8968.055 - 6509.507) + .500 \cdot 8968.055]$$

$$= 0.14414 + 05 \leq 0.32550 + 05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N4-Tube inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8968.055, 0.500 \cdot 8968.055 + 8968.055 - 14413.7 / (\pi \cdot 244.602^3))$$

$$= \min(8968.055, 10377.816)$$

$$= 8968.055 \text{ KPa.}$$

Evaluating Flange on Nozzle [Tube outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(454) + 4 \cdot \text{abs}(1147)192.151 \leq \pi \cdot 192.151^3 [(8968.055 - 6505.586) + .500 \cdot 8968.055]$$

$$= 0.81476 + 04 \leq 0.15789 + 05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle Tube outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8968.055, 0.500 \cdot 8968.055 + 8968.055 - 8147.6 / (\pi \cdot 192.151^3))$$

$$= \min(8968.055, 9867.415)$$

$$= 8968.055 \text{ KPa.}$$



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Nozzle Flange MAWP:

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Evaluating Flange on Nozzle [Shell outlet]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(129) + 4 \cdot \text{abs}(612) 111.125 \leq \pi \cdot 111.125^3 [(6308.068 - 1092.123) + .500 \cdot 6308.068] \\ &= 0.23388 + 04 \leq 0.36797 + 04 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle Shell outlet [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(6308.068, 0.500 \cdot 6308.068 + 6308.068 - 2338.8 / (\pi \cdot 111.125^3)) \\ &= \min(6308.068, 4142.275) \\ &= 4142.275 \text{ KPa.} \end{aligned}$$

Evaluating Flange on Nozzle [Shell Inlet]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(202) + 4 \cdot \text{abs}(765) 138.176 \leq \pi \cdot 138.176^3 [(4064.047 - 1096.044) + .500 \cdot 4064.047] \\ &= 0.36585 + 04 \leq 0.42260 + 04 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle Shell Inlet [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(4064.047, 0.500 \cdot 4064.047 + 4064.047 - 3658.5 / (\pi \cdot 138.176^3)) \\ &= \min(4064.047, 1767.404) \\ &= 1767.404 \text{ KPa.} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2020 Metric Edition

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Wind Load Calculation:

Step: 16 9:31am Sep 9,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$\begin{aligned}
 &= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar}))) \\
 &= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.99)/(1 + 1.7 \cdot 3.4 \cdot 0.23))) \\
 &= \min(0.85, 0.92) \\
 &= 0.850
 \end{aligned}$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.598
Structure Height to Diameter ratio	6.860

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Z_g = 274320 \text{ mm}$

Effective Height [z]:

$$\begin{aligned}
 &= \text{Centroid Height} + \text{Vessel Base Elevation} \\
 &= 600 + 300 = 900 \text{ mm} \\
 &= 2.95 \text{ ft. Imperial Units}
 \end{aligned}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\begin{aligned}
 &\text{Because } z (2.95 \text{ ft.}) < 15 \text{ ft.} \\
 &= 2.01(15 / Z_g)^{2/\alpha} \\
 &= 2.01(15/900)^{2/9.5} \\
 &= 0.85
 \end{aligned}$$

Type of Hill: No Hill



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Wind Load Calculation:

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Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0 • 0 • 0)²

= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 0.85 • 1 • 1 • 1 • 60.4²)

= 16 psf [78.1] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16 • 0.85 • 0.6 • 1.48

= 12.1 lbs. [5.47] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Channel head	900.0	0.000	0.000	0.000	0.849	1.000	78.120
Channel shell	900.0	0.000	0.000	0.000	0.849	1.000	78.120
Channel flng	900.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell flng	900.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell	900.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell head	900.0	0.000	0.000	0.000	0.849	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Partition we	0.00	900.00	78.12	0.00
Fire proofing	0.00	300.00	78.12	0.00
Fire proofing	0.00	300.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	900	752.64	1379.26	78.12	6.02115
20	30	900	758.4	4550.4	78.12	19.8648
30	40	900	714.475	993.12	78.12	4.33547
40	50	900	660.174	911.04	78.12	3.97715
50	60	900	796.8	23904	78.12	104.353
60	70	900	774.48	1484.42	78.12	6.48022

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation: Step: 17 9:31am Sep 9,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hnx$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(2.690.75)$
 $= 0.042$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33)$
 $= 0.030$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.042, 0.03) \text{ per 12.8.2}$
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.16 \cdot 3323 \cdot 1.25$
 $= 195.408 \text{ Kg}$

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.16 \cdot 3323$
 $= 104.2 \text{ Kg}$

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kg	Element Ope Load Kg
10	20	220	415.409	24.5306



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Earthquake Load Calculation:

Step: 17

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20	30	220	415.409	24.5306
30	40	220	415.409	24.5306
40	50	220	415.409	24.5306
50	Sad1	220	415.409	24.5306
Sad1	60	220	415.409	24.5306
50	60	220	415.409	24.5306
60	70	212.5	415.409	23.6943

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 18 9:31am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	2540.523 mm
Center of Gravity of the Liquid	1957.667 mm
Center of Gravity of the Insulation	2146.899 mm
Center of Gravity of the Nozzles	753.693 mm
Center of Gravity of the Added Weights (Operating)	1826.567 mm
Center of Gravity of the Added Weights (Empty)	1826.567 mm
Center of Gravity of the Tubesheet(s)	845.500 mm
Center of Gravity of the Tubes	2294.500 mm
Center of Gravity of the Baffles	2377.070 mm
Center of Gravity of Bare Shell New and Cold	1544.856 mm
Center of Gravity of Bare Shell Corroded	1478.685 mm
Vessel CG in the Operating Condition	1792.982 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	1716.917 mm
Vessel CG in the Test Condition	1764.040 mm

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Saddle Calcs: Bundle Pull New: Step: 19 9:31am Sep 9,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, New Condition

Note:

Wear Pad Width (365.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(285 + 1.56 \cdot \sqrt{226 \cdot 12}, 2 \cdot 620)$$

$$= 366.2399 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	226.00	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	3188.00	mm
Distance from Saddle to Vessel tangent	a1 or a	620.00	mm
Saddle Width	b1 or b	285.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		136.80	N./mm ²
Head Allowable Stress used in Calculation		131.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	600.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		1100.02	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		2505.86	Kgf
Pressure used in Analysis	P	0.000	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-2.36	-234.01
Long. Stress at Top of Midspan		2.36	136.80
Long. Stress at Bottom of Midspan		2.36	136.80
Long. Stress at Top of Saddles		13.48	136.80
Long. Stress at Bottom of Saddles		-7.47	-234.01
Long. Stress at Bottom of Saddles		7.47	136.80
Tangential Shear in Shell		6.48	109.44
Circ. Stress at Horn of Saddle		14.93	205.20
Circ. Compressive Stress in Shell		0.43	136.80

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:

$$= \text{Bundle Pull Out Force} \cdot \text{Impact Factor} \cdot B / L_s$$

$$= 1100 \cdot 1.5 \cdot 600 / 1860$$

$$= 532.3 \text{ Kgf}$$

Load Combination Results for Tube Bundle Pull Out [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$



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Saddle Calcs: Bundle Pull New: Step: 19 9:31am Sep 9, 2026

$$= 1974 + \max(532, 0, 0, 0)$$

$$= 2505.9 \text{ Kgf}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	2597.04	Kgf
Transverse Shear Load	0.00	Kgf
Longitudinal Shear Load	1650.03	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(440/2 + 12 + 15)\sin(120/2), 2 \cdot 226)$$

$$= 427.817 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h m^2)/(2a \cdot L)) / (1 + (4h m^2)/3L)]$$

$$= -2506 \cdot 620 [1 - (1 - 620/3188 + (226^2 - 0^2)/(2 \cdot 620 \cdot 3188)) / (1 + (4 \cdot 0)/(3 \cdot 3188))]$$

$$= -282.1 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 (1 + 2(R_m^2 - h m^2)/(L^2)) / (1 + (4h m^2)/(3L)) - 4a/L$$

$$= 2506 \cdot 3188 / 4 (1 + 2(226^2 - 0^2)/(3188^2)) / (1 + (4 \cdot 0)/(3 \cdot 3188)) - 4 \cdot 620 / 3188$$

$$= 463.6 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 t)$$

$$= 0 \cdot 226 / (2 \cdot 12) - 464 / (\pi \cdot 226^2 \cdot 12)$$

$$= -2.36 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 226 / (2 \cdot 12) + 464 / (\pi \cdot 226^2 \cdot 12)$$

$$= 2.36 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 0 \cdot 226 / (2 \cdot 12) - 282 / (0.11 \cdot \pi \cdot 226^2 \cdot 12)$$

$$= 13.48 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M1 / (K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 226 / (2 \cdot 12) + 282 / (0.19 \cdot \pi \cdot 226^2 \cdot 12)$$

$$= -7.47 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h m / 3))$$

$$= 2506 (3188 - 2 \cdot 620) / (3188 + (4 \cdot 0 / 3))$$

$$= 1531.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 1531 / (226 \cdot 12)$$

$$= 6.48 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:



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Saddle Calcs: Bundle Pull New: Step: 19 9:31am Sep 9,2026

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(226 \cdot 12)}$$

$$= 40.620 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 2506 \cdot 0.1 / (12(285 + 40.6 + 40.6))$$

$$= -0.43 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -2506 / (4 \cdot 12(285 + 40.6 + 40.6)) - 3 \cdot 0.053 \cdot 2506 / (2 \cdot 12^2)$$

$$= -14.93 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(285 + 1.56 \cdot \sqrt{(226 \cdot 12)}, 2 \cdot 620)$$

$$= 366.24 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	560.0000	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	310.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	262.0	N./
Height of Web at Center	Hw,c	390.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(220 \cdot 12)} + 365)^3) / 12$$

$$= 6.410 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 6.41 + 53.4(123 - 6)^2$$

$$= 7291.767 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	445.2	12.0	6.0	53.4	32051.1	0.729E+04
Wearplate	365.0	15.0	19.5	54.8	106762.5	0.585E+04
Web	10.0	341.0	197.5	34.1	673475.0	0.521E+04
BasePlate	310.0	12.0	374.0	37.2	1391280.0	0.235E+05
Totals	...	...	...	179.5	2203568.5	0.418E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2203568 / 179$$

$$= 122.783 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$



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$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 2506$$

$$= 509.9971 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.3968 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 600 - 220 \cdot \sin(120/2) / 1.05$$

$$= 418.061 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 213.2144 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.6137 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(2506 + 91.2) \cdot 310) / (4 \cdot 560 \cdot 175)}$$

$$= 7.781 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (220 + 12 + 15)$$

$$= 427.817 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 428 / (3 - 1)$$

$$= 213.908 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 214 \cdot 310 / 2$$

$$= 331.558 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 2597 / (560 \cdot 310)$$

$$= 1.496 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 332 \cdot 1.5$$

$$= 496.008 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 27.5 + 10.7$$

$$= 38.195 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 496 / 38.2$$

$$= 1.274 \text{ N./mm}^2$$

Check of Outside Ribs:



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Saddle Calcs: Bundle Pull New: Step: 19 9:31am Sep 9, 2026

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	285.0	...	28.5	...	0.193E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 285 - 10

= 275.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/38.2

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 285/2

= 142.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{(1929/38.2)}$

= 71.068 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 600 - cos(120/2)(220 + 12 + 15) - 12

= 464.500 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \cdot \pi^2 \cdot 199943392/262)}$

= 122.739

Slenderness ratio [KL/r]:

= KL/r

= 1 • 464/71.1

= 6.536

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 1650 / (2 • 560) • 214 • 464/2

= 73.192 Kg-m.

Compressive Allowable, KL/r < Cc (6.54 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (6.54)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (6.54) / (8 \cdot 123) - (6.54)^3 / (8 \cdot 123^3))$

= 155.1 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 1.27/155 + (73.2 • 142/19290936) / 175

= 0.039

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	275.0	0.0	27.5	0.0	0.193E+04
Web	213.9	10.0	0.0	21.4	0.0	1.78
Totals	...	...	...	48.9	...	0.193E+04



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Saddle Calcs: Bundle Pull New: Step: 19 9:31am Sep 9, 2026

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/48.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 285/2$$

$$= 142.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 600 - \cos(247/2)(12 + 0 + 0) - 0$$

$$= 341.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1931/48.9)}$$

$$= 62.843 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 341/62.8$$

$$= 5.426$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 1650 / (2 \cdot 560)$$

$$= 1.473 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.47 \cdot 214 \cdot 341$$

$$= 107.464 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 107/135494$$

$$= 7.778 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($5.43 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.43)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (5.43) / (8 \cdot 123) - (5.43)^3 / (8 \cdot 123^3))$$

$$= 155.5 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 1.99/155 + (107 \cdot 142/1931)/175$$

$$= 0.057$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-307 A	
Bolt Allowable Stress	Stba	138.00	N./mm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.0	mm
User Defined Area in a Single Bolt	UserBoltArea	1.5500	cm ²
BasePlate Allowable Stress	S	138.00	N./mm ²
Area Available in a Single Bolt	BltArea	0.9592	cm ²
Saddle Load Q0 (Weight)	Q0	2597.0	Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1650.0	Kgf



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Saddle Calcs: Bundle Pull New: Step: 19 9:31am Sep 9,2026

Maximum Transverse Force	Ft	0.0	Kgf
Maximum Longitudinal Force	F1	1650.0	Kgf
Saddle Bolted to Steel Foundation		No	

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $1650 / (0.96 \cdot 4)$

= 42.2 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $0 / (0.96 \cdot 4)$

= 0.0 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F1 / (\text{BoltShearAllowable} \cdot \text{Nbols})$

= $1650 / (82.8 \cdot 4)$

= 0.4886 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + \text{Sum of X Moments}$

= $600 \cdot 0 + 0$

= 0.00 Kg-m.

Eccentricity (e):

= $Rmom / QO$

= $0 / 2597$

= 0.00 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$

= $\max[0, 0.49, 0]$

= 0.4886 cm²

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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Saddle Calcs: Bundle Pull Cor: Step: 20 9:31am Sep 9,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, Corroded Condition

Input and Calculated Values:

Vessel Mean Radius	Rm	227.50	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	3188.00	mm
Distance from Saddle to Vessel tangent	a1 or a	620.00	mm
Saddle Width	b1 or b	285.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	365.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	15.0	mm
Wear Plate Allowable Stress	fw or Sr	136.80	N./mm ²
Shell Allowable Stress used in Calculation		136.80	N./mm ²
Head Allowable Stress used in Calculation		131.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	600.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		1100.02	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		2432.48	Kgf
Pressure used in Analysis	P	0.000	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-3.02	-234.01
Long. Stress at Top of Midspan	3.02	136.80
Long. Stress at Bottom of Midspan	3.02	136.80
Long. Stress at Top of Saddles	17.20	136.80
Long. Stress at Bottom of Saddles	-9.53	-234.01
Long. Stress at Bottom of Saddles	9.53	136.80
Tangential Shear in Shell	8.33	109.44
Circ. Stress at Horn of Saddle	3.98	205.20
Circ. Compressive Stress in Shell	0.21	136.80

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:

= Bundle Pull Out Force • Impact Factor • B / Ls

= 1100 • 1.5 • 600/1860

= 532.3 Kgf

Load Combination Results for Tube Bundle Pull Out [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 1900 + max(532, 0, 0, 0)

= 2432.5 Kgf

Summary of Loads at the Base of this Saddle:



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Vertical Load (including saddle weight)	2523.66	Kgf
Transverse Shear Load	0.00	Kgf
Longitudinal Shear Load	1650.03	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(440/2 + 12 + 15)\sin(120/2), 2 \cdot 227)$$

$$= 427.817 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m^2)/3L)]$$

$$= -2432 \cdot 620 [1 - (1 - 620/3188 + (227^2 - 0^2)/(2 \cdot 620 \cdot 3188))/(1 + (4 \cdot 0)/(3 \cdot 3188))]$$

$$= -273.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L$$

$$= 2432 \cdot 3188/4(1 + 2(227^2 - 0^2)/(3188^2))/(1 + (4 \cdot 0)/(3 \cdot 3188)) - 4 \cdot 620/3188$$

$$= 450.3 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 0 \cdot 227/(2 \cdot 9) - 450/(\pi \cdot 227^2 \cdot 9)$$

$$= -3.02 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 227/(2 \cdot 9) + 450/(\pi \cdot 227^2 \cdot 9)$$

$$= 3.02 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 0 \cdot 227/(2 \cdot 9) - 274/(0.11 \cdot \pi \cdot 227^2 \cdot 9)$$

$$= 17.20 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m/(2t) + M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 227/(2 \cdot 9) + 274/(0.19 \cdot \pi \cdot 227^2 \cdot 9)$$

$$= -9.53 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 2432(3188 - 2 \cdot 620)/(3188 + (4 \cdot 0/3))$$

$$= 1486.3 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 1486 / (227 \cdot 9)$$

$$= 8.33 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$



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$$= 0.78 \cdot \sqrt{(227 \cdot 9)}$$

$$= 35.294 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(285 + 1.56 \cdot \sqrt{(227 \cdot 9)}, 2 \cdot 620)$$

$$= 355.59 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, bundle pull case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 2432 \cdot 0.1 / (356 (9 + 1 \cdot 15))$$

$$= -0.21 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -2432 / (4(9 + 1 \cdot 15)356) -$$

$$3 \cdot 0.053 \cdot 2432 / (2(9 + 1 \cdot 15)^2)$$

$$= -3.98 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	560.0000	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	310.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	262.0	N./
Height of Web at Center	Hw,c	390.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(223 \cdot 9)} + 365)9^3) / 12$$

$$= 2.642 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 2.64 + 39.1 (130 - 4.5)^2$$

$$= 6190.162 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	434.9	9.0	4.5	39.1	17612.9	0.619E+04
Wearplate	365.0	15.0	16.5	54.8	90337.5	0.709E+04
Web	10.0	341.0	194.5	34.1	663244.9	0.471E+04
BasePlate	310.0	12.0	371.0	37.2	1380119.9	0.216E+05
Totals	...	...	...	165.2	2151315.2	0.396E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2151315 / 165$$

$$= 130.233 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$



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Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 2432$$

$$= 495.0615 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.3852 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 600 - 223 \cdot \sin(120/2) / 1.05$$

$$= 415.580 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 205.7420 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.6641 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(2432 + 91.2)310 / (4 \cdot 560 \cdot 175)}$$

$$= 7.670 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (220 + 12 + 15)$$

$$= 427.817 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 428 / (3 - 1)$$

$$= 213.908 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 214 \cdot 310 / 2$$

$$= 331.558 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 2524 / (560 \cdot 310)$$

$$= 1.454 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 332 \cdot 1.45$$

$$= 481.992 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 27.5 + 10.7$$

$$= 38.195 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 482 / 38.2$$

$$= 1.238 \text{ N./mm}^2$$



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Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	285.0	...	28.5	...	0.193E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
= 285 - 10
= 275.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
= 0/38.2
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
= 285/2
= 142.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{1929/38.2}$
= 71.068 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
= 600 - cos(120/2)(220 + 12 + 15) - 12
= 464.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 199943392/262}$
= 122.739

Slenderness ratio [KL/r]:

= KL/r
= 1 • 464/71.1
= 6.536

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
= 1650 / (2 • 560) • 214 • 464/2
= 73.192 Kg-m.

Compressive Allowable, KL/r < Cc (6.54 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (6.54)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (6.54) / (8 \cdot 123) - (6.54^3) / (8 \cdot 123^3))$
= 155.1 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
= 1.24/155 + (73.2 • 142/19290936) / 175
= 0.038

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	275.0	0.0	27.5	0.0	0.193E+04



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Web	213.9	10.0	0.0	21.4	0.0	1.78
Totals	...	...	...	48.9	...	0.193E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/48.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 285/2$$

$$= 142.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 600 - \cos(247/2)(12 + 0 + 0) - 0$$

$$= 341.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1931/48.9)}$$

$$= 62.843 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 341/62.8$$

$$= 5.426$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 1650/(2 \cdot 560)$$

$$= 1.473 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.47 \cdot 214 \cdot 341$$

$$= 107.464 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 107/135494$$

$$= 7.778 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($5.43 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.43)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (5.43) / (8 \cdot 123) - (5.43)^3 / (8 \cdot 123^3))$$

$$= 155.5 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 1.94/155 + (107 \cdot 142/1931) / 175$$

$$= 0.057$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-307 A	
Bolt Allowable Stress	Stba	138.00	N./mm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.0	mm
User Defined Area in a Single Bolt	UserBoltArea	1.5500	cm ²
BasePlate Allowable Stress	S	138.00	N./mm ²
Area Available in a Single Bolt	BltArea	0.9592	cm ²



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Saddle Load Q0 (Weight)	Q0	2523.7	Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1650.0	Kgf
Maximum Transverse Force	Ft	0.0	Kgf
Maximum Longitudinal Force	F1	1650.0	Kgf
Saddle Bolted to Steel Foundation		No	

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

= $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $1650 / (0.96 \cdot 4)$

= 42.2 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

= $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $0 / (0.96 \cdot 4)$

= 0.0 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F1 / (\text{BoltShearAllowable} \cdot \text{Nbols})$

= $1650 / (82.8 \cdot 4)$

= 0.4886 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + \text{Sum of X Moments}$

= $600 \cdot 0 + 0$

= 0.00 Kg-m.

Eccentricity (e):

= $Rmom / Q0$

= $0 / 2524$

= 0.00 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$

= $\max[0, 0.49, 0]$

= 0.4886 cm²

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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Saddle Calcs: Road Transport: Step: 21 9:31am Sep 9,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Note:

Wear Pad Width (365.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(285 + 1.56 \cdot \sqrt{226 \cdot 12}, 2 \cdot 620)$$

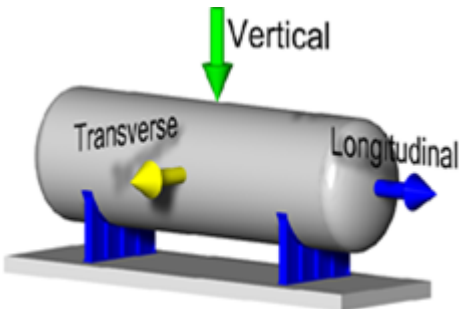
$$= 366.2399 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	226.00	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	3188.00	mm
Distance from Saddle to Vessel tangent	a1 or a	620.00	mm
Saddle Width	b1 or b	285.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		136.80	N./mm ²
Head Allowable Stress used in Calculation		131.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	600.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		4922.26	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	1.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	0.0	Km/hr



Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top	of Midspan	-4.64	-234.01
Long. Stress at Top	of Midspan	4.64	136.80
Long. Stress at Bottom	of Midspan	4.64	136.80
Long. Stress at Top	of Saddles	26.47	136.80
Long. Stress at Bottom	of Saddles	-14.67	-234.01
Long. Stress at Bottom	of Saddles	14.67	136.80



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Tangential Shear in Shell	12.73	109.44
Circ. Stress at Horn of Saddle	29.32	205.20
Circ. Compressive Stress in Shell	0.84	136.80

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf$$

$$= 35.8 \cdot 0.00256 \cdot 0^2 \cdot 1$$

$$= 0.0 \text{ lbf}$$

$$= 0.000 \text{ Kg}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult}$$

$$= \pi \cdot 2.05^2/4 \cdot 0.00256 \cdot 0^2 \cdot 1 \cdot 1.2$$

$$= 0.0 \text{ lbf}$$

$$= 0.0 \text{ Kg}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$= gz \cdot \text{Shiprtng Weight} \cdot gw + Ftw$$

$$= 0.1 \cdot 2301 \cdot 1 + 0$$

$$= 230.1 \text{ Kg}$$

Longitudinal Force due to Axial Acceleration [Fl]:

$$= gx \cdot \text{Shiprtng Weight} \cdot gw + Flw$$

$$= 0.1 \cdot 2301 \cdot 1 + 0$$

$$= 230.1 \text{ Kg}$$

Saddle Reaction Force due to Transverse Load [Fwt]:

$$= Ft/\text{Num of Saddles} \cdot B / E$$

$$= 230/2 \cdot 600/428$$

$$= 161.4 \text{ Kg}$$

Saddle Reaction Force due to Longitudinal Load [Fwl]:

$$= Fl \cdot B / Ls$$

$$= 230 \cdot 600/1860$$

$$= 74.2 \text{ Kg}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [Fsl]:

$$= \text{Shiprtng Weight} \cdot gy$$

$$= 2301 \cdot 1.05$$

$$= 2416.397 \text{ Kg}$$

Load Combination Results due to Transport Loading [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl)$$

$$= 2506 + \max(74.2, 161, 2416)$$

$$= 4922.3 \text{ Kg}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	5013.44	Kgf
Transverse Shear Load	115.07	Kgf
Longitudinal Shear Load	230.13	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(440/2 + 12 + 15)\sin(120/2), 2 \cdot 226)$$

$$= 427.817 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405



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Saddle Calcs: Road Transport: Step: 21 9:31am Sep 9, 2026

$$K9 = 0.2711 \quad K10 = 0.0581 \quad K1* = 0.1923$$

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned} &= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm^2)/3L)] \\ &= -4922 \cdot 620 [1 - (1 - 620/3188 + (226^2 - 0^2) / (2 \cdot 620 \cdot 3188)) / (1 + (4 \cdot 0) / (3 \cdot 3188))] \\ &= -554.1 \text{ Kg-m.} \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned} &= Q \cdot L / 4 (1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L \\ &= 4922 \cdot 3188 / 4 (1 + 2(226^2 - 0^2)/(3188^2)) / (1 + (4 \cdot 0) / (3 \cdot 3188)) - 4 \cdot 620 / 3188 \\ &= 910.7 \text{ Kg-m.} \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned} &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\ &= 0 \cdot 226 / (2 \cdot 12) - 911 / (\pi \cdot 226^2 \cdot 12) \\ &= -4.64 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned} &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\ &= 0 \cdot 226 / (2 \cdot 12) + 911 / (\pi \cdot 226^2 \cdot 12) \\ &= 4.64 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned} &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ &= 0 \cdot 226 / (2 \cdot 12) - 554 / (0.11 \cdot \pi \cdot 226^2 \cdot 12) \\ &= 26.47 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned} &= P \cdot Rm / (2t) + M1 / (K1^* \cdot \pi \cdot Rm^2 \cdot t) \\ &= 0 \cdot 226 / (2 \cdot 12) + 554 / (0.19 \cdot \pi \cdot 226^2 \cdot 12) \\ &= -14.67 \text{ N./mm}^2 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned} &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\ &= 4922 (3188 - 2 \cdot 620) / (3188 + (4 \cdot 0 / 3)) \\ &= 3007.7 \text{ Kgf} \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned} &= K2 \cdot T / (Rm \cdot t) \\ &= 1.17 \cdot 3008 / (226 \cdot 12) \\ &= 12.73 \text{ N./mm}^2 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned} &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\ &= 0.78 \cdot \sqrt{(226 \cdot 12)} \\ &= 40.620 \text{ mm} \end{aligned}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$\begin{aligned} &= -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\ &= -0.76 \cdot 4922 \cdot 0.1 / (12(285 + 40.6 + 40.6)) \\ &= -0.84 \text{ N./mm}^2 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$\begin{aligned} &= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2) \\ &= -4922 / (4 \cdot 12(285 + 40.6 + 40.6)) - 3 \cdot 0.053 \cdot 4922 / (2 \cdot 12^2) \\ &= -29.32 \text{ N./mm}^2 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned} &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\ &= \min(285 + 1.56 \cdot \sqrt{(226 \cdot 12)}, 2 \cdot 620) \end{aligned}$$



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= 366.24 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	560.0000	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	310.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	262.0	N./
Height of Web at Center	Hw,c	390.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth} })^3) / 12$$

$$= ((1.56 \cdot \sqrt{(220 \cdot 12) + 365 })^3) / 12$$

$$= 6.410 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{shell}} (C1 - Y)^2$$

$$= 6.41 + 53.4 (123 - 6)^2$$

$$= 7291.767 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	445.2	12.0	6.0	53.4	32051.1	0.729E+04
Wearplate	365.0	15.0	19.5	54.8	106762.5	0.585E+04
Web	10.0	341.0	197.5	34.1	673475.0	0.521E+04
BasePlate	310.0	12.0	374.0	37.2	1391280.0	0.235E+05
Totals	...	...	...	179.5	2203568.5	0.418E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2203568 / 179$$

$$= 122.783 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 4922$$

$$= 1001.7862 \text{ Kgf}$$

$$\text{Tension Stress, St} = (Fh / A_s) = 0.7794 \text{ N./mm}^2$$

$$\text{Allowed Stress, Sa} = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 600 - 220 \cdot \sin(120 / 2) / 1.05$$

$$= 418.061 \text{ mm}$$

$$\text{Bending Moment, M} = Fh \cdot d = 418.8165 \text{ Kg-m.}$$



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Bending Stress, $S_b = (M * C_1 / I) = 1.2055 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 * \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{ (3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{ (3(4922 + 91.2) 310 / (4 * 560 * 175))}$
 $= 10.811 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120/2) (220 + 12 + 15)$
 $= 427.817 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 428 / (3 - 1)$
 $= 213.908 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 214 * 310 / 2$
 $= 331.558 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 5013 / (560 * 310)$
 $= 2.888 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p * B_p$
 $= 332 * 2.89$
 $= 957.514 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 27.5 + 10.7$
 $= 38.195 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 958 / 38.2$
 $= 2.458 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	285.0	...	28.5	...	0.193E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 285 - 10$
 $= 275.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 38.2$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$



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$$= 285/2$$

$$= 142.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{1929/38.2}$$

$$= 71.068 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 600 - \cos(120/2) (220 + 12 + 15) - 12$$

$$= 464.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 199943392/262}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 464/71.1$$

$$= 6.536$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 230 / (2 \cdot 560) \cdot 214 \cdot 464/2$$

$$= 10.208 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($6.54 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (6.54)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (6.54) / (8 \cdot 123) - (6.54)^3 / (8 \cdot 123^3))$$

$$= 155.1 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 2.46/155 + (10.2 \cdot 142/19290936) / 175$$

$$= 0.020$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	275.0	0.0	27.5	0.0	0.193E+04
Web	213.9	10.0	0.0	21.4	0.0	1.78
Totals	...	...	...	48.9	...	0.193E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/48.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 285/2$$

$$= 142.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 600 - \cos(247/2) (12 + 0 + 0) - 0$$

$$= 341.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$



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$$= \sqrt{(1931/48.9)}$$

$$= 62.843 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 341/62.8$$

$$= 5.426$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 230 / (2 \cdot 560)$$

$$= 0.205 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.21 \cdot 214 \cdot 341$$

$$= 14.988 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 15/135494$$

$$= 1.085 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($5.43 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.43)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (5.43) / (8 \cdot 123) - (5.43)^3 / (8 \cdot 123^3))$$

$$= 155.5 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 3.84/155 + (15 \cdot 142/1931) / 175$$

$$= 0.031$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (365.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(285 + 1.56 \cdot \sqrt{226 \cdot 12}, 2 \cdot 600)$$

$$= 366.2399 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	226.00	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	3188.00	mm
Distance from Saddle to Vessel tangent	a1 or a	600.00	mm
Saddle Width	b1 or b	285.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	111.35	mm
Shell Allowable Stress used in Calculation		136.80	N./mm ²
Head Allowable Stress used in Calculation		136.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	600.00	mm
Coefficient of Friction	mu	0.30	



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Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		2420.93	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	1.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	0.0	Km/hr

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-2.06	-234.01
Long. Stress at Top of Midspan		2.06	136.80
Long. Stress at Bottom of Midspan		2.06	136.80
Long. Stress at Top of Saddles		14.90	136.80
Long. Stress at Bottom of Saddles		-8.26	-234.01
Long. Stress at Bottom of Saddles		8.26	136.80
Tangential Shear in Shell		6.11	109.44
Circ. Stress at Horn of Saddle		14.42	205.20
Circ. Compressive Stress in Shell		0.41	136.80

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 35.8 \cdot 0.00256 \cdot 0^2 \cdot 1 \\
 &= 0.0 \text{ lbf} \\
 &= 0.000 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 2.05^2/4 \cdot 0.00256 \cdot 0^2 \cdot 1 \cdot 1.2 \\
 &= 0.0 \text{ lbf} \\
 &= 0.0 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$\begin{aligned}
 &= gz \cdot \text{Shipping Weight} \cdot gw + Ftw \\
 &= 0.1 \cdot 2301 \cdot 1 + 0 \\
 &= 230.1 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [Fl]:

$$\begin{aligned}
 &= gx \cdot \text{Shipping Weight} \cdot gw + Flw \\
 &= 0.1 \cdot 2301 \cdot 1 + 0 \\
 &= 230.1 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [Fwt]:

$$\begin{aligned}
 &= Ft/\text{Num of Saddles} \cdot B / E \\
 &= 230/2 \cdot 600/419 \\
 &= 164.7 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [Fwl]:

$$\begin{aligned}
 &= Fl \cdot B / Ls \\
 &= 230 \cdot 600/1860 \\
 &= 74.2 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [Fsl]:

$$\begin{aligned}
 &= \text{Shipping Weight} \cdot gy \\
 &= 2301 \cdot 1.05 \\
 &= 2416.397 \text{ Kgf}
 \end{aligned}$$



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Load Combination Results due to Transport Loading [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl})$$

$$= 4.53 + \max(74.2, 165, 2416)$$

$$= 2420.9 \text{ Kg}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	2502.92	Kgf
Transverse Shear Load	115.07	Kgf
Longitudinal Shear Load	230.13	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(440/2 + 12 + 10)\sin(120/2), 2 \cdot 226)$$

$$= 419.156 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m^2)/3L)]$$

$$= -2421 \cdot 600 [1 - (1 - 600/3188 + (226^2 - 111^2)/(2 \cdot 600 \cdot 3188))/(1 + (4 \cdot 111)/(3 \cdot 3188))]$$

$$= -311.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L$$

$$= 2421 \cdot 3188/4(1 + 2(226^2 - 111^2)/(3188^2))/(1 + (4 \cdot 111)/(3 \cdot 3188)) - 4 \cdot 600/3188$$

$$= 405.1 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 0 \cdot 226/(2 \cdot 12) - 405/(\pi \cdot 226^2 \cdot 12)$$

$$= -2.06 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 226/(2 \cdot 12) + 405/(\pi \cdot 226^2 \cdot 12)$$

$$= 2.06 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 0 \cdot 226/(2 \cdot 12) - 312/(0.11 \cdot \pi \cdot 226^2 \cdot 12)$$

$$= 14.90 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 226/(2 \cdot 12) + 312/(0.19 \cdot \pi \cdot 226^2 \cdot 12)$$

$$= -8.26 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 2421(3188 - 2 \cdot 600)/(3188 + (4 \cdot 111/3))$$

$$= 1442.5 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 1442 / (226 \cdot 12)$$

$$= 6.11 \text{ N./mm}^2$$



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Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(226 \cdot 12)}$$

$$= 40.620 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 2421 \cdot 0.1 / (12(285 + 40.6 + 40.6))$$

$$= -0.41 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -2421 / (4 \cdot 12(285 + 40.6 + 40.6)) - 3 \cdot 0.053 \cdot 2421 / (2 \cdot 12^2)$$

$$= -14.42 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(285 + 1.56 \cdot \sqrt{(226 \cdot 12)}, 2 \cdot 600)$$

$$= 366.24 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	560.0000	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	310.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	262.0	N./
Height of Web at Center	Hw,c	390.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(220 \cdot 12)} + 365)^3) / 12$$

$$= 6.410 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 6.41 + 53.4(134 - 6)^2$$

$$= 8699.846 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	445.2	12.0	6.0	53.4	32051.1	0.870E+04
Wearplate	365.0	10.0	17.0	36.5	62050.0	0.496E+04
Web	10.0	346.0	195.0	34.6	674699.9	0.476E+04
BasePlate	310.0	12.0	374.0	37.2	1391279.9	0.215E+05
Totals	...	...	...	161.7	2160081.0	0.399E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2160081 / 162$$

$$= 133.570 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$



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Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 2421$$

$$= 492.7114 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.4462 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 600 - 220 \cdot \sin(120/2) / 1.05$$

$$= 418.061 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 205.9877 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.6758 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(2421 + 82) \cdot 310) / (4 \cdot 560 \cdot 175)}$$

$$= 7.638 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot \text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}$$

$$= 2 \cdot \cos(90 - 120/2) \cdot (220 + 12 + 10)$$

$$= 419.156 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 419 / (3 - 1)$$

$$= 209.578 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 210 \cdot 310 / 2$$

$$= 324.846 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 2503 / (560 \cdot 310)$$

$$= 1.442 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 325 \cdot 1.44$$

$$= 468.355 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 27.5 + 10.5$$

$$= 37.979 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 468 / 38$$

$$= 1.209 \text{ N./mm}^2$$



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Saddle Calcs: Road Transport: Step: 21 9:31am Sep 9, 2026

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	285.0	...	28.5	...	0.193E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 285 - 10

= 275.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/38

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 285/2

= 142.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{(1929/38)}$

= 71.270 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 600 - cos(120/2)(220 + 12 + 10) - 12

= 467.000 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \cdot \pi^2 \cdot 199943008/262)}$

= 122.739

Slenderness ratio [KL/r]:

= KL/r

= 1 • 467/71.3

= 6.553

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 230 / (2 • 560) • 210 • 467/2

= 10.055 Kg-m.

Compressive Allowable, KL/r < Cc (6.55 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (6.55)^2 / (2 \cdot 123^2)) 262 /$

$(5/3 + 3 \cdot (6.55) / (8 \cdot 123) - (6.55^3) / (8 \cdot 123^3))$

= 155.1 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 1.21/155 + (10.1 • 142/19290936) / 175

= 0.012

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
--	---	---	---	---	----	----



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Saddle Calcs: Road Transport: Step: 21 9:31am Sep 9, 2026

Rib	10.0	275.0	0.0	27.5	0.0	0.193E+04
Web	209.6	10.0	0.0	21.0	0.0	1.75
Totals	...	...	...	48.5	...	0.193E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/48.5$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 285/2$$

$$= 142.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 600 - \cos(242/2)(12 + 0 + 0) - 0$$

$$= 346.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1931/48.5)}$$

$$= 63.122 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 346/63.1$$

$$= 5.481$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 230/(2 \cdot 560)$$

$$= 0.205 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.21 \cdot 210 \cdot 346$$

$$= 14.900 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 14.9/135492$$

$$= 1.078 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($5.48 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.48)^2 / (2 \cdot 123^2)) 262 /$$

$$(5/3 + 3 \cdot (5.48) / (8 \cdot 123) - (5.48^3) / (8 \cdot 123^3))$$

$$= 155.5 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 1.9/155 + (14.9 \cdot 142/1931) / 175$$

$$= 0.018$$

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Operating Case: Step: 22 9:31am Sep 9,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	227.50	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	3188.00	mm
Distance from Saddle to Vessel tangent	a1 or a	620.00	mm
Saddle Width	b1 or b	285.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	365.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	15.0	mm
Wear Plate Allowable Stress	fw or Sr	136.80	N./mm ²
Shell Allowable Stress used in Calculation		136.80	N./mm ²
Head Allowable Stress used in Calculation		131.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	600.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		3609.80	Kgf
Pressure used in Analysis	P	521.764	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	2.12	136.80
Long. Stress at Bottom of Midspan	11.07	136.80
Long. Stress at Top of Saddles	32.11	136.80
Long. Stress at Bottom of Saddles	-7.55	-187.92
Long. Stress at Bottom of Saddles	7.55	136.80
Tangential Shear in Shell	12.37	109.44
Circ. Stress at Horn of Saddle	5.91	171.00
Circ. Compressive Stress in Shell	0.32	136.80

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (145/2 + 0) \cdot 600/428$
 $= 305.1 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(37.8, 0) \cdot 600/1860$
 $= 12.2 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(195, 0) \cdot 600/1860$
 $= 63.0 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:



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$$\begin{aligned}
 &= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E \\
 &= 3 (195/2 + 0) \cdot 600/428 \\
 &= 411.1 \text{ Kg}
 \end{aligned}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$\begin{aligned}
 &= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst}) \\
 &= 3199 + \max(12.2, 305, 63, 411) \\
 &= 3609.8 \text{ Kg}
 \end{aligned}$$

Longitudinal Wind Force [F]:

$$\begin{aligned}
 &= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2)) \\
 &= 1.1 \cdot 766 (0 + \pi/4 (0.62 \cdot 1.2)^2) \\
 &= 371.100 \text{ N}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	3700.98	Kgf
Transverse Shear Load	97.70	Kgf
Longitudinal Shear Load	195.41	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m) \\
 &= \min(2(440/2 + 12 + 15) \sin(120/2), 2 \cdot 227) \\
 &= 427.817 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m^2)/3L)] \\
 &= -3610 \cdot 620 [1 - (1 - 620/3188 + (227^2 - 0^2) / (2 \cdot 620 \cdot 3188)) / (1 + (4 \cdot 0) / (3 \cdot 3188))] \\
 &= -406.0 \text{ Kg-m}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L / 4 (1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L \\
 &= 3610 \cdot 3188 / 4 (1 + 2(227^2 - 0^2)/(3188^2)) / (1 + (4 \cdot 0) / (3 \cdot 3188)) - 4 \cdot 620 / 3188 \\
 &= 668.2 \text{ Kg-m}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 \cdot t) \\
 &= 522 \cdot 227 / (2 \cdot 9) - 668 / (\pi \cdot 227^2 \cdot 9) \\
 &= 2.12 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t) \\
 &= 522 \cdot 227 / (2 \cdot 9) + 668 / (\pi \cdot 227^2 \cdot 9) \\
 &= 11.07 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 522 \cdot 227 / (2 \cdot 9) - 406 / (0.11 \cdot \pi \cdot 227^2 \cdot 9) \\
 &= 32.11 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$



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$$= 522 \cdot 227 / (2 \cdot 9) + 406 / (0.19 \cdot \pi \cdot 227^2 \cdot 9)$$

$$= -7.55 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 3610 (3188 - 2 \cdot 620) / (3188 + (4 \cdot 0 / 3))$$

$$= 2205.7 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 2206 / (227 \cdot 9)$$

$$= 12.37 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(227 \cdot 9)}$$

$$= 35.294 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(285 + 1.56 \cdot \sqrt{(227 \cdot 9)}, 2 \cdot 620)$$

$$= 355.59 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(137 / 137, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 3610 \cdot 0.1 / (356(9 + 1 \cdot 15))$$

$$= -0.32 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -3610 / (4(9 + 1 \cdot 15)356) -$$

$$3 \cdot 0.053 \cdot 3610 / (2(9 + 1 \cdot 15)^2)$$

$$= -5.91 \text{ N./mm}^2$$

Distance between Saddle Supports [L_s]:

$$= 1860.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.13139 \text{E-04} \cdot 1860 (280 - 21.1)$$

$$= 6.327 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	560.0000	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	310.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	262.0	N./
Height of Web at Center	Hw,c	390.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(223 \cdot 9)} + 365)^3) / 12$$

$$= 2.642 \text{ cm}^4$$



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Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 2.64 + 39.1 (130 - 4.5)^2$$

$$= 6190.162 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	434.9	9.0	4.5	39.1	17612.9	0.619E+04
Wearplate	365.0	15.0	16.5	54.8	90337.5	0.709E+04
Web	10.0	341.0	194.5	34.1	663244.9	0.471E+04
BasePlate	310.0	12.0	371.0	37.2	1380119.9	0.216E+05
Totals	...	...	...	165.2	2151315.2	0.396E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2151315/165$$

$$= 130.233 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 3610$$

$$= 734.6722 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.5716 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 600 - 223 \cdot \sin(120/2) / 1.05$$

$$= 415.580 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 305.3215 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.9856 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(3610 + 91.2)310 / (4 \cdot 560 \cdot 175))}$$

$$= 9.288 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (220 + 12 + 15)$$

$$= 427.817 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 428 / (3 - 1)$$

$$= 213.908 \text{ mm}$$



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 214 \cdot 310 / 2$$

$$= 331.558 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 3701 / (560 \cdot 310)$$

$$= 2.132 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 332 \cdot 2.13$$

$$= 706.848 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 27.5 + 10.7$$

$$= 38.195 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 707 / 38.2$$

$$= 1.815 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	285.0	...	28.5	...	0.193E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 285 - 10$$

$$= 275.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 38.2$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 285 / 2$$

$$= 142.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (1929 / 38.2) }$$

$$= 71.068 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 600 - \cos(120 / 2) (220 + 12 + 15) - 12$$

$$= 464.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199943392 / 262) }$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 464/71.1$$

$$= 6.536$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 195 / (2 \cdot 560) \cdot 214 \cdot 464 / 2$$

$$= 8.668 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($6.54 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (6.54)^2 / (2 \cdot 123^2)) 262 /$$

$$(5/3 + 3 \cdot (6.54) / (8 \cdot 123) - (6.54^3) / (8 \cdot 123^3))$$

$$= 155.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 1.81/155 + (8.67 \cdot 142/19290936) / 175$$

$$= 0.015$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	275.0	0.0	27.5	0.0	0.193E+04
Web	213.9	10.0	0.0	21.4	0.0	1.78
Totals	...	...	...	48.9	...	0.193E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/48.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 285/2$$

$$= 142.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 600 - \cos(247/2) (12 + 0 + 0) - 0$$

$$= 341.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1931/48.9) }$$

$$= 62.843 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 341/62.8$$

$$= 5.426$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 195 / (2 \cdot 560)$$

$$= 0.174 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.17 \cdot 214 \cdot 341$$

$$= 12.727 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, \text{base}, c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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$$= 12.7/135494$$

$$= 0.921 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($5.43 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r)^4 / (8 \cdot C_c^4) - (KL/r)^6 / (8 \cdot C_c^6))$$

$$= (1 - (5.43)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (5.43)^4 / (8 \cdot 123^4) - (5.43)^6 / (8 \cdot 123^6))$$

$$= 155.5 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 2.84/155 + (12.7 \cdot 142/1931) / 175$$

$$= 0.024$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-307 A	
Bolt Allowable Stress	Stba	138.00	N./mm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.0	mm
User Defined Area in a Single Bolt	UserBoltArea	1.5500	cm ²
BasePlate Allowable Stress	S	138.00	N./mm ²
Area Available in a Single Bolt	BlArea	0.9592	cm ²
Saddle Load QO (Weight)	QO	3289.9	Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	63.0	Kgf
Maximum Transverse Force	Ft	97.7	Kgf
Maximum Longitudinal Force	Fl	195.4	Kgf
Saddle Bolted to Steel Foundation		No	

Friction force due to thermal growth, [Ff]:

$$= 93.2 \text{ Kgf}$$

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= \max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 195 / (0.96 \cdot 4)$$

$$= 5.0 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 97.7 / (0.96 \cdot 4)$$

$$= 2.5 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 195 / (82.8 \cdot 4)$$

$$= 0.0579 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 600 \cdot 97.7 + 0$$

$$= 58.62 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 58.6/3290$$

$$= 17.82 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$



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Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0, 0.058, 0]
= 0.0579 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 12, tr = 2.5, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.28, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	227.50	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	3188.00	mm
Distance from Saddle to Vessel tangent	a1 or a	600.00	mm
Saddle Width	b1 or b	285.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	365.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	136.80	N./mm ²
Inside or Mean Depth of Head	Hi or hm	111.35	mm
Shell Allowable Stress used in Calculation		136.80	N./mm ²
Head Allowable Stress used in Calculation		136.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	600.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 648.35 Kgf
Pressure used in Analysis P 521.764 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	5.87	136.80
Long. Stress at Bottom of Midspan	7.32	136.80
Long. Stress at Top of Saddles	11.84	136.80
Long. Stress at Bottom of Saddles	3.69	136.80
Tangential Shear in Shell	2.17	109.44
Circ. Stress at Horn of Saddle	1.63	171.00
Circ. Compressive Stress in Shell	0.07	136.80

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:



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Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (145/2 + 0) \cdot 600/419$$

$$= 311.4 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(37.8, 0) \cdot 600/1860$$

$$= 12.2 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(195, 0) \cdot 600/1860$$

$$= 63.0 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (195/2 + 0) \cdot 600/419$$

$$= 419.6 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 229 + \max(12.2, 311, 63, 420)$$

$$= 648.3 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(0.62 \cdot 1.2)^2)$$

$$= 371.100 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	730.34	Kgf
Transverse Shear Load	97.70	Kgf
Longitudinal Shear Load	195.41	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(440/2 + 12 + 10)\sin(120/2), 2 \cdot 227)$$

$$= 419.156 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm^2)/3L)]$$

$$= -648 \cdot 600 [1 - (1 - 600/3188 + (227^2 - 111^2)/(2 \cdot 600 \cdot 3188))/(1 + (4 \cdot 111)/(3 \cdot 3188))]$$

$$= -83.4 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4(1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L$$

$$= 648 \cdot 3188/4(1 + 2(227^2 - 111^2)/(3188^2))/(1 + (4 \cdot 111)/(3 \cdot 3188)) - 4 \cdot 600/3188$$

$$= 108.6 \text{ Kg-m.}$$



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Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 522 \cdot 227 / (2 \cdot 9) - 109 / (\pi \cdot 227^2 \cdot 9)$$

$$= 5.87 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 522 \cdot 227 / (2 \cdot 9) + 109 / (\pi \cdot 227^2 \cdot 9)$$

$$= 7.32 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 522 \cdot 227 / (2 \cdot 9) - 83.4 / (0.11 \cdot \pi \cdot 227^2 \cdot 9)$$

$$= 11.84 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 522 \cdot 227 / (2 \cdot 9) + 83.4 / (0.19 \cdot \pi \cdot 227^2 \cdot 9)$$

$$= 3.69 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 648 (3188 - 2 \cdot 600) / (3188 + (4 \cdot 111 / 3))$$

$$= 386.3 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 386 / (227 \cdot 9)$$

$$= 2.17 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(227 \cdot 9)}$$

$$= 35.294 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(285 + 1.56 \cdot \sqrt{(227 \cdot 9)}, 2 \cdot 600)$$

$$= 355.59 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(137 / 137, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 648 \cdot 0.1 / (356(9 + 1 \cdot 10))$$

$$= -0.07 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -648 / (4(9 + 1 \cdot 10)356) -$$

$$3 \cdot 0.053 \cdot 648 / (2(9 + 1 \cdot 10)^2)$$

$$= -1.63 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	560.0000	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	310.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	



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Saddle Yield Stress	Sy	262.0	N./
Height of Web at Center	Hw, c	390.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(223 \cdot 9)} + 365)^3) / 12$$

$$= 2.642 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 2.64 + 39.1 (143 - 4.5)^2$$

$$= 7545.606 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	434.9	9.0	4.5	39.1	17612.9	0.755E+04
Wearplate	365.0	10.0	14.0	36.5	51100.0	0.611E+04
Web	10.0	346.0	192.0	34.6	664319.9	0.427E+04
BasePlate	310.0	12.0	371.0	37.2	1380119.9	0.193E+05
Totals	...	...	...	147.4	2113153.0	0.372E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2113153 / 147$$

$$= 143.323 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 648$$

$$= 131.9523 \text{ Kg}$$

Tension Stress, St = (Fh/As)	=	0.1195	N./mm ²
Allowed Stress, Sa = 0.6 * Yield Str	=	157.1972	N./mm ²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 600 - 223 \cdot \sin(120/2) / 1.05$$

$$= 415.580 \text{ mm}$$

Bending Moment, M = Fh * d	=	54.8379	Kg-m.
----------------------------	---	---------	-------

Bending Stress, Sb = (M * C1 / I)	=	0.2071	N./mm ²
Allowed Stress, Sa = 2/3 * Yield Str	=	174.6635	N./mm ²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(648 + 82)310 / (4 \cdot 560 \cdot 175)}$$

$$= 4.126 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:



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Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (220 + 12 + 10)$$

$$= 419.156 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 419 / (3 - 1)$$

$$= 209.578 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 210 \cdot 310 / 2$$

$$= 324.846 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 730 / (560 \cdot 310)$$

$$= 0.421 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 325 \cdot 0.42$$

$$= 136.664 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 27.5 + 10.5$$

$$= 37.979 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 137 / 38$$

$$= 0.353 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	285.0	...	28.5	...	0.193E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 285 - 10$$

$$= 275.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 38$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 285 / 2$$

$$= 142.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1929 / 38)}$$

$$= 71.270 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$



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$$= 600 - \cos(120/2)(220 + 12 + 10) - 12$$

$$= 467.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 262)}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 467 / 71.3$$

$$= 6.553$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 195 / (2 \cdot 560) \cdot 210 \cdot 467 / 2$$

$$= 8.538 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($6.55 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (6.55)^2 / (2 \cdot 123^2)) 262 /$$

$$(5/3 + 3 \cdot (6.55) / (8 \cdot 123) - (6.55)^3 / (8 \cdot 123^3))$$

$$= 155.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 0.35 / 155 + (8.54 \cdot 142 / 19290936) / 175$$

$$= 0.006$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	275.0	0.0	27.5	0.0	0.193E+04
Web	209.6	10.0	0.0	21.0	0.0	1.75
Totals	...	...	...	48.5	...	0.193E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 48.5$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 285 / 2$$

$$= 142.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{\text{pthk}}$$

$$= 600 - \cos(242/2)(12 + 0 + 0) - 0$$

$$= 346.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1931 / 48.5)}$$

$$= 63.122 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 346 / 63.1$$

$$= 5.481$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$



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$$= 195 / (2 \cdot 560)$$

$$= 0.174 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.17 \cdot 210 \cdot 346$$

$$= 12.652 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 12.7 / 135492$$

$$= 0.916 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($5.48 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.48)^2 / (2 \cdot 123^2)) 262 /$$

$$(5/3 + 3 \cdot (5.48) / (8 \cdot 123) - (5.48^3) / (8 \cdot 123^3))$$

$$= 155.5 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 0.55 / 155 + (12.7 \cdot 142 / 1931) / 175$$

$$= 0.009$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-307 A	
Bolt Allowable Stress	Stba	138.00	N./mm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.0	mm
User Defined Area in a Single Bolt	UserBoltArea	1.5500	cm ²
BasePlate Allowable Stress	S	138.00	N./mm ²
Area Available in a Single Bolt	BltArea	0.9592	cm ²
Saddle Load Q0 (Weight)	Q0	310.8	Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	63.0	Kgf
Maximum Transverse Force	Ft	97.7	Kgf
Maximum Longitudinal Force	F1	195.4	Kgf
Saddle Bolted to Steel Foundation		No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 195 / (0.96 \cdot 4)$$

$$= 5.0 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 97.7 / (0.96 \cdot 4)$$

$$= 2.5 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 195 / (82.8 \cdot 4)$$

$$= 0.0579 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$



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$$= 600 \cdot 97.7 + 0$$

$$= 58.62 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 58.6/311$$

$$= 188.64 \text{ mm} > B_{plen}/6 \rightarrow \text{Uplift in Transverse direction}$$

$$f = B_{plen} / 2 - Edgedis$$

$$= 560/2 - 50$$

$$= 229.99 \text{ mm}$$

Modular Ratio Of Steel/Concrete (n1):

$$= E_S / E_C$$

$$= 203402/21526$$

$$= 9.45$$

$$K1 = 3 (e - 0.5 \cdot B_{plen})$$

$$= 3 (189 - 0.5 \cdot 560)$$

$$= -274.08 \text{ mm}$$

$$K2 = 6 \cdot n1 \cdot A_t / B_{pwid} \cdot (f + e)$$

$$= 6 \cdot 9.45 \cdot 1.92/310 \cdot (230 + 189)$$

$$= 14686.78 \text{ mm}^2$$

$$K3 = -K2 \cdot (0.5 \cdot B_{plen} + f)$$

$$= -14687 \cdot (0.5 \cdot 560 + 230)$$

$$= -7490072.61 \text{ mm}^3$$

Iteratively Solving for the Effective Bearing Length:

$$Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$$

$$Y^3 + -274 \cdot Y^2 + 14687 \cdot Y + -7490073 = 0$$

$$Y = 306.06 \text{ mm}$$

$$Num = (B_{plen} / 2 - Y / 3 - e)$$

$$= (560/2 - 306/3 - 189)$$

$$= -10.66$$

$$Denom = (B_{plen} / 2 - Y / 3 + f)$$

$$= (560/2 - 306/3 + 230)$$

$$= 407.97$$

Total Bolt Tension Force [Tforce]:

$$= -QO \cdot Num / Denom$$

$$= -311 \cdot -10.7/408$$

$$= 8.12 \text{ Kg}$$

Bolt Area Required due to Transverse Load [Bltareart]:

$$= Tforce / (Stba \cdot Nbt)$$

$$= 8.12 / (138 \cdot 2)$$

$$= 0.0029 \text{ cm}^2$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.058, 0.0029]$$

$$= 0.0579 \text{ cm}^2$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$= 2(QO + Tforce) / (Y \cdot B_{pwid})$$

$$= 2(311 + 8.12) / (306 \cdot 310)$$

$$= 65.92 \text{ KPa.}$$



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Distance from Baseplate Edge to the Web [ADIST]:

$$= (B_{plen} - WebLength) / 2$$

$$= (560 - 509) / 2$$

$$= 25.4000 \text{ mm}$$

Overturning Moment due To Bolt Tension [Mt]:

$$= T_{force} \cdot Adist$$

$$= 8.12 \cdot 25.4$$

$$= 0.21 \text{ Kg-m.}$$

Equivalent Bearing Pressure (f1):

$$= f_c \cdot (Y - Adist) / Y$$

$$= 65.9 \cdot (306 - 25.4) / 306$$

$$= 60.45 \text{ KPa.}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$= (Adist^2 \cdot Bp_{wid} / 6) \cdot (f1 + 2 \cdot f_c)$$

$$= (25.4^2 \cdot 310 / 6) \cdot (60.4 + 2 \cdot 65.9)$$

$$= 0.65 \text{ Kg-m.}$$

Baseplate Required Thickness [Treq]:

$$= \sqrt{(6 \cdot \max(Mt, Mc) / (Bp_{wid} \cdot S_{ba}))}$$

$$= \sqrt{(6 \cdot \max(0.21, 0.65) / (310 \cdot 207))}$$

$$= 0.7742 \text{ mm}$$

MDMT Calculations for: Sdl 2 Fr40, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 10, c = 3 mm, E* = 1

Thickness Ratio = tr * E* / (tg - c) = 0.28, Temp. Reduction = 78 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	227.50	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	3188.00	mm
Distance from Saddle to Vessel tangent	a1 or a	620.00	mm
Saddle Width	b1 or b	285.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	365.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	15.0	mm
Wear Plate Allowable Stress	fw or Sr	136.80	N./mm ²
Shell Allowable Stress used in Calculation		234.01	N./mm ²
Head Allowable Stress used in Calculation		234.01	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	600.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		3144.14	Kgf
Pressure used in Analysis	P	678.230	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	4.67	234.01
Long. Stress at Bottom of Midspan	12.47	234.01
Long. Stress at Top of Saddles	30.80	234.01
Long. Stress at Bottom of Saddles	-3.75	-234.01
Long. Stress at Bottom of Saddles	3.75	234.01
Tangential Shear in Shell	10.77	138.59
Circ. Stress at Horn of Saddle	5.15	351.01
Circ. Compressive Stress in Shell	0.27	234.01

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (72.5/2 + 0) \cdot 600/428$$

$$= 152.6 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(18.9, 0) \cdot 600/1860$$

$$= 6.1 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 2992 + \max(6.1, 153, 0, 0)$$

$$= 3144.1 \text{ Kgf}$$

Longitudinal Wind Force [FI]:



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$$\begin{aligned}
 &= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2)) \\
 &= 1.1 \cdot 766 (0 + \pi/4(0.62 \cdot 1.2)^2) \\
 &= 371.100 \text{ N}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	3235.32	Kgf
Transverse Shear Load	36.26	Kgf
Longitudinal Shear Load	18.92	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m) \\
 &= \min(2(440/2 + 12 + 15)\sin(120/2), 2 \cdot 227) \\
 &= 427.817 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m^2)/3L)] \\
 &= -3144 \cdot 620 [1 - (1 - 620/3188 + (227^2 - 0^2)/(2 \cdot 620 \cdot 3188))/(1 + (4 \cdot 0)/(3 \cdot 3188))] \\
 &= -353.6 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/3L) - 4a/L \\
 &= 3144 \cdot 3188/4(1 + 2(227^2 - 0^2)/(3188^2))/(1 + (4 \cdot 0)/(3 \cdot 3188)) - 4 \cdot 620/3188 \\
 &= 582.0 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t) \\
 &= 678 \cdot 227/(2 \cdot 9) - 582/(\pi \cdot 227^2 \cdot 9) \\
 &= 4.67 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t) \\
 &= 678 \cdot 227/(2 \cdot 9) + 582/(\pi \cdot 227^2 \cdot 9) \\
 &= 12.47 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t) \\
 &= 678 \cdot 227/(2 \cdot 9) - 354/(0.11 \cdot \pi \cdot 227^2 \cdot 9) \\
 &= 30.80 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M1/(K1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 678 \cdot 227/(2 \cdot 9) + 354/(0.19 \cdot \pi \cdot 227^2 \cdot 9) \\
 &= -3.75 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a)/(L + (4 \cdot h_m/3)) \\
 &= 3144(3188 - 2 \cdot 620)/(3188 + (4 \cdot 0/3)) \\
 &= 1921.2 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$



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$$= 1.17 \cdot 1921 / (227 \cdot 9)$$

$$= 10.77 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(227 \cdot 9)}$$

$$= 35.294 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(285 + 1.56 \cdot \sqrt{(227 \cdot 9)}, 2 \cdot 620)$$

$$= 355.59 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ6,r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 3144 \cdot 0.1 / (356(9 + 1 \cdot 15))$$

$$= -0.27 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L ≥ 8Rm (4.15.28) [σ7,r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -3144 / (4(9 + 1 \cdot 15)356) -$$

$$3 \cdot 0.053 \cdot 3144 / (2(9 + 1 \cdot 15)^2)$$

$$= -5.15 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	560.0000	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	310.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	262.0	N./
Height of Web at Center	Hw,c	390.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(223 \cdot 9)} + 365)9^3) / 12$$

$$= 2.642 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 2.64 + 39.1(130 - 4.5)^2$$

$$= 6190.162 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	434.9	9.0	4.5	39.1	17612.9	0.619E+04
Wearplate	365.0	15.0	16.5	54.8	90337.5	0.709E+04
Web	10.0	341.0	194.5	34.1	663244.9	0.471E+04
BasePlate	310.0	12.0	371.0	37.2	1380119.9	0.216E+05
Totals	...	...	...	165.2	2151315.2	0.396E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2151315 / 165$$

$$= 130.233 \text{ mm}$$



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Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120/2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.2 \cdot 3144 \\
 &= 639.8998 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 0.4978 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 600 - 223 \cdot \sin(120/2) / 1.05 \\
 &= 415.580 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 265.9352 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 0.8585 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{(3(3144 + 91.2)310) / (4 \cdot 560 \cdot 175)} \\
 &= 8.684 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120/2) (220 + 12 + 15) \\
 &= 427.817 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 428 / (3 - 1) \\
 &= 213.908 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 214 \cdot 310 / 2 \\
 &= 331.558 \text{ cm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 3235 / (560 \cdot 310) \\
 &= 1.864 \text{ Kgf/cm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= Ap \cdot Bp \\
 &= 332 \cdot 1.86 \\
 &= 617.911 \text{ Kgf}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 27.5 + 10.7 \\
 &= 38.195 \text{ cm}^2
 \end{aligned}$$



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Compressive Stress [Sc]:

= P/A_r

= 618/38.2

= 1.587 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	285.0	...	28.5	...	0.193E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 285 - 10

= 275.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/38.2

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 285/2

= 142.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{(1929/38.2)}$

= 71.068 mm

Length of Outer Rib [L]:

= Saddle Height - $\cos(\theta/2) \cdot (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$

= 600 - $\cos(120/2) \cdot (220 + 12 + 15) - 12$

= 464.500 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \cdot \pi^2 \cdot 199943392/262)}$

= 122.739

Slenderness ratio [KL/r]:

= KL/r

= 1 • 464/71.1

= 6.536

Bending Moment [Rm]:

= $Fl / (2 \cdot B_{pln}) \cdot e \cdot L / 2$

= 18.9/(2 • 560) • 214 • 464/2

= 0.839 Kg-m.

Compressive Allowable, $KL/r < C_c$ (6.54 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$

= $(1 - (6.54)^2 / (2 \cdot 123^2)) 262 /$

$(5/3 + 3 \cdot (6.54) / (8 \cdot 123) - (6.54^3) / (8 \cdot 123^3))$

= 155.1 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= $Sc/Sca + (R_m \cdot C_1 / I) / Sba$

= 1.59/155 + (0.84 • 142/19290936)/175

= 0.011

Check of Inside Ribs:



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Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	275.0	0.0	27.5	0.0	0.193E+04
Web	213.9	10.0	0.0	21.4	0.0	1.78
Totals	...	...	...	48.9	...	0.193E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/48.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 285/2$$

$$= 142.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 600 - \cos(247/2)(12 + 0 + 0) - 0$$

$$= 341.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1931/48.9)}$$

$$= 62.843 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 341/62.8$$

$$= 5.426$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 18.9 / (2 \cdot 560)$$

$$= 0.017 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.017 \cdot 214 \cdot 341$$

$$= 1.232 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1.23/135494$$

$$= 0.089 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($5.43 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.43)^2 / (2 \cdot 123^2)) 262 /$$

$$(5/3 + 3 \cdot (5.43) / (8 \cdot 123) - (5.43^3) / (8 \cdot 123^3))$$

$$= 155.5 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 2.48/155 + (1.23 \cdot 142/1931) / 175$$

$$= 0.016$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification	SA-307 A	
Bolt Allowable Stress	Stba	138.00 N./mm ²



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Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.0	mm
User Defined Area in a Single Bolt	UserBoltArea	1.5500	cm ²
BasePlate Allowable Stress	S	138.00	N./mm ²
Area Available in a Single Bolt	BlArea	0.9592	cm ²
Saddle Load QO (Weight)	QO	3082.8	Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	6.1	Kgf
Maximum Transverse Force	Ft	36.3	Kgf
Maximum Longitudinal Force	Fl	18.9	Kgf
Saddle Bolted to Steel Foundation		No	

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 18.9 / (0.96 \cdot 4)$$

$$= 0.5 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 36.3 / (0.96 \cdot 4)$$

$$= 0.9 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 18.9 / (82.8 \cdot 4)$$

$$= 0.0056 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 600 \cdot 36.3 + 0$$

$$= 21.76 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 21.8 / 3083$$

$$= 7.06 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.0056, 0]$$

$$= 0.0056 \text{ cm}^2$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	227.50	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	3188.00	mm
Distance from Saddle to Vessel tangent	a1 or a	600.00	mm
Saddle Width	b1 or b	285.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	365.00	mm



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Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	136.80	N./mm ²
Inside or Mean Depth of Head	Hi or hm	111.35	mm
Shell Allowable Stress used in Calculation		234.01	N./mm ²
Head Allowable Stress used in Calculation		234.01	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	600.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		251.78	Kgf
Pressure used in Analysis	P	678.230	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	8.29	234.01
Long. Stress at Bottom of Midspan	8.86	234.01
Long. Stress at Top of Saddles	10.61	234.01
Long. Stress at Bottom of Saddles	7.44	234.01
Tangential Shear in Shell	0.84	138.59
Circ. Stress at Horn of Saddle	0.63	351.01
Circ. Compressive Stress in Shell	0.03	234.01

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (72.5/2 + 0) \cdot 600/419$
 $= 155.7 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(18.9, 0) \cdot 600/1860$
 $= 6.1 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 96.1 + \max(6.1, 156, 0, 0)$
 $= 251.8 \text{ Kgf}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4 (0.62 \cdot 1.2)^2)$
 $= 371.100 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	333.77	Kgf
Transverse Shear Load	36.26	Kgf
Longitudinal Shear Load	18.92	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(440/2 + 12 + 10) \sin(120/2), 2 \cdot 227)$
 $= 419.156 \text{ mm}$



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The Computed K values from Table 4.15.1:

$K1 = 0.1066$ $K2 = 1.1707$ $K3 = 0.8799$ $K4 = 0.4011$
 $K5 = 0.7603$ $K6 = 0.0529$ $K7 = 0.0529$ $K8 = 0.3405$
 $K9 = 0.2711$ $K10 = 0.0581$ $K1^* = 0.1923$ $K6p = 0.0434$
 $K7p = 0.0434$

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm^2)}{3L} \right) \right] \\
 &= -252 \cdot 600 \left[1 - \left(1 - \frac{600}{3188} + \frac{(227^2 - 111^2)}{(2 \cdot 600 \cdot 3188)} \right) / \left(1 + \frac{(4 \cdot 111^2)}{(3 \cdot 3188)} \right) \right] \\
 &= -32.4 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L / 4 \left(1 + 2 \frac{(Rm^2 - hm^2)}{(L^2)} \right) / \left(1 + \frac{(4hm^2)}{(3L)} \right) - 4a/L \\
 &= 252 \cdot 3188 / 4 \left(1 + 2 \frac{(227^2 - 111^2)}{(3188^2)} \right) / \left(1 + \frac{(4 \cdot 111^2)}{(3 \cdot 3188)} \right) - 4 \cdot 600 / 3188 \\
 &= 42.2 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t) \\
 &= 678 \cdot 227 / (2 \cdot 9) - 42.2 / (\pi \cdot 227^2 \cdot 9) \\
 &= 8.29 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 678 \cdot 227 / (2 \cdot 9) + 42.2 / (\pi \cdot 227^2 \cdot 9) \\
 &= 8.86 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t) \\
 &= 678 \cdot 227 / (2 \cdot 9) - 32.4 / (0.11 \cdot \pi \cdot 227^2 \cdot 9) \\
 &= 10.61 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1^* \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 678 \cdot 227 / (2 \cdot 9) + 32.4 / (0.19 \cdot \pi \cdot 227^2 \cdot 9) \\
 &= 7.44 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 252 (3188 - 2 \cdot 600) / (3188 + (4 \cdot 111 / 3)) \\
 &= 150.0 \text{ Kg}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.17 \cdot 150 / (227 \cdot 9) \\
 &= 0.84 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(227 \cdot 9)} \\
 &= 35.294 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(285 + 1.56 \cdot \sqrt{(227 \cdot 9)}, 2 \cdot 600) \\
 &= 355.59 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

= 1.0000 Materials are the same, test case



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Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 252 \cdot 0.1 / (356(9 + 1 \cdot 10))$$

$$= -0.03 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -252 / (4(9 + 1 \cdot 10)356) -$$

$$3 \cdot 0.053 \cdot 252 / (2(9 + 1 \cdot 10)^2)$$

$$= -0.63 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	560.0000	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	310.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	262.0	N./
Height of Web at Center	Hw,c	390.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(223 \cdot 9) + 365})9^3) / 12$$

$$= 2.642 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 2.64 + 39.1(143 - 4.5)^2$$

$$= 7545.606 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	434.9	9.0	4.5	39.1	17612.9	0.755E+04
Wearplate	365.0	10.0	14.0	36.5	51100.0	0.611E+04
Web	10.0	346.0	192.0	34.6	664319.9	0.427E+04
BasePlate	310.0	12.0	371.0	37.2	1380119.9	0.193E+05
Totals	...	...	...	147.4	2113153.0	0.372E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2113153 / 147$$

$$= 143.323 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 252$$

$$= 51.2417 \text{ Kgf}$$



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Tension Stress, $St = (Fh/As) = 0.0464 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 * \text{Yield Str} = 157.1972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 600 - 223 \cdot \sin(120/2) / 1.05$
 $= 415.580 \text{ mm}$

Bending Moment, $M = Fh * d = 21.2955 \text{ Kg-m.}$

Bending Stress, $Sb = (M * C1 / I) = 0.0804 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 * \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{(3(252 + 82)310 / (4 * 560 * 175))}$
 $= 2.789 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120/2) (220 + 12 + 10)$
 $= 419.156 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (N_{\text{ribs}} - 1)$
 $= 419 / (3 - 1)$
 $= 209.578 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * B_{\text{pwid}} / 2$
 $= 210 * 310 / 2$
 $= 324.846 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 334 / (560 * 310)$
 $= 0.192 \text{ Kg/cm}^2$

Axial Load [P]:
 $= Ap * Bp$
 $= 325 * 0.19$
 $= 62.456 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 27.5 + 10.5$
 $= 37.979 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P/Ar$
 $= 62.5/38$
 $= 0.161 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	285.0	...	28.5	...	0.193E+04

Rib dimension [D]:



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= Saddle Width - Web Thickness

= 285 - 10

= 275.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/38

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 285/2

= 142.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{1929/38}$

= 71.270 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 600 - cos(120/2)(220 + 12 + 10) - 12

= 467.000 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199943008/262}$

= 122.739

Slenderness ratio [KL/r]:

= KL/r

= 1 • 467/71.3

= 6.553

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 18.9 / (2 • 560) • 210 • 467/2

= 0.827 Kg-m.

Compressive Allowable, KL/r < Cc (6.55 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (6.55)^2 / (2 \cdot 123^2)) 262 /$

$(5/3 + 3 \cdot (6.55) / (8 \cdot 123) - (6.55^3) / (8 \cdot 123^3))$

= 155.1 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 0.16/155 + (0.83 • 142/19290936) / 175

= 0.001

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	275.0	0.0	27.5	0.0	0.193E+04
Web	209.6	10.0	0.0	21.0	0.0	1.75
Totals	...	...	...	48.5	...	0.193E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/48.5

= 0.000 mm

Distance to Centroid [C1]:



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= Saddle Width / 2
= 285/2
= 142.500 mm

Length of Inner Rib [L]:
= Saddle Height - Outside Radius - Bpthk
= 600 - cos(242/2)(12 + 0 + 0) - 0
= 346.000 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{1931/48.5}$
= 63.122 mm

Slenderness ratio [KL/r]:
= KL/r
= 1 • 346/63.1
= 5.481

Unit Force [Force,u]:
= FI / (2 • Baseplate Length)
= 18.9/(2 • 560)
= 0.017 Kg/mm

Moment at base of inner Rib [Mbase,c]:
= Unit Force • e • L
= 0.017 • 210 • 346
= 1.225 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
= Bending Moment / Section Modulus
= 1.23/135492
= 0.089 N./mm²

Compressive Allowable, KL/r < Cc (5.48 < 123) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (5.48)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (5.48) / (8 \cdot 123) - (5.48^3) / (8 \cdot 123^3))$
= 155.5 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
= Sc/Sca + (Mbase,c • C1/I)/Sba
= 0.25/155 + (1.23 • 142/1931)/175
= 0.002

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-307 A	
Bolt Allowable Stress	Stba	138.00	N./mm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.0	mm
User Defined Area in a Single Bolt	UserBoltArea	1.5500	cm ²
BasePlate Allowable Stress	S	138.00	N./mm ²
Area Available in a Single Bolt	BltArea	0.9592	cm ²
Saddle Load Q0 (Weight)	Q0	178.1	Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	6.1	Kgf
Maximum Transverse Force	Ft	36.3	Kgf
Maximum Longitudinal Force	F1	18.9	Kgf
Saddle Bolted to Steel Foundation		No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
= FI / (Bolt Area • Number of Bolts)
= 18.9/(0.96 • 4)



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= 0.5 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= Ft / (Bolt Area • Number of Bolts)

= 36.3 / (0.96 • 4)

= 0.9 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= FI / (BoltShearAllowable • Nbolts)

= 18.9 / (82.8 • 4)

= 0.0056 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= B • Ft + Sum of X Moments

= 600 • 36.3 + 0

= 21.76 Kg-m.

Eccentricity (e):

= Rmom / QO

= 21.8 / 178

= 122.17 mm > Bplen / 6 --> Uplift in Transverse direction

f = Bplen / 2 - Edgedis

= 560 / 2 - 50

= 229.99 mm

Modular Ratio Of Steel/Concrete (n1):

= ES / EC

= 203402 / 21526

= 9.45

K1 = 3 (e - 0.5 • Bplen)

= 3 (122 - 0.5 • 560)

= -473.48 mm

K2 = 6 • n1 • At / Bpwid • (f + e)

= 6 • 9.45 • 1.92 / 310 • (230 + 122)

= 12354.91 mm²

K3 = -K2 • (0.5 • Bplen + f)

= -12355 • (0.5 • 560 + 230)

= -6300846.17 mm³

Iteratively Solving for the Effective Bearing Length:

$Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$

$Y^3 + -473 \cdot Y^2 + 12355 \cdot Y + -6300846 = 0$

Y = 475.38 mm

Num = (Bplen / 2 - Y / 3 - e)

= (560 / 2 - 475 / 3 - 122)

= -0.63

Denom = (Bplen / 2 - Y / 3 + f)

= (560 / 2 - 475 / 3 + 230)

= 351.53

Total Bolt Tension Force [Tforce]:

= - QO • Num / Denom



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$$= -178 \cdot -0.63/352$$

$$= 0.32 \text{ Kgf}$$

Bolt Area Required due to Transverse Load [Bltareart]:

$$= T_{\text{force}} / (S_{\text{tba}} \cdot N_{\text{bt}})$$

$$= 0.32 / (138 \cdot 2)$$

$$= 0.0001 \text{ cm}^2$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[B_{\text{ltarearl}}, B_{\text{ltarears}}, B_{\text{ltareart}}]$$

$$= \max[0, 0.0056, 0.00011]$$

$$= 0.0056 \text{ cm}^2$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$= 2(QO + T_{\text{force}}) / (Y \cdot B_{\text{pwid}})$$

$$= 2(178 + 0.32) / (475 \cdot 310)$$

$$= 23.74 \text{ KPa.}$$

Distance from Baseplate Edge to the Web [ADIST]:

$$= (B_{\text{plen}} - W_{\text{eblngth}}) / 2$$

$$= (560 - 509) / 2$$

$$= 25.4000 \text{ mm}$$

Overturning Moment due To Bolt Tension [Mt]:

$$= T_{\text{force}} \cdot A_{\text{dist}}$$

$$= 0.32 \cdot 25.4$$

$$= 0.01 \text{ Kg-m.}$$

Equivalent Bearing Pressure (f1):

$$= f_c \cdot (Y - A_{\text{dist}}) / Y$$

$$= 23.7 \cdot (475 - 25.4) / 475$$

$$= 22.47 \text{ KPa.}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$= (A_{\text{dist}}^2 \cdot B_{\text{pwid}} / 6) \cdot (f_1 + 2 \cdot f_c)$$

$$= (25.4^2 \cdot 310 / 6) \cdot (22.5 + 2 \cdot 23.7)$$

$$= 0.24 \text{ Kg-m.}$$

Baseplate Required Thickness [Treq]:

$$= \sqrt{(6 \cdot \max(M_t, M_c) / (B_{\text{pwid}} \cdot S_{\text{ba}}))}$$

$$= \sqrt{(6 \cdot \max(0.0081, 0.24) / (310 \cdot 207))}$$

$$= 0.4670 \text{ mm}$$

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	227.50	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	3188.00	mm
Distance from Saddle to Vessel tangent	a1 or a	620.00	mm
Saddle Width	b1 or b	285.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	365.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	15.0	mm
Wear Plate Allowable Stress	fw or Sr	136.80	N./mm ²
Shell Allowable Stress used in Calculation		136.80	N./mm ²
Head Allowable Stress used in Calculation		131.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	600.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		3609.80	Kgf
Pressure used in Analysis	P	-101.300	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-5.76	-187.92
Long. Stress at Top of Midspan		5.76	136.80
Long. Stress at Bottom of Midspan		3.20	136.80
Long. Stress at Top of Saddles		24.24	136.80
Long. Stress at Bottom of Saddles		-15.42	-187.92
Long. Stress at Bottom of Saddles		15.42	136.80
Tangential Shear in Shell		12.37	109.44
Circ. Stress at Horn of Saddle		5.91	171.00
Circ. Compressive Stress in Shell		0.32	136.80

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (145/2 + 0) \cdot 600/428$
 $= 305.1 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(37.8, 0) \cdot 600/1860$
 $= 12.2 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(195, 0) \cdot 600/1860$
 $= 63.0 \text{ Kgf}$



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Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (195/2 + 0) \cdot 600/428$$

$$= 411.1 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 3199 + \max(12.2, 305, 63, 411)$$

$$= 3609.8 \text{ Kg}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(0.62 \cdot 1.2)^2)$$

$$= 371.100 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	3700.98	Kgf
Transverse Shear Load	97.70	Kgf
Longitudinal Shear Load	195.41	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(440/2 + 12 + 15) \sin(120/2), 2 \cdot 227)$$

$$= 427.817 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm^2)/3L)]$$

$$= -3610 \cdot 620 [1 - (1 - 620/3188 + (227^2 - 0^2) / (2 \cdot 620 \cdot 3188)) / (1 + (4 \cdot 0) / (3 \cdot 3188))]$$

$$= -406.0 \text{ Kg-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 (1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L$$

$$= 3610 \cdot 3188 / 4 (1 + 2(227^2 - 0^2)/(3188^2)) / (1 + (4 \cdot 0) / (3 \cdot 3188)) - 4 \cdot 620 / 3188$$

$$= 668.2 \text{ Kg-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= -101 \cdot 227 / (2 \cdot 9) - 668 / (\pi \cdot 227^2 \cdot 9)$$

$$= -5.76 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= -101 \cdot 227 / (2 \cdot 9) + 668 / (\pi \cdot 227^2 \cdot 9)$$

$$= 3.20 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= -101 \cdot 227 / (2 \cdot 9) - 406 / (0.11 \cdot \pi \cdot 227^2 \cdot 9)$$

$$= 24.24 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:



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$$\begin{aligned}
 &= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= -101 \cdot 227 / (2 \cdot 9) + -406 / (0.19 \cdot \pi \cdot 227^2 \cdot 9) \\
 &= -15.42 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L-2a) / (L + (4 \cdot h_m / 3)) \\
 &= 3610 (3188 - 2 \cdot 620) / (3188 + (4 \cdot 0 / 3)) \\
 &= 2205.7 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (R_m \cdot t) \\
 &= 1.17 \cdot 2206 / (227 \cdot 9) \\
 &= 12.37 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(R_m \cdot t)} \\
 &= 0.78 \cdot \sqrt{(227 \cdot 9)} \\
 &= 35.294 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a) \\
 &= \min(285 + 1.56 \cdot \sqrt{(227 \cdot 9)}, 2 \cdot 620) \\
 &= 355.59 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$\begin{aligned}
 &= \min(S_r / S, 1) \\
 &= \min(137 / 137, 1) \\
 &= 1.0000
 \end{aligned}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot t_r)) \\
 &= -0.76 \cdot 3610 \cdot 0.1 / (356(9 + 1 \cdot 15)) \\
 &= -0.32 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot t_r)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot t_r)^2) \\
 &= -3610 / (4(9 + 1 \cdot 15)356) - \\
 &\quad 3 \cdot 0.053 \cdot 3610 / (2(9 + 1 \cdot 15)^2) \\
 &= -5.91 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	560.0000	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	310.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	262.0	N./
Height of Web at Center	Hw, c	390.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$\begin{aligned}
 &= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12 \\
 &= ((1.56 \cdot \sqrt{(223 \cdot 9)} + 365)^3) / 12 \\
 &= 2.642 \text{ cm}^4
 \end{aligned}$$

Resolved Inertia for the Shell [I, shell]:

$$\begin{aligned}
 &= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2 \\
 &= 2.64 + 39.1(130 - 4.5)^2 \\
 &= 6190.162 \text{ cm}^4
 \end{aligned}$$

B

D

Y

A

AY

I + AD⁽²⁾



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	mm	mm	mm	cm ²	mm ³	cm ⁴
Shell	434.9	9.0	4.5	39.1	17612.9	0.619E+04
Wearplate	365.0	15.0	16.5	54.8	90337.5	0.709E+04
Web	10.0	341.0	194.5	34.1	663244.9	0.471E+04
BasePlate	310.0	12.0	371.0	37.2	1380119.9	0.216E+05
Totals	...	...	...	165.2	2151315.2	0.396E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2151315 / 165$$

$$= 130.233 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 3610$$

$$= 734.6722 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.5716 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 600 - 223 \cdot \sin(120/2) / 1.05$$

$$= 415.580 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 305.3215 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.9856 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(3610 + 91.2) \cdot 310) / (4 \cdot 560 \cdot 175)}$$

$$= 9.288 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120 / 2) (220 + 12 + 15)$$

$$= 427.817 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 428 / (3 - 1)$$

$$= 213.908 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 214 \cdot 310 / 2$$

$$= 331.558 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$



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$$= 3701 / (560 \cdot 310)$$

$$= 2.132 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 332 \cdot 2.13$$

$$= 706.848 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 27.5 + 10.7$$

$$= 38.195 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 707 / 38.2$$

$$= 1.815 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	285.0	...	28.5	...	0.193E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 285 - 10$$

$$= 275.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 38.2$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 285 / 2$$

$$= 142.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1929 / 38.2)}$$

$$= 71.068 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 600 - \cos(120/2) (220 + 12 + 15) - 12$$

$$= 464.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943392 / 262)}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 464 / 71.1$$

$$= 6.536$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 195 / (2 \cdot 560) \cdot 214 \cdot 464 / 2$$

$$= 8.668 \text{ Kg-m.}$$



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Compressive Allowable, $KL/r < C_c$ ($6.54 < 123$) per AISC E2-1 [Sca]:

$$= \left(1 - \frac{(Kl/r)^2}{(2 \cdot C_c^2)} \right) F_y / \left(\frac{5}{3} + 3 \cdot \frac{(Kl/r)}{(8 \cdot C_c)} - \frac{(Kl/r)^3}{(8 \cdot C_c^3)} \right)$$

$$= \left(1 - \frac{(6.54)^2}{(2 \cdot 123^2)} \right) 262 / \left(\frac{5}{3} + 3 \cdot \frac{(6.54)}{(8 \cdot 123)} - \frac{(6.54^3)}{(8 \cdot 123^3)} \right)$$

$$= 155.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 1.81 / 155 + (8.67 \cdot 142 / 19290936) / 175$$

$$= 0.015$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	275.0	0.0	27.5	0.0	0.193E+04
Web	213.9	10.0	0.0	21.4	0.0	1.78
Totals	...	...	...	48.9	...	0.193E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 48.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 285 / 2$$

$$= 142.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 600 - \cos(247/2) (12 + 0 + 0) - 0$$

$$= 341.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1931 / 48.9)}$$

$$= 62.843 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 341 / 62.8$$

$$= 5.426$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 195 / (2 \cdot 560)$$

$$= 0.174 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.17 \cdot 214 \cdot 341$$

$$= 12.727 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 12.7 / 135494$$

$$= 0.921 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($5.43 < 123$) per AISC E2-1 [Sca]:

$$= \left(1 - \frac{(Kl/r)^2}{(2 \cdot C_c^2)} \right) F_y / \left(\frac{5}{3} + 3 \cdot \frac{(Kl/r)}{(8 \cdot C_c)} - \frac{(Kl/r)^3}{(8 \cdot C_c^3)} \right)$$

$$= \left(1 - \frac{(5.43)^2}{(2 \cdot 123^2)} \right) 262 / \left(\frac{5}{3} + 3 \cdot \frac{(5.43)}{(8 \cdot 123)} - \frac{(5.43^3)}{(8 \cdot 123^3)} \right)$$



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= 155.5 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 2.84/155 + (12.7 • 142/1931)/175

= 0.024

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	227.50	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	3188.00	mm
Distance from Saddle to Vessel tangent	a1 or a	600.00	mm
Saddle Width	b1 or b	285.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	365.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	136.80	N./mm ²
Inside or Mean Depth of Head	Hi or hm	111.35	mm
Shell Allowable Stress used in Calculation		136.80	N./mm ²
Head Allowable Stress used in Calculation		136.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	600.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		648.35	Kgf
Pressure used in Analysis	P	-101.300	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-2.01	-187.92
Long. Stress at Top of Midspan		2.01	136.80
Long. Stress at Bottom of Midspan		-0.55	-187.92
Long. Stress at Bottom of Midspan		0.55	136.80
Long. Stress at Top of Saddles		3.96	136.80
Long. Stress at Bottom of Saddles		-4.19	-187.92
Long. Stress at Bottom of Saddles		4.19	136.80
Tangential Shear in Shell		2.17	109.44
Circ. Stress at Horn of Saddle		1.63	171.00
Circ. Compressive Stress in Shell		0.07	136.80

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3 (145/2 + 0) • 600/419

= 311.4 Kgf

Longitudinal Saddle Reaction Force [Fwl]:

= max(Fl,wind; Sum of X Forces) • B / Ls

= max(37.8, 0) • 600/1860



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Saddle Calcs: Ext. Pressure: Step: 24 9:32am Sep 9, 2026

= 12.2 Kgf

Saddle Reaction Force due to Earthquake FI [Fsl]:

= max(FI,eq; Sum of X Forces) • B / Ls

= max(195, 0) • 600/1860

= 63.0 Kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3 (195/2 + 0) • 600/419

= 419.6 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 229 + max(12.2, 311, 63, 420)

= 648.3 Kgf

Longitudinal Wind Force [FI]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))

= 1.1 • 766 (0 + $\pi/4$ (0.62 • 1.2)²)

= 371.100 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	730.34	Kgf
Transverse Shear Load	97.70	Kgf
Longitudinal Shear Load	195.41	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)

= min(2(440/2 + 12 + 10)sin(120/2), 2•227)

= 419.156 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1- a/L + (Rm²-hm²)/(2a•L))/(1+(4hm²)/3L)]

= -648•600[1 - (1-600/3188+(227²-111²)/

(2•600•3188))/(1+(4•111)/(3•3188))]]

= -83.4 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4(1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L

= 648•3188/4(1+2(227²-111²)/(3188²))/(1+(4•111)/

(3•3188))-4•600/3188

= 108.6 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)

= -101 • 227/(2•9) - 109/(π •227²•9)

= -2.01 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= -101 • 227/(2 • 9) + 109/(π • 227² • 9)

= -0.55 N./mm²



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Saddle Calcs: Ext. Pressure:

Step: 24

9:32am

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Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -101 \cdot 227 / (2 \cdot 9) - 83.4 / (0.11 \cdot \pi \cdot 227^2 \cdot 9)$$

$$= 3.96 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -101 \cdot 227 / (2 \cdot 9) + 83.4 / (0.19 \cdot \pi \cdot 227^2 \cdot 9)$$

$$= -4.19 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 648 (3188 - 2 \cdot 600) / (3188 + (4 \cdot 111 / 3))$$

$$= 386.3 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [$\tau2$]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 386 / (227 \cdot 9)$$

$$= 2.17 \text{ N./mm}^2$$

Decay Length (4.15.20) [$x1, x2$]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(227 \cdot 9)}$$

$$= 35.294 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(285 + 1.56 \cdot \sqrt{(227 \cdot 9)}, 2 \cdot 600)$$

$$= 355.59 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(137 / 137, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma6, r$]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 648 \cdot 0.1 / (356(9 + 1 \cdot 10))$$

$$= -0.07 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [$\sigma7, r$]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -648 / (4(9 + 1 \cdot 10)356) - 3 \cdot 0.053 \cdot 648 / (2(9 + 1 \cdot 10)^2)$$

$$= -1.63 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	560.0000	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	310.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	262.0	N./
Height of Web at Center	Hw,c	390.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(223 \cdot 9)} + 365)9^3) / 12$$



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Saddle Calcs: Ext. Pressure: Step: 24 9:32am Sep 9, 2026

$$= 2.642 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 2.64 + 39.1 (143 - 4.5)^2$$

$$= 7545.606 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	434.9	9.0	4.5	39.1	17612.9	0.755E+04
Wearplate	365.0	10.0	14.0	36.5	51100.0	0.611E+04
Web	10.0	346.0	192.0	34.6	664319.9	0.427E+04
BasePlate	310.0	12.0	371.0	37.2	1380119.9	0.193E+05
Totals	...	...	...	147.4	2113153.0	0.372E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2113153/147$$

$$= 143.323 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 648$$

$$= 131.9523 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.1195 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 600 - 223 \cdot \sin(120/2) / 1.05$$

$$= 415.580 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 54.8379 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.2071 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(648 + 82)310) / (4 \cdot 560 \cdot 175)}$$

$$= 4.126 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (220 + 12 + 10)$$

$$= 419.156 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 419 / (3 - 1)$$

$$= 209.578 \text{ mm}$$



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Saddle Calcs: Ext. Pressure: Step: 24 9:32am Sep 9, 2026

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 210 \cdot 310 / 2$$

$$= 324.846 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 730 / (560 \cdot 310)$$

$$= 0.421 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 325 \cdot 0.42$$

$$= 136.664 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 27.5 + 10.5$$

$$= 37.979 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 137 / 38$$

$$= 0.353 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	285.0	...	28.5	...	0.193E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 285 - 10$$

$$= 275.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 38$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 285 / 2$$

$$= 142.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (1929 / 38) }$$

$$= 71.270 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 600 - \cos(120 / 2) (220 + 12 + 10) - 12$$

$$= 467.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199943008 / 262) }$$

$$= 122.739$$

Slenderness ratio [KL/r]:



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Saddle Calcs: Ext. Pressure: Step: 24 9:32am Sep 9,2026

$$= KL/r$$

$$= 1 \cdot 467/71.3$$

$$= 6.553$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 195 / (2 \cdot 560) \cdot 210 \cdot 467/2$$

$$= 8.538 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($6.55 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (6.55)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (6.55) / (8 \cdot 123) - (6.55^3) / (8 \cdot 123^3))$$

$$= 155.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 0.35/155 + (8.54 \cdot 142/19290936) / 175$$

$$= 0.006$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	275.0	0.0	27.5	0.0	0.193E+04
Web	209.6	10.0	0.0	21.0	0.0	1.75
Totals	...	...	...	48.5	...	0.193E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/48.5$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 285/2$$

$$= 142.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 600 - \cos(242/2) (12 + 0 + 0) - 0$$

$$= 346.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ (1931/48.5) }$$

$$= 63.122 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 346/63.1$$

$$= 5.481$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 195 / (2 \cdot 560)$$

$$= 0.174 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.17 \cdot 210 \cdot 346$$

$$= 12.652 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:



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Saddle Calcs: Ext. Pressure: Step: 24 9:32am Sep 9, 2026

= Bending Moment / Section Modulus

= 12.7/135492

= 0.916 N./mm²

Compressive Allowable, $KL/r < Cc$ (5.48 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (5.48)^2 / (2 \cdot 123^2)) 262 /$

$(5/3 + 3 \cdot (5.48) / (8 \cdot 123) - (5.48^3) / (8 \cdot 123^3))$

= 155.5 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$

= $0.55/155 + (12.7 \cdot 142/1931) / 175$

= 0.009

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: N4-Tube inlet

Nozl: 13

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Input, Nozzle Desc: N4-Tube inlet

From: 20

Pressure for Reinforcement Calculations	P	6400.000	KPa.
Temperature for Internal Pressure	Temp	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	440.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		350.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	37.3087	mm
Height of Beveled Transition	L'	27.3087	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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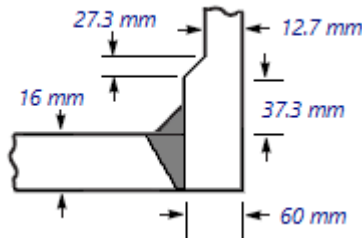
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Nozzle Calcs.: N4-Tube inlet

Nozl: 13

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Reinforcement CALCULATION, Description: N4-Tube inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	7.625 in.
Actual Thickness Used in Calculation	0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6400 \cdot 223 / (138 \cdot 1 - 0.6 \cdot 6400)$
 $= 10.6389 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6400 \cdot 99.8 / (138 \cdot 1 - 0.6 \cdot 6400)$
 $= 4.7630 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 47.3 / 0.5773$
 $= 81.9260 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	399.3500	mm
Parallel to Vessel Wall, opening length	d	199.6750	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	32.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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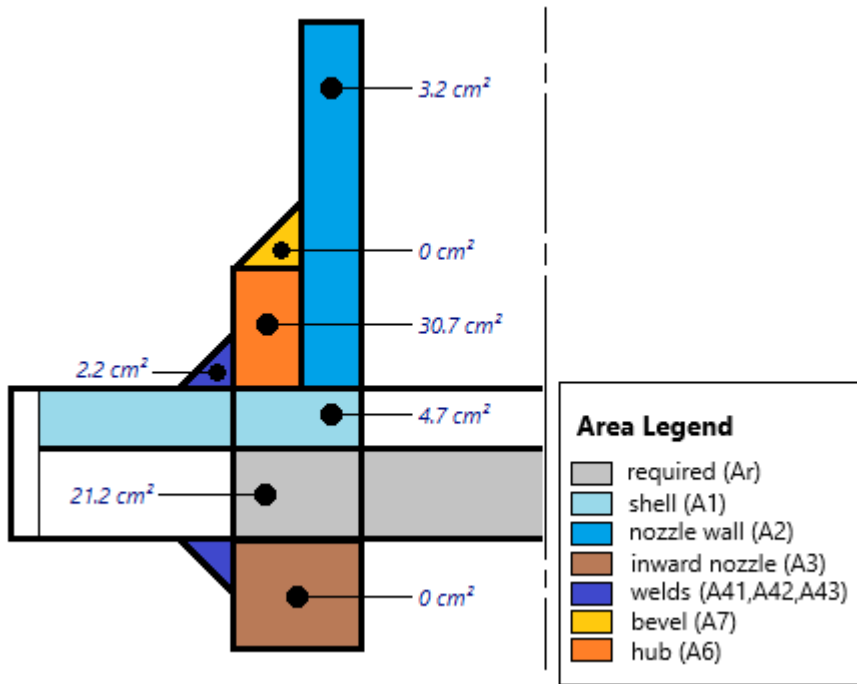
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Nozzle Calcs.: N4-Tube inlet

Noz1: 13

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	21.243	NA	NA
Area in Shell	A1	4.715	NA	NA
Area in Nozzle Wall	A2	3.209	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.250	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	30.745	NA	NA
TOTAL AREA AVAILABLE	Atot	40.919	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (200 \cdot 10.6 \cdot 1 + 2 \cdot 57 \cdot 10.6 \cdot 1 \cdot (1 - 1))$$

$$= 21.243 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 200(1 \cdot 13 - 1 \cdot 10.6) - 2 \cdot 57$$

$$(1 \cdot 13 - 1 \cdot 10.6) \cdot (1 - 1)$$

$$= 4.715 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 32.5)(9.7 - 4.76)1$$

$$= 3.209 \text{ cm}^2$$



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Nozzle Calcs.: N4-Tube inlet

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 15^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 2.250 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(32.5, 200, 37.3)) \cdot (60 - 12.7) \cdot 1$$

$$= 30.745 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 7.7630 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.6389 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.6389 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 10.1600 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[10.2, \max(13.6, 5.5)]$$

$$= 10.1600 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(7.76, 10.2)$$

$$= 10.1600 \text{ mm}$$

Minimum wall thk. of nozzle necks including pressure, bending and shear loads [tBndUG-22]:

$$= \max(tUG-45, trB31.3)$$

$$= \max(10.2, 6.9)$$

$$= 10.1600 \text{ mm}$$

The value, trB31.3 was determined by iteration. This value is the required nozzle thickness considering sustained, occasional, expansion, pressure, bending and shear loads. The rules from ASME B31.3 have been applied.

Available Nozzle Neck Thickness = $0.875 \cdot 12.700 = 11.113 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 53.8,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 291.2 N./mm ²	Passed
Occasional	: 31.4,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 17.8,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 11.1, tr = 4.76, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.59$, Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-28 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 16, tr = 10.6, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.82$, Temp. Reduction = 10 °C



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Nozzle Calcs.: N4-Tube inlet Nozl: 13 9:32am Sep 9, 2026

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -24 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -39 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6400.00/10210.00 = 0.627$

Weld Size Calculations, Description: N4-Tube inlet

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 10.6050 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (21.2 - 4.71 + 2 \cdot 57 \cdot 1 \cdot (1 \cdot 13 - 10.6)) \cdot 138)$
 $= 27045.96 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot f_r2) \cdot S_v$
 $= (34 + 2.25 - 0 \cdot 1) \cdot 138$
 $= 50944.74 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (34 + 0 + 2.25 + (14.8)) \cdot 138$
 $= 71798.80 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (34 + 0 + 2.25 + 0 + (14.8)) \cdot 138$
 $= 71798.80 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 314 \cdot 15 \cdot 0.49 \cdot 138$
 $= 50960. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 128) \cdot (60 - 3) \cdot 0.7 \cdot 138$
 $= 226370. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 314 \cdot (16 - 3) \cdot 0.74 \cdot 138$



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Nozzle Calcs.: N4-Tube inlet

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= 66699. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= (SONW + SNW) = (50960 + 226370) = 277330 Kgf

Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (50960 + 0 + 66699 + 0) = 117659 Kgf

Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (50960 + 66699 + 0) = 117659 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 277330 Kgf, must exceed W = 27045 Kgf or W1 = 50944 Kgf

Path 2-2 = 117658 Kgf, must exceed W = 27045 Kgf or W2 = 71798 Kgf

Path 3-3 = 117658 Kgf, must exceed W = 27045 Kgf or W3 = 71798 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 7768 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 65.7210 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 281.7210 mm

Input Echo, WRC297 Item 1, Description: N4-Tube inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	440.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	193.675	mm
Nozzle Thickness	60.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	6400.00	KPa.
Radial Load	P	-1529.68	Kgf
Circumferential Shear	Vc	1019.78	Kgf
Longitudinal Shear	Vl	1019.78	Kgf



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Nozzle Calcs.: N4-Tube inlet Nozl: 13 9:32am Sep 9, 2026

Circumferential Moment	Mc	570.86	Kg-m.
Longitudinal Moment	Ml	570.86	Kg-m.
Torsional Moment	Mt	815.51	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Dv (.665) is > 0.5
check the limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (5.228) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 313.675 + 2 \cdot 1.65 \cdot \sqrt{229.5(16.0 - 3.0)}$$

$$= 493.926 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	459.000 mm
Nozzle Outside Diameter	(d) =	313.675 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	57.000 mm
T / t	=	0.228
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.061$$

Nr/P	=	0.049
Mr/P	=	0.080
M0/P	=	0.026
N0/P	=	0.055
MrD/Mc	=	0.255
NrDL/Mc	=	0.089
M0d/Mc	=	0.076
N0DL/Mc	=	0.084
MrD/Ml	=	0.068
NrDL/Ml	=	0.047
M0D/Ml	=	0.021
N0DL/Ml	=	0.058

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-6	-6	6	6
Outplane Bending (Ml)	-13	13	13	-13
Normal Pressure Stress	106	113	106	113
Outplane Stress Summary	106	111	144	97

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4



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Nozzle Calcs.: N4-Tube inlet

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Inplane Bending (P)	42	-42	42	-42
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-43	43	43	-43
Inplane Pressure Stress	53	53	53	53
Inplane Stress Summary	52	53	148	-23

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)		1	1	-1	-1
Outplane Shear (Vl)		0	0	0	0
Outplane Shear (Mt)		3	3	3	3
Shear Stress Summary		5	5	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress		106	111	149	120

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)		4	4	4	4
Outplane Bending (P)		13	-13	13	-13
Outplane Membrane (Mc)		-8	-8	8	8
Outplane Bending (Mc)		-48	48	48	-48
Outplane Membrane (ML)		0	0	0	0
Outplane Bending (ML)		0	0	0	0
Normal Pressure Stress		53	53	53	53
Outplane Stress Summary		14	83	129	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)		4	4	4	4
Inplane Bending (P)		42	-42	42	-42
Inplane Membrane (Mc)		-9	-9	9	9
Inplane Bending (Mc)		-161	161	161	-161
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		106	113	106	113
Inplane Stress Summary		-16	226	324	-77

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-1	-1	1	1
Torsional Shear (Mt)		3	3	3	3



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Nozzle Calcs.: N4-Tube inlet

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Shear Stress Summary	2	2	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	31	226	324	83

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-6	-6	6	6
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	18	10	18
Hoop Stress Summary	9	16	21	29

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	3	3	10	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	9	17	21	29

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-8	-8	8	8
Hoop Bending (Mc)	0	0	0	0



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Nozzle Calcs.: N4-Tube inlet Nozl: 13 9:32am Sep 9,2026

Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	18	10	18

Hoop Stress Summary	6	14	24	31
---------------------	---	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-6	6	6	-6
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	-2	8	16	0
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	1	1
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	8	14	24	32
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	106	113	106	113	106	113	106	113
Circ. Pl (SUS)	-2	-2	10	10	-5	-5	13	13
Circ. Q (SUS)	0	0	26	-26	-119	119	203	-203
Long. Pm (SUS)	53	53	53	53	53	53	53	53
Long. Pl (SUS)	0	0	8	8	-4	-4	12	12
Long. Q (SUS)	-1	1	85	-85	-35	35	61	-61
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	106.0	113.0	106.0	113.0	106.0	113.0	106.0	113.0
Pm+Pl (SUS)	104.0	111.0	116.0	123.0	101.0	108.0	119.0	126.0
Pm+Pl+Q (Total)	104.3	111.3	146.8	121.1	32.2	227.0	322.1	81.4

Vessel Stress Summation Comparison (N./mm²):



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Nozzle Calcs.: N4-Tube inlet

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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	113.00	138.00	Passed
Pm+Pl (SUS)	126.02	207.00	Passed
Pm+Pl+Q (TOTAL)	322.08	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10	18	10	18	10	18	10	18
Circ. Pl (SUS)	-2	-2	10	10	-4	-4	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	0	0	2	-2	-5	5	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	10.0	18.0	10.0	18.0	10.0	18.0	10.0	18.0
Pm+Pl (SUS)	8.0	16.0	20.0	28.0	6.0	14.0	22.0	30.0
Pm+Pl+Q (Total)	8.0	16.0	20.0	28.0	8.0	14.0	22.0	30.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	18.00	138.00	Passed
Pm+Pl (SUS)	30.00	207.00	Passed
Pm+Pl+Q (TOTAL)	30.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: Tube outlet

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Input, Nozzle Desc: Tube outlet

From: 20

Pressure for Reinforcement Calculations	P	6403.920	KPa.
Temperature for Internal Pressure	Temp	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	440.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		350.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	15.9854	mm
Height of Beveled Transition	L'	10.9854	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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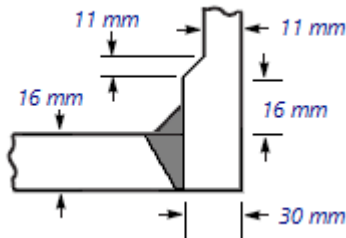
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Nozzle Calcs.: Tube outlet

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Reinforcement CALCULATION, Description: Tube outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.761 in.
Actual Thickness Used in Calculation	0.432 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6404 \cdot 223 / (138 \cdot 1 - 0.6 \cdot 6404)$
 $= 10.6456 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6404 \cdot 76.2 / (138 \cdot 1 - 0.6 \cdot 6404)$
 $= 3.6360 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 19 / 0.5773$
 $= 32.9561 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	304.6588	mm
Parallel to Vessel Wall, opening length	d	152.3294	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	32.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



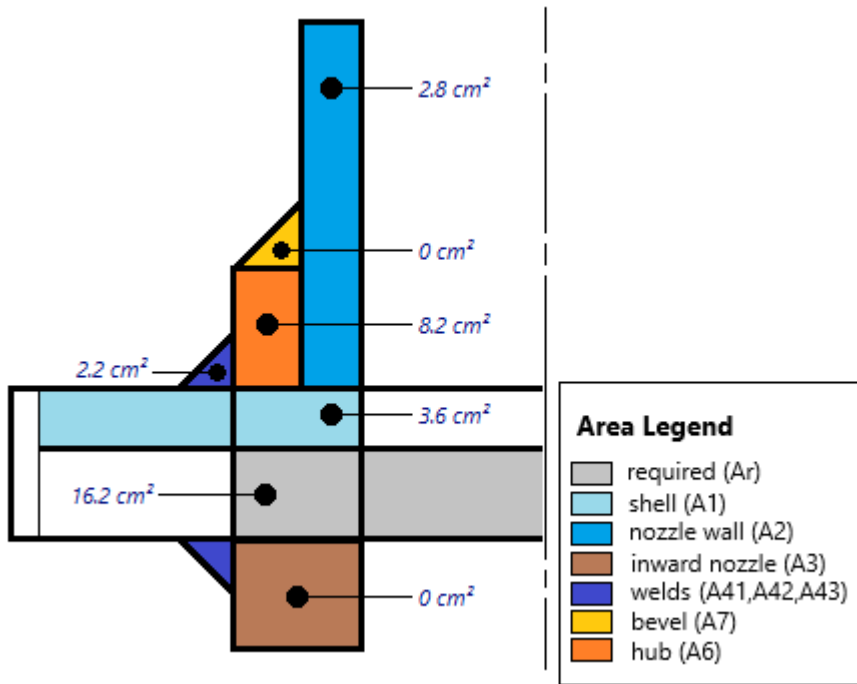
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Nozzle Calcs.: Tube outlet

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	16.216	NA	NA
Area in Shell	A1	3.586	NA	NA
Area in Nozzle Wall	A2	2.819	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.250	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	8.173	NA	NA
TOTAL AREA AVAILABLE	Atot	16.829	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (152 \cdot 10.6 \cdot 1 + 2 \cdot 27 \cdot 10.6 \cdot 1 \cdot (1 - 1))$$

$$= 16.216 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 152(1 \cdot 13 - 1 \cdot 10.6) - 2 \cdot 27(1 \cdot 13 - 1 \cdot 10.6) \cdot (1 - 1)$$

$$= 3.586 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 32.5)(7.97 - 3.64)1$$

$$= 2.819 \text{ cm}^2$$



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Nozzle Calcs.: Tube outlet

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - c_{an}/0.707)^2 \cdot fr_2$$

$$= 15^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 2.250 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(32.5, 200, 16)) \cdot (30 - 11) \cdot 1$$

$$= 8.173 \text{ cm}^2 \text{ Includes Bevel Area of } 2.090 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 6.6360 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.6456 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.6456 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 9.2200 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.22, \max(13.6, 5.5)]$$

$$= 9.2200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(6.64, 9.22)$$

$$= 9.2200 \text{ mm}$$

Minimum wall thk. of nozzle necks including pressure, bending and shear loads [tBndUG-22]:

$$= \max(tUG-45, trB31.3)$$

$$= \max(9.22, 6.3)$$

$$= 9.2200 \text{ mm}$$

The value, trB31.3 was determined by iteration. This value is the required nozzle thickness considering sustained, occasional, expansion, pressure, bending and shear loads. The rules from ASME B31.3 have been applied.

Available Nozzle Neck Thickness = $0.875 \cdot 10.973 = 9.601 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 55.2,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 289.7 N./mm ²	Passed
Occasional	: 29.1,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 21.0,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.6, tr = 3.64, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.55$, Temp. Reduction = 27 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-34 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 16, tr = 10.6, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.82$, Temp. Reduction = 10 °C



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Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -24 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6403.92/10210.00 = 0.627$

Weld Size Calculations, Description: Tube outlet

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	10.6050 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (16.2 - 3.59 + 2 \cdot 27 \cdot 1 \cdot (1 \cdot 13 - 10.6)) \cdot 138)$
 $= 19561.26 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (11 + 2.25 - 0 \cdot 1) \cdot 138$
 $= 18634.01 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (11 + 0 + 2.25 + (7.02)) \cdot 138$
 $= 28512.26 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (11 + 0 + 2.25 + 0 + (7.02)) \cdot 138$
 $= 28512.26 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 206 \cdot 15 \cdot 0.49 \cdot 138$
 $= 33521. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 89.7) \cdot (30 - 3) \cdot 0.7 \cdot 138$
 $= 74916. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 206 \cdot (16 - 3) \cdot 0.74 \cdot 138$



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= 43873. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= (SONW + SNW) = (33521 + 74916) = 108437 Kgf

Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (33521 + 0 + 43873 + 0) = 77394 Kgf

Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (33521 + 43873 + 0) = 77394 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 108436 Kgf, must exceed W = 19561 Kgf or W1 = 18634 Kgf

Path 2-2 = 77393 Kgf, must exceed W = 19561 Kgf or W2 = 28512 Kgf

Path 3-3 = 77393 Kgf, must exceed W = 19561 Kgf or W3 = 28512 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6510 KPa.

The Drop for this Nozzle is : 25.6883 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 241.6883 mm

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 62.27 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 152.33 mm

Input Echo, WRC297 Item 1, Description: Tube outlet :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	440.000	mm	
Shell Thickness	16.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	146.329	mm	
Nozzle Thickness	30.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-266 2		
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	6403.92	KPa.
Radial Load	P	-1147.26	Kgf



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Circumferential Shear	Vc	764.84	Kgf
Longitudinal Shear	Vl	764.84	Kgf
Circumferential Moment	Mc	321.11	Kg-m.
Longitudinal Moment	Ml	321.11	Kg-m.
Torsional Moment	Mt	458.73	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (6.878) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 206.329 + 2 \cdot 1.65 \cdot \sqrt{229.5(16.0 - 3.0)}$$

$$= 386.580 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	459.000 mm
Nozzle Outside Diameter	(d) =	206.329 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	27.000 mm
T / t	=	0.481
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.671$$

Nr/P	=	0.062
Mr/P	=	0.114
M0/P	=	0.036
N0/P	=	0.087
MrD/Mc	=	0.270
NrDL/Mc	=	0.101
M0d/Mc	=	0.082
N0DL/Mc	=	0.082
MrD/Ml	=	0.099
NrDL/Ml	=	0.053
M0D/Ml	=	0.028
N0DL/Ml	=	0.101

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-9	-9	9	9
Outplane Bending (Ml)	-15	15	15	-15
Normal Pressure Stress	106	113	106	113
Outplane Stress Summary	102	110	151	98

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4



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Inplane Bending (P)	45	-45	45	-45
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-53	53	53	-53
Inplane Pressure Stress	53	53	53	53
Inplane Stress Summary	44	60	161	-36

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1	
Outplane Shear (Vl)	0	0	0	0	
Outplane Shear (Mt)	5	5	5	5	
Shear Stress Summary	6	6	3	3	

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	103	111	162	135	

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	5	5	5	5	
Outplane Bending (P)	14	-14	14	-14	
Outplane Membrane (Mc)	-7	-7	7	7	
Outplane Bending (Mc)	-44	44	44	-44	
Outplane Membrane (ML)	0	0	0	0	
Outplane Bending (ML)	0	0	0	0	
Normal Pressure Stress	53	53	53	53	
Outplane Stress Summary	21	81	125	7	

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	4	4	4	4	
Inplane Bending (P)	45	-45	45	-45	
Inplane Membrane (Mc)	-9	-9	9	9	
Inplane Bending (Mc)	-146	146	146	-146	
Inplane Membrane (ML)	0	0	0	0	
Inplane Bending (ML)	0	0	0	0	
Inplane Pressure Stress	106	113	106	113	
Inplane Stress Summary	1	208	311	-65	

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0	
Outplane Shear (Vl)	-1	-1	1	1	
Torsional Shear (Mt)	5	5	5	5	



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Shear Stress Summary	3	3	6	6
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	22	209	311	74

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	18	26	18	26
Hoop Stress Summary	15	22	33	40

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-5	-5	5	5
Axial Bending (ML)	-9	9	9	-9
Axial Pressure Stress	9	9	9	9
Axial Stress Summary	3	6	31	-1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	3	3	1	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	15	23	34	42

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-7	-7	7	7
Hoop Bending (Mc)	0	0	0	0



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Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	18	26	18	26

Hoop Stress Summary	16	24	31	39
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	-5	-5	5	5
Axial Bending (Mc)	-27	27	27	-27
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	9	9	9	9

Axial Stress Summary	-15	24	50	-20
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	1	1	3	3
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	32	26	50	59
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	106	113	106	113	106	113	106	113
Circ. Pl (SUS)	-4	-4	14	14	-5	-5	13	13
Circ. Q (SUS)	-1	1	29	-29	-101	101	191	-191
Long. Pm (SUS)	53	53	53	53	53	53	53	53
Long. Pl (SUS)	0	0	8	8	-2	-2	12	12
Long. Q (SUS)	-8	8	98	-98	-30	30	58	-58
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	106.0	113.0	106.0	113.0	106.0	113.0	106.0	113.0
Pm+Pl (SUS)	102.0	109.0	120.0	127.0	101.0	108.0	119.0	126.0
Pm+Pl+Q (Total)	101.6	110.7	160.4	135.2	22.5	209.1	310.2	73.0

Vessel Stress Summation Comparison (N./mm²):



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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	113.00	138.00	Passed
Pm+Pl (SUS)	127.02	207.00	Passed
Pm+Pl+Q (TOTAL)	310.19	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	18	26	18	26	18	26	18	26
Circ. Pl (SUS)	-4	-4	14	14	-2	-2	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-5	-5	5	5	-5	-5	5	5
Long. Q (SUS)	-2	2	16	-16	-20	20	34	-34
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	18.0	26.0	18.0	26.0	18.0	26.0	18.0	26.0
Pm+Pl (SUS)	14.0	22.0	32.0	40.0	16.0	24.0	30.0	38.0
Pm+Pl+Q (Total)	14.3	22.2	33.2	42.2	32.2	26.0	48.2	58.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	26.00	138.00	Passed
Pm+Pl (SUS)	40.00	207.00	Passed
Pm+Pl+Q (TOTAL)	58.14	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: Shell outlet

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Input, Nozzle Desc: Shell outlet

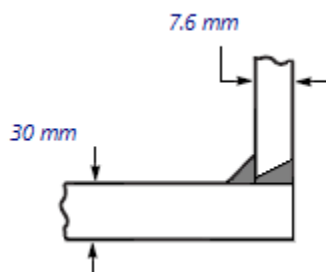
From: 50

Pressure for Reinforcement Calculations	P	523.921	KPa.
Temperature for Internal Pressure	Temp	280	°C
Design External Pressure	Pext	101.30	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	136.80	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	440.00	mm
Design Length of Section	L	3085.4167	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3889.85	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	7.6200	mm
Flange Class		600	
Flange Grade		GR 2.1	

The Pressure Design option was Design Pressure + static head.





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Nozzle Calcs.: Shell outlet

Nozl: 15

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Reinforcement CALCULATION, Description: Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.900 in.
Actual Thickness Used in Calculation 0.300 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $524 \cdot 223 / (137 \cdot 1 - 0.6 \cdot 524)$

= 0.8561 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $524 \cdot 39.8 / (118 \cdot 1 - 0.6 \cdot 524)$

= 0.1773 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3979 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 159.3200 mm

Parallel to Vessel Wall, opening length d 79.6600 mm

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 11.5500 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: Shell outlet.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 3.3979 mm

Wall Thickness per UG16(b), tr16b = 5.5000 mm

Wall Thickness, shell/head, internal pressure trb1 = 3.8561 mm

Wall Thickness tb1 = max(trb1, tr16b) = 5.5000 mm

Wall Thickness tb2 = max(trb2, tr16b) = 5.5000 mm

Wall Thickness per table UG-45 tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]

= min[7.8, max(5.5, 5.5)]

= 5.5000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)

= max(3.4, 5.5)

= 5.5000 mm

Minimum wall thk. of nozzle necks including pressure, bending and shear loads [tBndUG-22]:

= max(tUG-45, trB31.3)

= max(5.5, 5.1)

= 5.5000 mm

The value, trB31.3 was determined by iteration. This value is the required nozzle thickness considering sustained, occasional, expansion, pressure, bending and shear loads. The rules from ASME B31.3 have been applied.

Available Nozzle Neck Thickness = $0.875 \cdot 7.620 = 6.668$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):



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Nozzle Calcs.: Shell outlet Nozl: 15 9:32am Sep 9,2026

Sustained :	48.9,	Allowable :	118.0 N./mm ²	Passed
Expansion :	0.0,	Allowable :	246.1 N./mm ²	Passed
Occasional :	2.1,	Allowable :	156.9 N./mm ²	Passed
Shear :	37.6,	Allowable :	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 6.67$, $t_r = 0.18$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.048$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Note:

The flange pressure ratio is less than 0.35. Impact testing is not required due to low stress per UHA-51(g).

Design Pressure/Ambient Rating = $523.92/9930.00 = 0.053$

Weld Size Calculations, Description: Shell outlet

Intermediate Calc. for nozzle/shell Welds T_{min} 4.6200 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.2340 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 4.5372 mm

The Cut Length for this Nozzle is, Drop + H_o + H + T : 234.5373 mm

Input Echo, WRC297 Item 1, Description: Shell outlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	440.000	mm
Shell Thickness	30.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	S_{mc}	138.00 N./mm ²
Vessel Hot S.I. Allowable	S_{mh}	136.80 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	73.660	mm
Nozzle Thickness	7.6200	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	S_{Nmc}	118.00 N./mm ²
Nozzle Hot S.I. Allowable	S_{Nmh}	118.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.



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Nozzle Calcs.: Shell outlet

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Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	523.92	KPa.
Radial Load	P	-612.00	Kgf
Circumferential Shear	Vc	408.00	Kgf
Longitudinal Shear	Vl	408.00	Kgf
Circumferential Moment	Mc	91.34	Kg-m.
Longitudinal Moment	Ml	91.34	Kg-m.
Torsional Moment	Mt	130.48	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_v/T_v (16.667) is < 20
check the Limitations of WRC bulletin 297.

Warning - The ratio, D_n/T_v , (2.963) is < 5
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 88.9 + 2 \cdot 1.65 \cdot \sqrt{236.5(30.0 - 3.0)}$$

$$= 352.601 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	473.000 mm
Nozzle Outside Diameter	(d) =	88.900 mm
Vessel Thickness used	(T) =	27.000 mm
Nozzle Thickness used	(t) =	4.620 mm
T / t	=	5.844
d / t	=	19.242

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.787$$

Nr/P	=	0.095
Mr/P	=	0.061
M0/P	=	0.185
N0/P	=	0.324
MrD/Mc	=	0.098
NrDL/Mc	=	0.114
M0d/Mc	=	0.297
N0DL/Mc	=	0.124
MrD/Ml	=	0.096
NrDL/Ml	=	0.109
M0D/Ml	=	0.272
N0DL/Ml	=	0.269

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-3	-3	3	3
Outplane Bending (Ml)	-22	22	22	-22



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Normal Pressure Stress	4	4	4	4
------------------------	---	---	---	---

Outplane Stress Summary	-10	16	42	-20
-------------------------	-----	----	----	-----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-7	7	7	-7
Inplane Pressure Stress	2	2	2	2

Inplane Stress Summary	-3	6	15	-6
------------------------	----	---	----	----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	2	2
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress	12	18	42	21
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-24	24	24	-24
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	2	2	2	2

Outplane Stress Summary	-12	18	40	-27
-------------------------	-----	----	----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-8	8	8	-8
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	4	4	4	4



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Inplane Stress Summary	-1	8	17	-4
------------------------	----	---	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	4	4
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	13	19	41	28

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	5	5	5

Hoop Stress Summary	4	4	11	12
---------------------	---	---	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	23	-23	23	-23
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-36	-36	36	36
Axial Bending (ML)	-116	116	116	-116
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-122	63	183	-95
----------------------	------	----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	6	6	-6	-6
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	22	22	22	22

Shear Stress Summary	28	28	16	16
----------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	138	82	185	112



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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	5	5	5
Hoop Stress Summary	6	6	9	10

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	23	-23	23	-23
Axial Membrane (Mc)	-36	-36	36	36
Axial Bending (Mc)	-115	115	115	-115
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-121	63	182	-94

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-6	-6	6	6
Shear due to Torsion	22	22	22	22
Shear Stress Summary	16	16	28	28

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	131	67	187	119

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	4	4	4	4	4	4	4
Circ. Pl (SUS)	-1	-1	5	5	-1	-1	1	1
Circ. Q (SUS)	-13	13	31	-31	-5	5	11	-11
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-1	-1	1	1	1	1	3	3
Long. Q (SUS)	-4	4	10	-10	-15	15	33	-33
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1



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Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
Pm+Pl (SUS)	3.4	3.4	9.2	9.2	4.0	4.0	6.0	6.0
Pm+Pl+Q (Total)	11.8	17.3	40.1	22.3	12.4	18.4	38.7	28.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	4.00	136.80	Passed
Pm+Pl (SUS)	9.16	205.20	Passed
Pm+Pl+Q (TOTAL)	40.15	410.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	5	5	5	5	5	5	5
Circ. Pl (SUS)	-1	-1	5	5	1	1	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-32	-32	40	40	-32	-32	40	40
Long. Q (SUS)	-93	93	139	-139	-92	92	138	-138
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	6	6	-6	-6	-6	-6	6	6
Shear Q (SUS)	22	22	22	22	22	22	22	22
Pm (SUS)	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
Pm+Pl (SUS)	36.1	36.1	43.1	43.1	37.9	37.9	43.0	43.0
Pm+Pl+Q (Total)	138.8	81.3	182.5	111.7	131.9	66.2	184.4	118.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	118.00	Passed
Pm+Pl (SUS)	43.09	177.00	Passed
Pm+Pl+Q (TOTAL)	184.44	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: Shell Inlet

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Input, Nozzle Desc: Shell Inlet

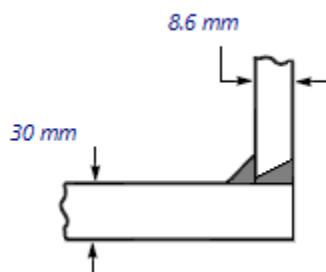
From: 50

Pressure for Reinforcement Calculations	P	520.000	KPa.
Temperature for Internal Pressure	Temp	280	°C
Design External Pressure	Pext	101.30	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	136.80	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	440.00	mm
Design Length of Section	L	3085.4167	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1339.85	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.5598	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





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Nozzle Calcs.: Shell Inlet

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Reinforcement CALCULATION, Description: Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.826 in.
Actual Thickness Used in Calculation	0.337 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $520 \cdot 223 / (137 \cdot 1 - 0.6 \cdot 520)$

= 0.8497 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $520 \cdot 51.6 / (118 \cdot 1 - 0.6 \cdot 520)$

= 0.2280 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4111 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	206.3608	mm
--	----	----------	----

Parallel to Vessel Wall, opening length	d	103.1804	mm
---	---	----------	----

Normal to Vessel Wall (Thickness Limit), no pad	T1np	13.8995	mm
---	------	---------	----

Weld Strength Reduction Factor [fr1]:

= 1.000

Weld Strength Reduction Factor [fr2]:

= min(1, S_n / S_v)

= min(1, 118/137)

= 0.863

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(0.86, 1)

= 0.863



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Nozzle Calcs.: Shell Inlet

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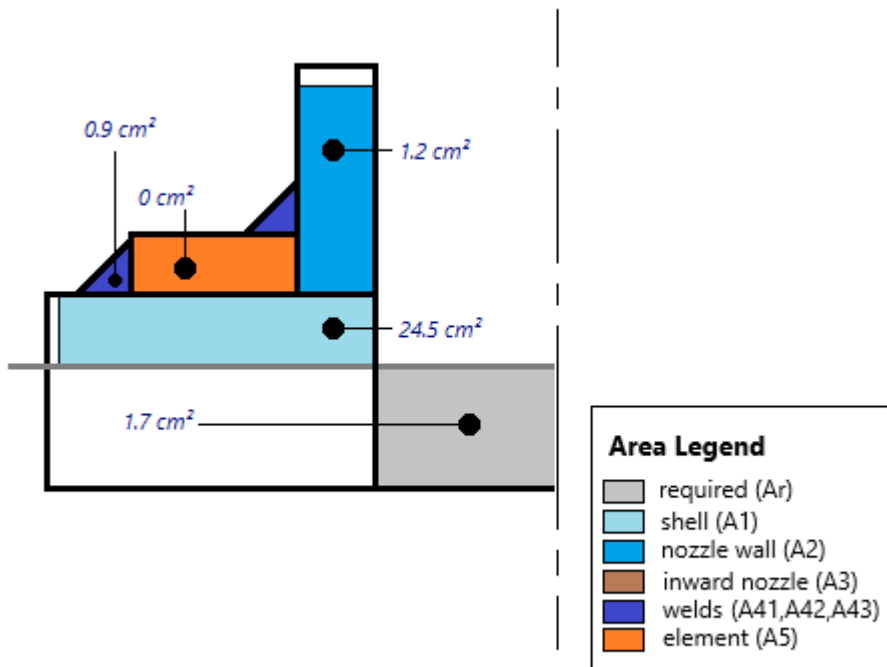


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	0.877	1.658	NA
Area in Shell	A1	26.982	24.543	NA
Area in Nozzle Wall	A2	1.278	1.235	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.863	0.863	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	29.123	26.640	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1)) \text{ per UG-37(d)}$$

$$= 0.5(103 \cdot 3.21 \cdot 1 + 2 \cdot 5.56 \cdot 3.21 \cdot 1(1 - 1))$$

$$= 1.658 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 103(1 \cdot 27 - 1 \cdot 3.21) - 2 \cdot 5.56$$

$$(1 \cdot 27 - 1 \cdot 3.21) \cdot (1 - 1)$$

$$= 24.543 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 13.9)(5.56 - 0.41)0.86$$

$$= 1.235 \text{ cm}^2$$



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Nozzle Calcs.: Shell Inlet

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$$

$$= 0.863 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.4111 \text{ mm}$

Wall Thickness per UG16(b), $tr_{16b} = 5.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $tr_{b1} = 3.8497 \text{ mm}$

Wall Thickness $tb_1 = \max(tr_{b1}, tr_{16b}) = 5.5000 \text{ mm}$

Wall Thickness $tb_2 = \max(tr_{b2}, tr_{16b}) = 5.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb_3 = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[8.26, \max(5.5, 5.5)]$$

$$= 5.5000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.41, 5.5)$$

$$= 5.5000 \text{ mm}$$

Minimum wall thk. of nozzle necks including pressure, bending and shear loads [tBndUG-22]:

$$= \max(tUG-45, tr_{B31.3})$$

$$= \max(5.5, 4.98)$$

$$= 5.5000 \text{ mm}$$

The value, trB31.3 was determined by iteration. This value is the required nozzle thickness considering sustained, occasional, expansion, pressure, bending and shear loads. The rules from ASME B31.3 have been applied.

Available Nozzle Neck Thickness = $0.875 \cdot 8.560 = 7.490 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	38.6,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	256.4 N./mm ²	Passed
Occasional	:	2.3,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	29.4,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 7.49$, $t_r = 0.23$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.051$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A -8°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C

Min Metal Temp. w/o impact per UG-20(f) -29°C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 7.49$, $t_r = 0.23$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.051$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve B -29°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C



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Nozzle Calcs.: Shell Inlet

Nozl: 16

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Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 520.00/5110.00 = 0.102

Weld Size Calculations, Description: Shell Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 5.5598 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	3.8919 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 = (Ar - A1) • Sv
 = (1.66 - 24.5) • 137
 = 0.00 Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 = (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
 = (1.23 + 0 + 0.86 - 0 • 0.86) • 137
 = 2925.36 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 = (π/2) • Dlo • Wo • 0.49 • Snw
 = (3.14/2.0) • 114 • 10 • 0.49 • 118
 = 10585. Kgf

Shear, Nozzle Groove Weld [Sngw]:
 = (π/2) • (Dlr+Wgnva) • (Wgnva-Can) • 0.60 • Sng
 = (3.14/2) • (103 + 8.56) • (8.56 - 3) • 0.6 • 137
 = 8168. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 = (SONW + SNGW) = (10585 + 8168) = 18753 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 18752 Kgf, must exceed W = 0 Kgf or W1 = 2925 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5386 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 7.5527 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 187.5527 mm



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Input Echo, WRC297 Item 1, Description: Shell Inlet :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	440.000	mm	
Shell Thickness	30.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	136.80	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	97.180	mm	
Nozzle Thickness	8.5598	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-106 B		
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	520.00	KPa.
Radial Load	P	-765.00	Kgf
Circumferential Shear	Vc	510.00	Kgf
Longitudinal Shear	Vl	510.00	Kgf
Circumferential Moment	Mc	143.00	Kg-m.
Longitudinal Moment	Ml	143.00	Kg-m.
Torsional Moment	Mt	204.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_v/T_v (16.667) is < 20
check the limitations of WRC bulletin 297.

Warning - The ratio, D_n/T_v , (3.810) is < 5
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{236.5(30.0 - 3.0)}$$

$$= 378.001 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	473.000 mm
Nozzle Outside Diameter	(d) =	114.300 mm
Vessel Thickness used	(T) =	27.000 mm
Nozzle Thickness used	(t) =	5.560 mm
T / t	=	4.856
d / t	=	20.558



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Nozzle Calcs.: Shell Inlet

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$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.011$$

Nr/P	=	0.087
Mr/P	=	0.062
M0/P	=	0.141
N0/P	=	0.304
MrD/Mc	=	0.104
NrDL/Mc	=	0.120
M0d/Mc	=	0.274
N0DL/Mc	=	0.105
MrD/M1	=	0.097
NrDL/M1	=	0.112
M0D/M1	=	0.236
N0DL/M1	=	0.320

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-23	23	23	-23
Normal Pressure Stress	4	4	4	4
Outplane Stress Summary	-13	17	45	-19

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-9	9	9	-9
Inplane Pressure Stress	2	2	2	2
Inplane Stress Summary	-4	7	18	-8

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	15	19	45	20

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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Nozzle Calcs.: Shell Inlet

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(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-27	27	27	-27
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	2	2	2	2

Outplane Stress Summary	-15	22	43	-29
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-10	10	10	-10
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	4	4	4	4

Inplane Stress Summary	-3	10	21	-6
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	16	22	44	30
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	6	5	6

Hoop Stress Summary	3	3	13	14
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	3	3	3	3
Axial Bending (P)	26	-26	26	-26
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-28	-28	28	28
Axial Bending (ML)	-97	97	97	-97
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-93	49	159	-88
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	5	5	-5	-5
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	17	17	17	17
Shear Stress Summary	22	22	12	12

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	106	64	160	106

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	6	5	6
Hoop Stress Summary	6	7	10	10

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	26	-26	26	-26
Axial Membrane (Mc)	-28	-28	28	28
Axial Bending (Mc)	-105	105	105	-105
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-101	57	167	-96

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-5	-5	5	5
Shear due to Torsion	17	17	17	17



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Shear Stress Summary 12 12 22 22

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Stress Intensities Cu Cl Du D1
Outside Inside Outside Inside
Two * Max Shear Stress 110 60 170 116

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	4	4	4	4	4	4	4
Circ. Pl (SUS)	-2	-2	8	8	-2	-2	2	2
Circ. Q (SUS)	-15	15	31	-31	-7	7	13	-13
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-1	-1	1	1	2	2	4	4
Long. Q (SUS)	-6	6	12	-12	-19	19	35	-35
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
Pm+Pl (SUS)	2.6	2.6	12.1	12.1	4.4	4.4	7.0	7.0
Pm+Pl+Q (Total)	14.7	18.4	43.1	19.4	15.4	23.3	41.7	29.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	4.00	136.80	Passed
Pm+Pl (SUS)	12.11	205.20	Passed
Pm+Pl+Q (TOTAL)	43.14	410.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	6	5	6	5	6	5	6
Circ. Pl (SUS)	-2	-2	8	8	2	2	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-25	-25	31	31	-25	-25	31	31
Long. Q (SUS)	-71	71	123	-123	-79	79	131	-131
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	17	17	17	17	17	17	17	17



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Pm (SUS)	5.0	6.0	5.0	6.0	5.0	6.0	5.0	6.0
Pm+Pl (SUS)	27.9	28.8	34.2	34.2	31.6	32.6	34.0	34.0
Pm+Pl+Q (Total)	106.5	62.2	157.0	106.7	111.6	58.8	167.1	116.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	118.00	Passed
Pm+Pl (SUS)	34.24	177.00	Passed
Pm+Pl+Q (TOTAL)	167.06	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Nozzle Calcs.: Vent

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Input, Nozzle Desc: Vent

From: 50

Pressure for Reinforcement Calculations	P	520.000	KPa.
Temperature for Internal Pressure	Temp	280	°C
Design External Pressure	Pext	101.30	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	136.80	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	440.00	mm
Design Length of Section	L	3085.4167	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2539.85	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		1.6120	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	9.6901	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	136.80	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	100.0000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		19.8375	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



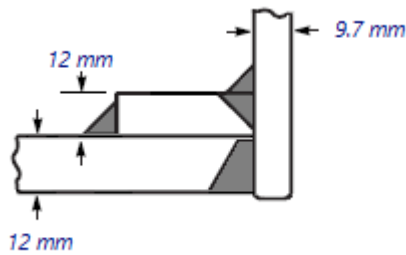
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Nozzle Calcs.: Vent

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Reinforcement CALCULATION, Description: Vent

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.612 in.
Actual Thickness Used in Calculation	0.382 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 520 \cdot 223 / (137 \cdot 1 - 0.6 \cdot 520)$
 $= 0.8497$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 520 \cdot 23.5 / (118 \cdot 1 - 0.6 \cdot 520)$
 $= 0.1037$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3158 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	93.8896 mm
Parallel to Vessel Wall, opening length	d	46.9448 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		22.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: Vent.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.3158 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 3.8497 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 5.5000 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(5.5, 5.5)]$
 $= 5.5000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.32, 5.5)$
 $= 5.5000$ mm



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Nozzle Calcs.: Vent

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Available Nozzle Neck Thickness = 9.6901 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.69, tr = 0.1, c = 3 mm, E* = 1

Thickness Ratio = $tr * E* / (tg - c) = 0.016$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.69, tr = 0.1, c = 3 mm, E* = 1

Thickness Ratio = $tr * E* / (tg - c) = 0.016$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12, tr = 0.84, c = 3 mm, E* = 1

Thickness Ratio = $tr * E* / (tg - c) = 0.094$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.69, tr = 0.1, c = 3 mm, E* = 1

Thickness Ratio = $tr * E* / (tg - c) = 0.016$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C

Gov. MDMT of the Reinforcement Pad : -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $520.00 / 1960.01 = 0.265$

Weld Size Calculations, Description: Vent

Intermediate Calc. for nozzle/shell Welds Tmin 6.6901 mm

Intermediate Calc. for pad/shell Welds TminPad 9.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness

Nozzle Weld $4.6831 = 0.7 * t_{min}$ 14.1400 = $0.7 * W_o$ mm

Pad Weld $4.5000 = 0.5 * t_{minPad}$ 7.0700 = $0.7 * W_p$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

**An Error occurred while iterating for the Mawp of this Nozzle.
Please check the input for this nozzle or contact PV Elite technical support.**



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Nozzle Calcs.: Vent

Nozl: 17

9:32am

Sep 9, 2026

The Drop for this Nozzle is : 2.0775 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.0775 mm

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Nozzle Calcs.: Drain

Nozl: 18 9:32am Sep 9,2026

Input, Nozzle Desc: Drain

From: 50

Pressure for Reinforcement Calculations	P	523.921	KPa.
Temperature for Internal Pressure	Temp	280	°C
Design External Pressure	Pext	101.30	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	136.80	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	440.00	mm
Design Length of Section	L	3085.4167	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2539.85	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		1.6120	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	9.6901	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	136.80	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	100.0000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		19.8375	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



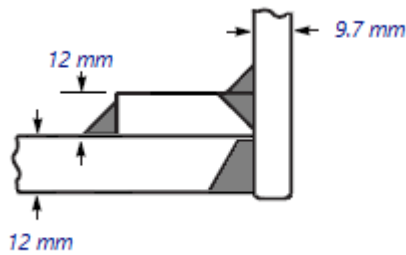
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Nozzle Calcs.: Drain

Nozl: 18 9:32am Sep 9,2026



Reinforcement CALCULATION, Description: Drain

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.612 in.
Actual Thickness Used in Calculation	0.382 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 524 \cdot 223 / (137 \cdot 1 - 0.6 \cdot 524)$
 $= 0.8561$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 524 \cdot 23.5 / (118 \cdot 1 - 0.6 \cdot 524)$
 $= 0.1045$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3158 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	93.8896 mm
Parallel to Vessel Wall, opening length	d	46.9448 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		22.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: Drain.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.3158 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 3.8561 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 5.5000 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(5.5, 5.5)]$
 $= 5.5000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.32, 5.5)$
 $= 5.5000$ mm



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Nozzle Calcs.: Drain

Nozl: 18 9:32am

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Available Nozzle Neck Thickness = 9.6901 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.69, tr = 0.1, c = 3 mm, E* = 1

Thickness Ratio = $tr * E* / (tg - c) = 0.016$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.69, tr = 0.1, c = 3 mm, E* = 1

Thickness Ratio = $tr * E* / (tg - c) = 0.016$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.69, tr = 0.1, c = 3 mm, E* = 1

Thickness Ratio = $tr * E* / (tg - c) = 0.016$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12, tr = 0.85, c = 3 mm, E* = 1

Thickness Ratio = $tr * E* / (tg - c) = 0.094$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.69, tr = 0.1, c = 3 mm, E* = 1

Thickness Ratio = $tr * E* / (tg - c) = 0.016$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $523.92 / 1960.01 = 0.267$

Weld Size Calculations, Description: Drain

Intermediate Calc. for nozzle/shell Welds	Tmin	6.6901 mm
Intermediate Calc. for pad/shell Welds	TminPad	9.0000 mm

Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Calcs.: Drain

Nozl: 18 9:32am Sep 9, 2026

Nozzle Weld $4.6831 = 0.7 * t_{min}$. $14.1400 = 0.7 * W_o$ mm

Pad Weld $4.5000 = 0.5 * T_{minPad}$ $7.0700 = 0.7 * W_p$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

***An Error occurred while iterating for the Mawp of this Nozzle.
Please check the input for this nozzle or contact PV Elite technical support.***

The Drop for this Nozzle is : 2.0775 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.0775 mm

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ASME TS Calc: DESIGN 1

Case: 1 9:32a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	520.00 KPa.
Shell Maximum Operating Pressure	Psox,max	520.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	12.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	440.000 mm
Shell Temperature for Internal Pressure	Ts	280.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	136.81 N./mm ²
Shell Allowable Stress at Ambient		138.01 N./mm ²

Channel Description: Channel shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	6400.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	6400.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	16.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	440.000 mm
Channel Design Temperature	TEMPC	160.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.01 N./mm ²
Channel Allowable Stress at Ambient		138.01 N./mm ²

Tube Data:

Number of Tube Holes	Nt	106
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	3000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		2898.00 mm
Design Temperature of the Tubes		280.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		92.10 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	143.20 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	199.046 mm
Largest Center-to-Center Tube Distance	U1	76.2000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm



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ASME TS Calc: DESIGN 1

Case: 1 9:32a Sep 9, 2026

Tube-side pass partition groove depth	hg	5.0000 mm
Area of Tube Layout	Ap	936.4 cm ²
Perimeter of Tube Layout	Cp	1155.6 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature	T	280.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	131.80 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	102.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	670.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	457.00 mm
Step 1 Depth on the Tubeside	ht1	6.00 mm
Step 2 Diameter on the Tubeside	dt2	397.00 mm
Step 2 Depth on the Tubeside	ht2	5.00 mm
Is the Tubesheet Flat on Shellside?		No
Calculated Tubesheet Diameter per Div 2, 4.18.4		670.00 mm

Note: Tubesheet diameter is now: 670.000 mm Div 2, 4.18.4.

Dimension G for the Channel Side	Gc	482.151 mm
Area of the Untubed Lanes	AL	203.2 cm ²

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	670.000 mm
Flange Inside Diameter	B	440.000 mm
Flange Face Outside Diameter	Fod	506.000 mm
Flange Face Inside Diameter	Fid	440.000 mm
Gasket Outside Diameter	Go	495.000 mm
Gasket Inside Diameter	Gi	469.000 mm
Small end Hub thk.	g0	12.0000 mm
Large end Hub thk.	g1	17.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch	Code Sketch 1a	
Column for Gasket Seating	Code Column II	
Gasket Thickness	tg	4.5000 mm
Full face Gasket Flange Option	Program Selects	
Length of Partition Gasket	lp	425.000 mm
Width of Partition Gasket	wp	10.0000 mm
Partition Gasket Factor,	mPart	3.0000
Partition Gasket Design Seating Stress	yPart	68.95 N./mm ²
Partition Gasket Facing Sketch	Code Sketch 1a	
Partition Gasket Column for Gasket Seating	Code Column II	

Bolting Information:

Diameter of Bolt Circle	C	590.000 mm
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ASME TS Calc: DESIGN 1

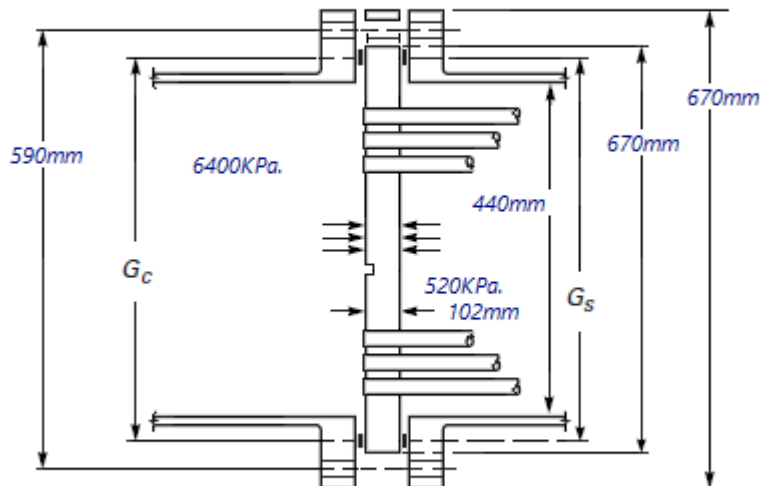
Case: 1 9:32a Sep 9, 2026

Nominal Bolt Diameter	dB	38.1000 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	20
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.01 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm
Alternate Flange Operating Bolt Load, Wm1		165696.47 Kgf
Alternate Flange Seating Bolt Load, Wm2		83360.32 Kgf
Alternate Flange Design Bolt Load, W		317955.00 Kgf

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with		None
Tubesheet Extended as Flange		Yes
Thickness of Extended Portion of Tubesheet	Tf	70.0000 mm
Is Bolt Load Transferred to the Tubesheet		Yes

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2023

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	280.0 °C	0.18624E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	160.0 °C	0.19457E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	280.0 °C	0.18624E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.



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ASME TS Calc: DESIGN 1

Case: 1 9:32a Sep 9, 2026

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	280.0 °C	0.18624E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (6400 \cdot (25.4/2 - 0)) / (92.1 \cdot 1 + 0.4 \cdot 6400)$$

$$= 0.8587 + 0.0000 = 0.8587 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	CS-1	at	280.00 °C
Elastic Modulus for Material			183055328.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	3097.05	12.04	50.0000	0.0075908	79.64

$$EMAWP = (4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 79.6) / (3 \cdot 12) = 8820 \text{ KPa.}$$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3989	25.40	3097.05	63.67	50.0000	0.0002713	24.83

$$EMAWP = (4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 24.8) / (3 \cdot 63.7) = 520 \text{ KPa.}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.8587 mm
Allowable Internal Pressure at Corroded thickness	16389.73 KPa.
Required Internal Design Pressure	6400.00 KPa.
Allowable External Pressure at Corroded thickness	8820.19 KPa.
Required External Design Pressure	520.00 KPa.
Required Thickness due to Shell Side pressure	0.3989 mm

Detailed Results for load Case D1 corr. (Psd,min + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2023

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	482.156 mm

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2023, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Compute the equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 199 + 25.4$$

$$= 423.492 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$



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ASME TS Calc: DESIGN 1

Case: 1 9:32a Sep 9, 2026

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Compute the ratio [ps]:

$$= G_s / D_o \text{ (Configurations d, e, f)}$$

$$= 482/423$$

$$= 1.1385$$

Compute the ratio [pc]:

$$= G_c / D_o \text{ (Configurations d)}$$

$$= 482/423$$

$$= 1.1385$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Compute the ratio [p]:

$$= l_{tx} / h$$

$$= 0/96$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Compute the effective tube hole Diameter [d*]:

$$= \max(dt - 2t_t (E_t/E) \cdot (StT/S) \cdot (p); dt - 2t_t)$$

$$= \max(25.4 - 2 \cdot 2.11 (186239296/186239296) \cdot$$

$$(92.1/132) \cdot (0); 25.4 - 2 \cdot 2.11)$$

$$= 25.4000 \text{ mm}$$

Compute the effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(203 ; 4 \cdot 423 \cdot 31.8) / (\pi \cdot 423^2)) }$$

$$= 34.3220 \text{ mm}$$

Compute the effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (34.3 - 25.4) / 34.3$$

$$= 0.2600$$

$h/p = 3.023622$; $\mu^* = 0.259950$; $E^* = 42696768 \text{ KPa}$.

$$E^*/E = 0.229258 ; v^* = 0.386557$$

Note: As h/p (3.024) is > 2, data values for $h/p = 2$ were used.

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 670/423$$

$$= 1.5821$$

Compute coefficient [F]:

$$= (1 - v^*) / E^* \cdot (E \cdot \ln(K))$$

$$= (1 - 0.39) / 42696768 \cdot (186239296 \cdot \ln(1.58))$$

$$= 1.2275$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2/16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 423^2/16 \cdot [(1.14 - 1) \cdot (1.14^2 + 1) \cdot 0 -$$

$$(1.14 - 1) \cdot (1.14^2 + 1) \cdot 6400]$$

$$= -22816584.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M^*]

$$= M_{ts} + W^* \cdot (G_c - G_s) / (2 \cdot \pi \cdot D_o)$$



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$$= -22816584 + 317955 \cdot (482 - 482) / (2 \cdot \pi \cdot 423)$$

$$= -22816584.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M_p - D_o^2/32 \cdot F(P_s - P_t)) / (1 + F)$$

$$= (-22816584 - 423^2/32 \cdot 1.23(0 - 6400)) / (1 + 1.23)$$

$$= 9523049.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2/64 \cdot (3 + \nu) \cdot (P_s - P_t)$$

$$= 9523049 + 423^2/64 \cdot (3 + 0.39) \cdot (0 - 6400)$$

$$= -51213328.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|9523049|; |-51213328|)$$

$$= 51213328.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 51213328 / (0.26(96 - 2)^2)$$

$$= 133.7870 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 132$$

$$= 263.60 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 51213328 / (0.26(69 - 2)^2)$$

$$= 263.4656 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 69 + 3 + 3$$

$$= 74.9843 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(75; 30.6)$$

$$= 74.9843 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$= |P_s - P_t|$$

$$= |0 - 6400|$$

$$= 6399.999 \text{ KPa}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 96 / 423 \cdot \min(0.8 \cdot 132; 0.533 \cdot 198)$$

$$= 19.122 \text{ N./mm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 105) \cdot (4 \cdot 936 / 1156) \cdot |0 - 6400| + 6$$



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= 30.5927 mm

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [At]:

= $\pi (d_o - t) t$

= $\pi (25.4 - 2.11) 2.11$

= 1.544 cm²

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

= $\pi \cdot t (d_o - t) S_a$

= $\pi \cdot 2.11 (25.4 - 2.11) 92.1$

= 1449.860 Kgf

Groove Weld Strength [Fg]:

= $0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not > Ft)

= $0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 92.1$

= 1418.8362 Kgf

Weld Strength Factor [fw]:

= S_a / S_w where $S_w = \min(S_a \text{ and } S_t)$

= 92.1/92.1

= 1.000

Design Strength Ratio [fd]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

= $0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > Ft)

= $0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 92.1$

= 918.0706 Kgf

Minimum Length of Weld [ar]:

= $2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d))} - 0.75 \cdot d_o \right)$

= $2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1 \cdot 1))} - 0.75 \cdot 25.4 \right)$

= 2.667 mm

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 1450 is >= tube strength 1450 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	133.8 <=	263.6 N./mm ²	Ok
Tubesheet shear stress	27.0 <=	105.4 N./mm ²	Ok

Thickness results for loadcase D1 corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	74.984	102.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok

Detailed Results for load Case D2 corr. (Psd,max + Ptd,min)



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Intermediate Calculations For Tubesheets Extended As Flanges:

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Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	13.000 mm
Basic Gasket Width,	$b_0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b_0) * 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = G_o - 2.0 * b$	482.156 mm

Required Tubesheet Flanged Extension thk. per 4.18.5 (operating) [h_{opr}]:

$$= \sqrt{\frac{1.9 \cdot W M_1 \cdot h_g}{(S \cdot G)}}$$

$$= \sqrt{\frac{1.9 \cdot 13453 \cdot 53.9}{(132 \cdot 482)}}$$

$$= 14.584 \text{ mm}$$

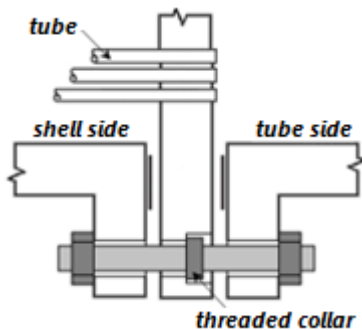
Required Tubesheet Flanged Extension thk. per 4.18.5 (seating) [h_{seat}]:

$$= \sqrt{\frac{1.9 \cdot W \cdot h_g}{(S_a \cdot G)}}$$

$$= \sqrt{\frac{1.9 \cdot 317955 \cdot 53.9}{(138 \cdot 482)}}$$

$$= 69.292 \text{ mm}$$

In this tubesheet configuration, it is assumed that the tubesheet flange extension is primarily influenced by the shell side loading as shown below:



Gasket Reaction Diameter (G_c or G_s as applicable): 482.156 mm

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2023, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Compute the equivalent outer tube limit circle diameter [D_o]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 199 + 25.4$$

$$= 423.492 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Compute the ratio [ρ_s]:

$$= G_s / D_o \text{ (Configurations d, e, f)}$$

$$= 482 / 423$$

$$= 1.1385$$

Compute the ratio [ρ_c]:



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= Gc / Do (Configurations d)
= 482/423
= 1.1385

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Compute the ratio [p]:

= ltx / h
= 0/96
= 0.000 (must be $0 \leq \rho \leq 1$)

Compute the effective tube hole Diameter [d*]:

= max(dt - 2tt(EtT/E) • (StT/S) • (p) ; dt - 2tt)
= max(25.4 - 2 • 2.11 (186239296/186239296) •
(92.1/132) • (0) ; 25.4 - 2 • 2.11)
= 25.4000 mm

Compute the effective tube pitch [p*]:

= p / $\sqrt{(1 - 4 \cdot \min(AI; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))}$
= 31.8 / $\sqrt{(1 - 4 \cdot \min(203; 4 \cdot 423 \cdot 31.8) / (\pi \cdot 423^2))}$
= 34.3220 mm

Compute the effective ligament efficiency for bending [μ^*]:

= (p* - d*) / p*
= (34.3 - 25.4) / 34.3
= 0.2600

h/p = 3.023622 ; μ^* = 0.259950 ; E* = 42696768 KPa.

E*/E = 0.229258 ; ν^* = 0.386557

Note: As h/p (3.024) is > 2, data values for h/p = 2 were used.

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

= A / Do
= 670/423
= 1.5821

Compute coefficient [F]:

= (1 - ν^*) / E* • (E • ln(K))
= (1 - 0.39) / 42696768 • (186239296 • ln(1.58))
= 1.2275

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

= Do²/16 • [(ps-1) • (ps²+1) • Ps - (pc-1) • (pc²+1) • Pt]
= 423²/16 • [(1.14 - 1) • (1.14² + 1) • 520 -
(1.14 - 1) • (1.14² + 1) • 0]
= 1854015.8750 KPa•mm²

Moment acting on unperforated tubesheet rim [M*]

= Mts + W* • (Gc - Gs) / (2 • π • Do)
= 1854015 + 317955 • (482 - 482) / (2 • π • 423)
= 1854015.8750 KPa•mm²

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

= ((M*) - Do²/32 • F(Ps - Pt)) / (1 + F)
= ((1854015) - 423²/32 • 1.23 (520 - 0)) / (1 + 1.23)
= -773671.8125 KPa•mm²



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Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= M_p + D_o^2/64 \cdot (3 + \nu) \cdot (P_s - P_t) \\
 &= -773672 + 423^2/64 \cdot (3 + 0.39) \cdot (520 - 0) \\
 &= 4161158.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= \max(|M_p|; |M_o|) \\
 &= \max(|-773672|; |4161158|) \\
 &= 4161158.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\
 &= 6 \cdot 4161158 / (0.26 \cdot (96 - 2)^2) \\
 &= 10.8704 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 \cdot S \\
 &= 2 \cdot 132 \\
 &= 263.60 \text{ N./mm}^2
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\
 &= 6 \cdot 4088594 / (0.26 \cdot (20.9 - 2)^2) \\
 &= 263.5779 \text{ N./mm}^2
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned}
 &= h + C_{ats} + C_{atc} \\
 &= 20.9 + 3 + 3 \\
 &= 26.9224 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(H_{reqB}; H_{reqS}) \\
 &= \max(26.9; 8) \\
 &= 26.9224 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$\begin{aligned}
 &= |P_s - P_t| \\
 &= |520 - 0| \\
 &= 520.000 \text{ KPa.}
 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned}
 &= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\
 &= 4 \cdot 0.2 \cdot 96 / 423 \cdot \min(0.8 \cdot 132; 0.533 \cdot 198) \\
 &= 19.122 \text{ N./mm}^2
 \end{aligned}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned}
 &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\
 &= 1 / (4 \cdot 0.2 \cdot 105) \cdot (4 \cdot 936 / 1156) \cdot |520 - 0| + 6 \\
 &= 7.9982 \text{ mm}
 \end{aligned}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [At]:

$$\begin{aligned}
 &= \pi (d_o - t) t \\
 &= \pi (25.4 - 2.11) 2.11 \\
 &= 1.544 \text{ cm}^2
 \end{aligned}$$



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ASME TS Calc: DESIGN 1

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Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t \cdot (d_o - t) S_a$$

$$= \pi \cdot 2.11 \cdot (25.4 - 2.11) 92.1$$

$$= 1449.860 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g \cdot (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 \cdot (25.4 + 0.67 \cdot 2.11) 92.1$$

$$= 1418.8362 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S_a / S_w \text{ where } S_w = \min(S_a \text{ and } S_t)$$

$$= 92.1 / 92.1$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f \cdot (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.11 \cdot (25.4 + 0.67 \cdot 2.11) 92.1$$

$$= 918.0706 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \cdot \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \cdot \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) \cdot 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$$

$$= 2.667 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 1450 is $\geq$ tube strength 1450 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D2 corr. (Psd,max + Ptd,min):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	10.9 $\leq$	263.6 N./mm ²	Ok
Tubesheet shear stress	2.2 $\leq$	105.4 N./mm ²	Ok

Thickness results for loadcase D2 corr. (Psd,max + Ptd,min):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	26.922	102.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok

Detailed Results for load Case D3 corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

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Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	13.000 mm
Basic Gasket Width,	$b_0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \sqrt{b_0} \cdot 2.5$	6.422 mm



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Gasket Reaction Diameter, $G = G_o - 2.0 \cdot b$ 482.156 mm

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2023, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Compute the equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 199 + 25.4 \\ &= 423.492 \text{ mm} \end{aligned}$$

Determine the basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (31.8 - 25.4) / 31.8 \\ &= 0.200 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Compute the ratio [ps]:

$$\begin{aligned} &= G_s / D_o \text{ (Configurations d, e, f)} \\ &= 482 / 423 \\ &= 1.1385 \end{aligned}$$

Compute the ratio [pc]:

$$\begin{aligned} &= G_c / D_o \text{ (Configurations d)} \\ &= 482 / 423 \\ &= 1.1385 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Compute the ratio [p]:

$$\begin{aligned} &= l_{tx} / h \\ &= 0 / 96 \\ &= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Compute the effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(dt - 2tt(E_t T / E) \cdot (St T / S) \cdot (\rho); dt - 2tt) \\ &= \max(25.4 - 2 \cdot 2.11 (186239296 / 186239296) \cdot \\ &\quad (92.1 / 132) \cdot (0); 25.4 - 2 \cdot 2.11) \\ &= 25.4000 \text{ mm} \end{aligned}$$

Compute the effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A / (4 \cdot D_o \cdot p); (\pi \cdot D_o^2))) } \\ &= 31.8 / \sqrt{ (1 - 4 \cdot \min(203 ; 4 \cdot 423 \cdot 31.8) / (\pi \cdot 423^2)) } \\ &= 34.3220 \text{ mm} \end{aligned}$$

Compute the effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (34.3 - 25.4) / 34.3 \\ &= 0.2600 \end{aligned}$$

$h/p = 3.023622$; $\mu^* = 0.259950$; $E^* = 42696768 \text{ KPa}$.

$E^*/E = 0.229258$; $v^* = 0.386557$

Note: As h/p (3.024) is > 2 , data values for $h/p = 2$ were used.

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$



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$$= 670/423$$

$$= 1.5821$$

Compute coefficient [F]:

$$= (1 - \nu) / E * (E * \ln(K))$$

$$= (1 - 0.39) / 42696768 * (186239296 * \ln(1.58))$$

$$= 1.2275$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2/16 * [(ps-1) * (ps^2+1) * Ps - (pc-1) * (pc^2+1) * Pt]$$

$$= 423^2/16 * [(1.14 - 1) * (1.14^2 + 1) * 520 - (1.14 - 1) * (1.14^2 + 1) * 6400]$$

$$= -20962568.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + W * (Gc - Gs) / (2 * \pi * Do)$$

$$= -20962568 + 317955 * (482 - 482) / (2 * \pi * 423)$$

$$= -20962568.0000 \text{ KPa} \cdot \text{mm}^2$$

Note: W is the maximum of the bolt loads between the shell and channel sides.*

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M) - Do^2/32 * F(Ps - Pt) / (1 + F)$$

$$= (-20962568) - 423^2/32 * 1.23(520 - 6400) / (1 + 1.23)$$

$$= 8749378.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 * (3 + \nu) * (Ps - Pt)$$

$$= 8749378 + 423^2/64 * (3 + 0.39) * (520 - 6400)$$

$$= -47052168.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|Mp|; |Mo|)$$

$$= \max(|8749378|; |-47052168|)$$

$$= 47052168.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 * M / (\mu * (h - hg')^2)$$

$$= 6 * 47052168 / (0.26(96 - 2)^2)$$

$$= 122.9166 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 * S$$

$$= 2 * 132$$

$$= 263.60 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 * M / (\mu * (h - hg')^2)$$

$$= 6 * 47052168 / (0.26(66.2 - 2)^2)$$

$$= 263.4380 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 66.2 + 3 + 3$$

$$= 72.2087 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB; HreqS)$$

$$= \max(72.2; 28.6)$$



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= 72.2087 mm

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

= $|P_s - P_t|$

= $|520 - 6400|$

= 5879.999 KPa.

Shear Stress check [τ_{limit}]:

= $4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$

= $4 \cdot 0.2 \cdot 96 / 423 \cdot \min(0.8 \cdot 132; 0.533 \cdot 198)$

= 19.122 N./mm²

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

= $1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$

= $1 / (4 \cdot 0.2 \cdot 105) \cdot (4 \cdot 936 / 1156) \cdot |520 - 6400| + 6$

= 28.5945 mm

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [A_t]:

= $\pi (d_o - t) t$

= $\pi (25.4 - 2.11) 2.11$

= 1.544 cm²

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:

= $\pi \cdot t (d_o - t) S_a$

= $\pi \cdot 2.11 (25.4 - 2.11) 92.1$

= 1449.860 Kgf

Groove Weld Strength [F_g]:

= $0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not > F_t)

= $0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 92.1$

= 1418.8362 Kgf

Weld Strength Factor [f_w]:

= S_a / S_w where $S_w = \min(S_a \text{ and } S_t)$

= $92.1 / 92.1$

= 1.000

Design Strength Ratio [f_d]:

= 1 for full strength welds

Fillet Weld Strength [F_f]:

= $0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > F_t)

= $0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 92.1$

= 918.0706 Kgf

Minimum Length of Weld [a_r]:

= $2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$

= $2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$

= 2.667 mm

Minimum Required Fillet Weld Leg [a_{fr}]: 1.3334 mm

Minimum Required Groove Weld Leg [a_{gr}]: 1.3334 mm

Tube-Tubesheet Jt allowable, 1450 is $\geq$ tube strength 1450 Kgf



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ASME TS Calc: DESIGN 1

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Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	122.9 <=	263.6 N./mm ²	Ok
Tubesheet shear stress	24.8 <=	105.4 N./mm ²	Ok

Thickness results for loadcase D3 corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	72.209	102.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2023

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	72.085 ...	126	264	25 105	Ps+Pt D1	Ok
D2uc	23.989 69.292	10	264	2 105	Ps+Pt D2	Ok
D3uc	69.324 ...	115	264	23 105	Ps+Pt D3	Ok
D1c	74.984 ...	134	264	27 105	Ps+Pt-c D1	Ok
D2c	26.922 69.292	11	264	2 105	Ps+Pt-c D2	Ok
D3c	72.209 ...	123	264	25 105	Ps+Pt-c D3	Ok
Max:	74.9843 69.292 mm	0.508	0.256	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	74.984	102.000	Ok
Tubesheet thickness flanged extension :	69.292	70.000	Ok
Tube thickness :	0.859	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok



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ASME TS Calc: DESIGN 1

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Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	12609.97	1.000	12609.74	1.000
Tubesheet Shear Stress	24983.00	1.000	24983.00	1.000
Tube Pressure Stress	16389.72	1.000	8820.18	1.000
Tubesheet Extension Stress	No Calc	No Calc	11979.16	...
Minimum MAWP	12609.97		8820.18	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	14059.01	1.000	14059.01	1.000
Tubesheet Shear Stress	27792.41	1.000	27792.40	1.000
Tube Pressure Stress	16443.07	1.000	10612.12	1.000
Tubesheet Extension Stress	No Calc	No Calc	12289.78	...
Minimum MAPnc	14059.01		10612.12	

(*) All load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Determine the governing MDMT considering the governing condition:

Governing thickness on the shell side per figure UCS-66.3 (c):

= tubesheet thickness/4

= 102/4

= 25.500 mm

Thickness Ratio = 0.46, Temperature Reduction per Fig. UCS 66.1 = 39 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 0 °C

Min Metal Temp. at Required thickness (UCS 66.1) -39 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:

= max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1

= max(6400/14059, 520/10612)

= 0.455

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Problems/Failures Summary:

Step: 36

9:32am

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: One or more Nozzles failed Code Requirements!**

**** Warning: A standard flange seems to be governing the Shellside MAWP!**

Please review all reports carefully!

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